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Synthetic self-amplifying mRNA molecules with secretion antigen and immunomodulator

Abstract

Lipid nanoparticle (LNP) encapsulating self-amplifying mRNA, compositions, and methods of using the novel nucleic acid constructs and compositions are disclosed. LNP constructs include novel ionizable lipid. Novel sa-mRNA constructs encode a modified SARS-COV-2 spike protein, wherein the polynucleotide has been truncated to not include nucleotides encoding a SARS-COV-2 transmembrane domain and short cytosolic domain amino acids and immunomodulators. Sa-mRNAs are useful in for use as a therapeutic, diagnostic and/or prophylactic agent to mammalian cells or organs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 18/316,033, filed May 11, 2023, which claims priority to U.S. Provisional Application No. 63/341,018, filed on May 12, 2022, and U.S. Provisional Application No. 63/393,688, filed on Jul. 29, 2022, the disclosures of each of which is incorporated by reference herein in its entirety for all purposes.

SEQUENCE LISTING STATEMENT

(1) The instant application contains a Sequence Listing in electronic format which has been submitted electronically. Said Sequence Listing, created on Nov. 1, 2024, is named “5292-102US2ST26.xml” and is 192871 bytes in size. The information in the electronic format of the Sequence Listing is part of the present application and is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure provides novel self-amplifying mRNA (sa-mRNA) constructs, compositions comprising such constructs, and methods to deliver one or more biologically active agents to a subject in need thereof.

BACKGROUND

(3) mRNA vaccine platforms are able to stimulate humoral and cellular immune responses to foreign antigen that are encoded, and are an improvement to traditional vaccines because they allow for rapid, scalable, and cell free manufacturing. However, achieving adequate antigen expression for protection or immunomodulation is a medical challenge for existing mRNA vaccines because the number of mRNA transcripts available in vivo is proportional to the number of mRNA transcripts successfully delivered during vaccination, thus existing mRNA vaccines require large

doses or repeated administrations. Large doses and repeated administrations of mRNA vaccines, circular mRNA, and sa-mRNA vaccines are undesirable because large doses of the mRNA, circular mRNA, or sa-mRNA can elicit undesirable immune responses and repeated administration can render subsequent administration of the same vaccine less effective.

(4) Sa-mRNA is a kind of mRNA with the ability to replicate itself in a cell and amplify the expression of cargo genes, e.g., a gene of interest. However, achieving sufficient production of the molecules and interferon responses of sa-mRNA remain difficult due to the large size of sa-mRNA and the immunogenicity due to its origination from alphaviruses. Thus, there remains a need to increase intracellular mRNA transcripts in vivo, to produce better immune response at lower doses and avoid safety challenges.

(5) Delivery of biologically active agents, including sa-mRNA, is also a medical challenge due to the inherent properties of RNA, including its highly negative charges and its size, which is much larger than modified mRNA and circular mRNA. In particular, the delivery of biologically active agents to cells is made difficult by the relative instability and low cell permeability of such molecules and safety concerns due to cytotoxicity. Ionizable lipids, one component of LNPs, are believed to play key role in uptake of LNPs by cells and the release of LNPs from the endosome. Thus, there exists a need to develop compounds, compositions, and methods for improved delivery of therapeutic, diagnostic and/or prophylactic molecules into cells or organs.

SUMMARY OF THE INVENTION

(6) The present invention addresses these needs by providing novel ionizable lipids, nanoparticle compositions and sa-mRNA, which improve the delivery of biologically active agents into cells or organs while reducing safety concerns associated with incompatibly high transcript levels and/or rapid decay of the biologically active agent leading to increased administrations. The present disclosure provides novel sa-mRNA and compositions and methods involving the same.

(7) In one aspect, the present disclosure provides a method of increasing transfection efficiency and decreasing cytotoxicity of a nanoparticle formulation by using a novel ionizable lipid in the LNP formulation. In some aspects, the method and compositions include the ionizable lipid Formula E6 (1,1',1''-(1,3,5-triazinane-1,3,5-triyl)tris(3-(ditridecylamino) propan-1-one)):

(8) ##STR00001##

(9) In some aspects, the method and compositions include the ionizable lipid Formula E2 (1,1',1''-(1,3,5-triazinane-1,3,5-triyl)tris(3-(dinonylamino) propan-1-one)):

(10) ##STR00002##

(11) In some aspects, the method and compositions include the ionizable lipid Formula P6 (N-(2-(cyclohex-1-en-1-ylamino)-1-(1-ethylpiperidin-4-yl)-2-oxoethyl)-N-(heptadecan-9-yl)palmitamide):

(12) ##STR00003##

(13) In one aspect, the sa-mRNA of the present disclosure is delivered to a host cell by an LNP formulated with an ionizable lipid, a helper lipid, a cholesterol, and/or a PEG-lipid. The present disclosure incorporates the ionizable lipid components of PCT Patent Application No.

PCT/US2023/017777, which is fully incorporated herein. In one aspect, the LNP has a molar ratio of about 2-60% ionizable lipid, about 5-40% helper lipid, about 30-80% cholesterol and about 0.5-30% PEG-lipid. The present disclosure incorporates any integer or fraction thereof within the recited ranges as if expressly written herein. In one aspect, the LNP has a molar ratio of about 5-50% or 8 to 40% or 10 to 30% ionizable lipid, about 10-30% or 13 to 25% or 15 to 20% helper lipid, about 40-70% or 45 to 65% or 50 to 60% cholesterol and about 1-20% or 3-15% or 5 to 10% PEG-lipid. In one aspect, the LNP has a molar ratio of about 2-10% ionizable lipid, about 5-15% helper lipid, about 40-80% cholesterol and about 0.5-3% PEG-lipid. In one aspect, the ionizable lipid is E6. In one aspect, the helper lipid is independently selected from DOPE (2-dioleoyl-sn-glycero-3-phosphoethanolamine), DSPC (1,2-distearoyl-sn-glycero-3-phosphocholine), and POPE (1-palmitoyl-2-oleoyl-sn-glycero-3-phosphoethanolamine). In one aspect, the LNP of the present

disclosure, the ionizable lipid is E6, the helper lipid is DOPE and the PEG-lipid is DMG-PEG2000. In one aspect of the disclosure, the LNP is composed of E6, DOPE, cholesterol, and DMG-PEG-2000. In one aspect, the LNP of the disclosure has a molar ratio of 50% ionizable lipid, 10% helper lipid, 38.5% cholesterol, and 1.5% PEG-lipid. In another aspect, the LNP of the disclosure has a molar ratio of 7.5% ionizable lipid, 15% helper lipid, 75% cholesterol, and 2.5% PEG-lipid. In another aspect, the LNP of the disclosure has a molar ratio of 5% ionizable lipid, 10% helper lipid, 50% cholesterol, and 1.5% PEG-lipid. In one aspect, the ionizable lipid is E6, the helper lipid is DOPE and the PEG-lipid is DMG-PEG2000.

(14) In one aspect, the biologically active agent is a nucleic acid molecule, and the nucleic acid molecule is RNA or DNA. In one aspect, the biologically active agent is RNA and the RNA is mRNA, tRNA, rRNA, siRNA, or snRNA. In one aspect of the present disclosure, the biologically active agent is sa-mRNA. In one aspect, the mRNA is chemically modified. In one aspect, the chemically modified mRNA is composed of nucleotides selected from the group 1-methyl-pseudouridine, 5-methyl-uridine, and 5-methyl-cytidine. In one aspect, the biologically active agent is a sa-mRNA encoding one or more antigens and one or more immunomodulators. In one aspect, the encoded antigen is a viral antigen. In one aspect, the encoded antigen is a modified SARS-COV-2 spike protein. In one aspect, the immunomodulator is a cytokine, a chemokine, or other immune stimulator or inhibitor.

(15) In one aspect, the present disclosure provides a method of increasing the copy number of a nucleic acid encoding two expression units comprising: i) an origin of replication sequence (Ori); ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter; and iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter, wherein the first expression unit encodes a selectable marker and the second expression unit encodes a self-amplifying mRNA; b) selecting cells that express the selectable marker; c) subculturing the selected cells to obtain a population of cells that express the selectable marker; and d) propagating the population of cells to increase the copy number of the nucleic acid. In some aspects, the nucleic acid is an RNA molecule. In one aspect, the nucleic acid molecule of the present disclosure is a recombinant DNA molecule. In one aspect, the nucleic acid molecule of the present disclosure is a closed circular molecule or a linear molecule. In one aspect, said nucleic acid molecule is a plasmid. In one aspect, the initial nucleic acid encoding two expression units is synthesized using standard synthetic techniques, e.g., using an automated DNA synthesizer.

(16) In one aspect, the nucleic acid includes a replication system allowing it to be maintained in the host for expression or for cloning and amplification. A nucleic acid may be present in the cell in either high or low copy number. Generally, about 5 to about 200 times of mRNA copies of house-keeping gene beta-Actins will be present within a host cell. A host cell containing a high copy number of mRNA transcripts will preferably contain at least about 10 to about 20 times mRNA copies of house-keeping gene beta-Actins. A host cell containing a low number of nucleic acid will preferably contain about 1 to 10, and usually about 1 to 4 times mRNA copies of house-keeping gene beta-Actins. The copy number of a nucleic acid including mRNA transcripts may be controlled by selection of different origins of replication according to methods known in the art. Sambrook and Russell, *Molecular Cloning: A Laboratory Manual*, 3rd edition (Jan. 15, 2001) Cold Spring Harbor Laboratory Press, ISBN: 0879695765.

(17) In one aspect, the first expression unit comprises the following operably linked nucleic acid sequence in a 5' to 3' direction or in a 3' to 5' direction: [Pr1]-[SM] wherein, Pr1 is a first promoter sequence, and SM is a selectable marker. In some aspects, the first promoter is a promoter that is recognized by bacterial machinery and drives transcription of the encoded selective marker. In some aspects, the first expression unit encodes a selectable marker to allow for the selection of bacterial host cells that have been transformed. Selectable markers can be expressed in the bacterial host cell and may include genes which render bacteria resistant to drugs

such as ampicillin, kanamycin (neomycin), chloramphenicol, erythromycin, and tetracycline (Davies et al., Ann. Rev. Microbiol., 32:469 (1978)).

(18) In one aspect, the second expression unit comprises the following operably linked nucleic acid sequence from 5' to 3': Pr2-5'UTR-nsP-SGP-GOI-3'UTR-PolyA

wherein, Pr2 is a second promoter sequence, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, GOI is one or more gene or genes of interest, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail (poly-A tail). In some aspects, when there is more than one GOI, each GOI is operably linked to its own SGP.

(19) In some aspects, the second promoter is a promoter that drives transcription of the encoded self-amplifying mRNA using the second expression unit as a template for in vitro transcription of nucleic acid, e.g. mRNA. Suitable promoters include, for example, T7 promoter, T3 promoter, SV40 promoter, SP6 promoter, T5 promoter, β -lactamase promoter, *E. coli* galactose promoter, arabinose promoter, alkaline phosphatase promoter, tryptophan (trp) promoter, lactose operon (lac) promoter, lacUV5 promoter, trc promoter, tac promoter, and the like, or mutants of these promoters. A sa-mRNA can be prepared by transcribing (e.g., in vitro transcription) a DNA that encodes the sa-mRNA using a suitable DNA-dependent RNA polymerase, such as: T7 phage RNA polymerase, SP6 phage RNA polymerase, T3 phage RNA polymerase, T5 phage RNA polymerase, RNA polymerase III, RNA polymerase II, Taq polymerase, Vent polymerase, and the like, or mutants of these polymerases. The transcription reaction will contain nucleotides, including modified nucleotides in some aspects, and other components that support the activity of the selected polymerase, such as a suitable buffer, and suitable salts. In some aspects, nucleotide analogs will be incorporated into a sa-mRNA to, for example, alter the stability of such RNA molecules, to increase resistance against RNases, to establish replication after introduction into appropriate host cells ("infectivity" of the RNA), and/or to induce or reduce innate and adaptive immune responses.

(20) In another aspect, the nucleic acid is engineered to express alphavirus nonstructural proteins. U.S. Pat. Nos. 7,045,335, 7,078,218, 7,425,337 and 7,442,381 describe numerous constructs for such alphavirus RNA replicons consisting of the 5' and 3' alphavirus replication recognition sequences, coding sequences for alphavirus nonstructural proteins, and a polyadenylation tract, and such constructs are incorporated herein by reference.

(21) In some aspects, at least one non-structural replicase domain sequence comprise sequences selected from Group IV RNA viruses, including Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiyama virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. In yet another aspect, at least one non-structural replicase domain sequence is obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). In some aspects, the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4) and, in some aspects, the sa-mRNA of the present disclosure contains a subgenomic promoter that directs expression of said proteins.

(22) In some aspects, a GOI can encode a therapeutic polypeptide, a prophylactic polypeptide, a diagnostic polypeptide, a reporter gene, an antigen, or a gene that encodes regulatory structures. In some aspects, a GOI can encode an infectious disease antigen, an allergic antigen or a tumor antigen. In some aspects, a GOI is a non-coding gene, which encodes regulatory structures such as small interfering RNA (siRNA), micro-RNA (miRNA), self-activating RNA (saRNA), transfer RNA (tRNA), guiding or guide RNA (gRNA) or long intergenic non-coding (lincRNA).

(23) In some aspects, the nucleic acid of the disclosure comprise a sequence that is 90%, 91%,

92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SAM001 (SEQ ID NO: 35), SAM002 (SEQ ID NO: 36), SAM003 (SEQ ID NO: 37), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), MOD001 (SEQ ID NO: 41), or T7-VEE-GFP (SEQ ID NO: 42).

(24) In some aspects, the nucleic acid molecule of the present disclosure is suitable, in particular after linearization, for in vitro transcription of RNA, in particular self-amplifying mRNA. Circular plasmids are generally linearized downstream of the poly-A tail of the second expression unit by type II restriction enzymes (recognition sequence corresponds to cleavage site), prior to in vitro transcription. The linearized plasmid can then be used as template for in vitro transcription, the resulting transcript ending in a poly-A sequence.

(25) Accordingly, in one aspect, it is preferred that the nucleic acid molecule of the present disclosure can be cleaved, preferably enzymatically or in another biochemical way, within the nucleic acid sequence in such a way that said cleavage results in a nucleic acid molecule which comprises, in the 5'.fwdarw.3' direction of transcription: L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-L4-3'UTR-PolyA

wherein, L1 is a first linker, Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker, Pr2 is a second promoter sequence, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, L3 is a third linker, GOI is one or more gene or genes of interest, L4 is a fourth linker, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail. The nucleic acid molecule of the present disclosure is preferably a closed circular molecule prior to cleavage and a linear molecule after cleavage. Preferably, cleavage is carried out with the aid of a restriction cleavage site which is preferably a restriction cleavage site for a type IIS restriction endonuclease. In one aspect, the recognition sequence for the type IIS restriction endonuclease is 5-26 base pairs. In aspect, restriction enzyme MluI is used at the end of the Poly A.

(26) In one aspect, the nucleic acid contains one or more linkers wherein each linker is independently selected from a nucleic acid sequence comprising

(27) TABLE-US-00001 (SEQ ID NO: 43)

CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCC
AGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT, (SEQ ID NO: 44)

CACATTTCCCCGAAAAGTGCCACCTGAGCTC, (SEQ ID NO: 45)

TTCGAAGGCGCGCCTCTAGAGCCACC, or (SEQ ID NO: 46)

CATCGATGATATCGCGGCCGCATACAGCAGC. In some aspects, L1 comprises
SEQ ID NO: 43

(CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGC

CAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT); L2 comprises SEQ ID NO: 44

(CACATTTCCCCGAAAAGTGCCACCTGAGCTC); L3 comprises SEQ ID NO: 45

(TTCGAAGGCGCGCCTCTAGAGCCACC); and L4 comprises SEQ ID NO: 46

(CATCGATGATATCGCGGCCGCATACAGCAGC).

(28) In one aspect, the present disclosure relates to a method of obtaining self-amplifying mRNA comprising: a) performing an in vitro transcription reaction using an initial amount of a nucleic acid molecule of the present disclosure, and b) producing a sa-mRNA by in vitro transcription, using the nucleic acid molecule as a template and RNA polymerase (e.g., T7 polymerase).

(29) In another aspect, the present disclosure relates to a nucleic acid molecule, preferably obtained by linearization of an above-described nucleic acid molecule by cleavage within the nucleic acid sequence, and to sa-mRNA obtainable by transcription, preferably in vitro transcription, with above-described nucleic acid molecules under the control of the second promoter.

(30) Thus, in one aspect, the present disclosure relates to sa-mRNA comprising in the 5'.fwdarw.3' direction:

(31) TABLE-US-00002 5'UTR-nsP-SGP-GOI-3'UTR-PolyA

wherein, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, GOI is one or more gene or genes of interest, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail. In some aspects, the RNA further comprises linkers before the nsP, and between the GOI and 3'UTR.

(32) The methods of the present disclosure may be performed in vitro or in vivo. In one aspect of any of the methods of the present disclosure, transcription is carried out in vitro.

(33) In one aspect, the present disclosure provides nucleic acids and modified regulatory elements, the use of which increases transcription efficiency while reducing the amount of truncated single-stranded ribonucleic acid (ssRNA) (e.g., sa-mRNA) transcript produced during an in vitro transcription (IVT) reaction. In a typical IVT reaction, greater than 50% (molarity) of the RNA transcripts produced are truncated abortive products (referred to herein as truncated ssRNA transcripts). Only a small fraction (e.g., 0.2-0.5%) of initiation events lead to full-length "run-off" ssRNA transcripts, which is inefficient and costly for large-scale IVT RNA synthesis systems. Sa-mRNA transcripts in particular are longer than conventional mRNA (larger than 7 kilo nucleotides) and are particularly susceptible to truncated abortive products. Thus, use of the IVT methods of the present disclosure (which include, for example, nucleic acids, modified promoters and/or modified 5'UTR), in some aspects, results in a sa-mRNA transcript yield that is at least 40% greater than the sa-mRNA transcript yield of an IVT method without the modified regulatory elements of the present disclosure.

(34) In one aspect, the present disclosure provide nucleic acid templates that comprise a modified T7 promoter operably linked to nucleic acid comprising a sequence that encodes a modified 5' untranslated region (UTR) a plurality of non-structural replicase domain sequences, one or more gene or genes of interest (GOI), a 3' UTR, and a poly-A tail, wherein the sequence that encodes the T7 promoter and the sequence that encodes the 5' UTR is modified to enhance the binding strength of T7 polymerase to the T7 promoter to increase transcript yield.

(35) In some aspects, a modified T7 promoter comprises at least one insertion at position at the 5' end of the wildtype T7 promoter nucleotide sequence. The modification may be, for example, insertion of a single guanine (G) at the 5' end of the wildtype T7 promoter. In some aspects, the modified T7 promoter comprises SEQ ID NO: 47 (TAATACGACTCACTATAGG).

(36) In some aspects, a modified 5'UTR comprises at least one insertion at position 3 relative to the 5' end of the wildtype 5'UTR nucleotide sequence. The modification may be, for example, insertion of a single adenine (A) at position 3 of the wildtype 5'UTR of wildtype T7-VEE-GFP (SEQ ID NO: 42). In some aspects, the modified 5'UTR comprises ATAGG.

(37) In one aspect, the present disclosure provides nucleic acids and modified regulatory elements, the use of which modulates, preferably decreases, the immunogenicity and/or immunostimulatory capacity of a mRNA (immune response against an mRNA), preferably a self-amplifying mRNA, which encodes at least one biologically active polypeptide or protein, by preferably increasing the adenine (A) content of the 3'UTR. In some aspects, use of the nucleic acids and modified regulatory elements of the present disclosure (which include, for example, nucleic acid constructs, and/or modified 3'UTR), results in interferon responses that are 2 times, 3 times, 4 times, or 5 times lower than the interferon response to self-amplifying mRNAs without the modified regulatory elements of the present disclosure after one day post-transfection. In one aspect, the nucleic acids and modified regulatory elements of the present disclosure is able to induce reduced interferon response without the use of modified nucleotides (e.g. N.sup.1-Methylpseudouridine-5'-Triphosphate).

(38) In some aspects, a modified 3'UTR comprises at least one modification at any one of positions 6, -1, or -2 relative to a conserved 19 nucleotide sequence SEQ ID NO: 49 (GGATTTTGTTTTAAATATTC). The modification may be, for example, a mutant 3'UTR of an alphavirus comprising point mutations at position 6 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. The modification may also be,

for example, a mutant 3'UTR of an alphavirus comprising point mutations at positions -1 and -2 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. The modification may also be, for example, a mutant 3'UTR of an alphavirus comprising point mutations at positions -1, -2 and 6 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. In some aspects, the modified 3'UTR conserved sequence comprise GGATTTTATTTTAAATATTC (SEQ ID NO: 50), AAATTTTGTTTTAAATATTC (SEQ ID NO: 51), or AAATTTTATTTTAAATATTC (SEQ ID NO: 52).

(39) In one aspect, the biologically active agent comprises a sa-mRNA containing a polynucleotide sequence selected from: a) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 1 (BA.1-1273); b) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 2 (BA.1-1273-S2P); c) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 3 (BA.2-1273); d) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 4 (BA.2-1273-S2P); e) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 5 (BA.1-1208); or f) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 6 (BA.1-1208-S2P); g) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 7 (BA.2-1208); or h) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 8 (BA.2-1208-S2P).

(40) In one aspect, the sa-mRNA of the present disclosure encodes two separated expression units, the nucleic acid comprising: i) a first expression unit comprising a polynucleotide encoding a modified antigen, wherein the polynucleotide encoding the modified antigen is truncated to not include nucleotides encoding a transmembrane domain and short cytosolic domain amino acids of the antigen, operably linked to a first subgenomic promoter; and ii) a second expression unit encoding immunomodulators (IM) that are operably linked to a second subgenomic promoter. In one aspect, the polynucleotide sequence encoding the modified antigen comprises replacement of a transmembrane domain of the antigen with a secretion antigen. In one aspect, the antigen is a modified SARS-COV-2 spike protein, wherein the polynucleotide has been truncated to not include nucleotides encoding a SARS-COV-2 transmembrane domain and short cytosolic domain amino acids. In one aspect, the polynucleotide sequence encoding a coronavirus spike protein truncated to not include nucleotides encoding a SARS-COV-2 transmembrane domain and short cytosolic domain amino acids corresponding to amino acids 1209-1273 of a polynucleotide is selected from the group SEQ ID NOs: 1 (BA.1-1273), and 3 (BA.2-1273).

(41) In one aspect, the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-Ag-SGP2-IM wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P). SGP2 is the second subgenomic promoter, and IM is the immunomodulator.

(42) In one aspect, the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-IM-SGP2-AG wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, IM is the immunomodulatory, SGP2 is the second subgenomic promoter, and Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P).

(43) In some aspects, the IM encodes one or more cytokines, chemokines, immune stimulators or inhibitors. In one aspect, the IM is selected from IL12 and IL21. In one aspect, the IM encodes one or more cytokines selected from SEQ ID NOs: 22 (hIL12-P40), 24 (hIL12-P35), 15 (mIL12 P40), 17 (mIL12-P35), and 19 (mIL21). In one aspect, SGP1 is SEQ ID NO: 9 (SGP1). In one aspect, SGP2 is SEQ ID NO: 11 (SGP2). In one aspect, IM is selected from SEQ ID NO: 13 (IM1), and SEQ ID NO: 20 (IM2).

(44) In another aspect, the present disclosure includes a sa-mRNA comprising the following operably linked nucleic acid sequence from 5' to 3': SP-IL12 P40-L1-IL12 P35-L2-IL21 wherein SP is a signal peptide, IL12-P40 is interleukin-12 comprising heavy chain p40, L1 is linker 1, IL12 P35 is interleukin-12 comprising light chain p35, L2 is linker 2, and IL21 is interleukin-21. In some aspects, SP is selected from SEQ ID NO: 14 (MSP) and SEQ ID NO: 21 (HSP). In some aspects, IL12-P40 is selected from SEQ ID NO: 15 (mIL12-P40) and SEQ ID NO: 22 (hIL12-P40). In some aspects, L1 is selected from SEQ ID NO: 16 (L(a)) and SEQ ID NO: 23 (L(c)). In some aspects, IL12-P35 is selected from SEQ ID NO: 17 (mIL12-P35) and SEQ ID NO: 24 (hIL12-P35). In some aspects, L2 is selected from SEQ ID NO: 18 (L(b)) and SEQ ID NO: 25 (L(d)). In some aspects, IL12-P40 is selected from SEQ ID NO: 19 (mIL21) and SEQ ID NO: 26 (hIL21).

(45) In some aspects, at least one non-structural replicase domain sequence comprise sequences selected from Group IV RNA viruses, selected from Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, and Astroviridae. In some aspects, at least one non-structural replicase domain sequence comprise sequences selected from Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiya virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. In yet another aspect, at least one non-structural replicase domain sequence is obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). In some aspects, the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4).

(46) In some aspects, SGP1 is a viral promoter recognized by viral RNA dependent RNA polymerase (RdRP). In some aspects, SGP2 is a viral promoter recognized by viral RNA dependent RNA polymerase (RdRP). In some aspects, SGP1 and SGP2 are different subgenomic promoters.

(47) In some aspects, the sa-mRNA of the disclosure comprise one or more linkers. In some aspects, the linkers are selected from the group SEQ ID NOs: 13 (L(a)), 15 (L(b)), 20 (L(c)), and 22 (L(d)).

(48) In some aspects, the sa-mRNA of the present disclosure comprise a polynucleotide encoding a modified SARS-COV-2 spike protein. In some aspects, the polynucleotide encoding a modified SARS-COV-2 spike protein comprising a nucleic sequence selected from SEQ ID NO: 1 (BA.1-1273), SEQ ID NO: 2 (BA.1-1273-S2P), SEQ ID NO: 3 (BA.2-1273), SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), and SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P).

(49) The molecules, platforms, methods and other aspects of the present disclosure may be utilized, for example, for increasing expression of certain biologically active proteins in cellular transcription and expression. In some aspects, the present disclosure may be used to induce expression of recombinant proteins. This includes, for example, recombinant antibodies, hormones, cytokines, enzymes, and the like.

(50) It is also possible to use the nucleic acid molecules of the present disclosure for gene therapy applications. Accordingly, in some aspects, a nucleic acid molecule of the present disclosure may be a gene therapy vector and used for expression of a transgene. Preferably, alphavirus vector systems may be used.

(51) It is also possible to use the nucleic acid molecules of the present disclosure for gene regulation applications. In some aspects, the present disclosure may be used to modulate, increase or decrease, transcription of certain genes for therapeutic purposes. This includes, for example, small interfering RNA (siRNA), guide RNA (gRNA), micro-RNA (miRNA), self-activating RNA (saRNA), transfer RNA (tRNA), long intergenic non-coding (lincRNA), and the like.

(52) Cells can be transfected with these nucleic acid molecules in vitro, for example in lymphocytes or dendritic cells, or else in vivo by direct administration.

(53) Sa-mRNA of the present disclosure (e.g. obtained using a nucleic acid molecule described herein as a transcription template) may be employed, for example, for transient expression or silencing of genes, with possible fields of application being self-amplifying mRNA-based vaccines which are transfected into cells in vitro or administered directly in vivo, transient expression of functional recombinant proteins in vitro, for example to study functions of proteins, and transient expression or silencing of functional proteins such as erythropoietin, hormones, coagulation inhibitors, etc., in vivo, in particular as pharmaceuticals.

(54) Sa-mRNAs of the present disclosure may be used for transfecting antigen-presenting cells and thus as a tool for delivering the antigen to be presented with said antigen corresponding to the peptide, protein, or protein fragment expressed from said self-amplifying mRNA or being derived therefrom, in particular by way of intracellular processing such as cleavage. Such antigen-presenting cells may be used for stimulating T cells, in particular CD4⁺ and/or CD8⁺ T cells.

(55) Accordingly, in a further aspect, the present disclosure relates to a use of the self-amplifying mRNAs of the present disclosure for transfecting a host cell. In one aspect, the host cell is an antigen-presenting cell, such as a dendritic cell, a monocyte or a macrophage.

(56) In a further aspect, the present disclosure relates to a use of sa-mRNAs of the present disclosure for therapy, in particular for vaccination.

(57) In a further aspect, the present disclosure relates to a pharmaceutical composition such as a vaccine composition comprising the sa-mRNAs of the present disclosure. In one aspect, the sa-mRNAs of the present disclosure is formulated within a pharmaceutically acceptable carrier. In some aspects, the pharmaceutically acceptable carrier is a sa-mRNA delivery system, preferably a nanoparticle composition. In some aspects, said nanoparticle composition comprises a cationic lipid, a PEG-modified lipid, a sterol and a non-cationic lipid.

(58) In one aspect, the present disclosure provides a method of delivering a peptide to a subject, comprising administering a pharmaceutical composition containing one or more self-amplifying mRNAs of the present disclosure to the subject, wherein the sa-mRNA(s) of the present disclosure produces a detectable amount of peptide in a tissue of the subject.

(59) In a further aspect, the present disclosure relates to a use of sa-mRNA of the present disclosure as a research tool, such transfecting a cell with the sa-mRNA of the present disclosure encoding a reporter gene, in particular a reporter gene encoding a Green fluorescent protein (GFP).

(60) Also provided herein are methods for generating a library of sa-mRNA using the sa-mRNA of the present disclosure as a reference.

(61) Each of the aspects of the present disclosure can encompass various elements of the present disclosure. It is, therefore, anticipated that each of the aspects of the present disclosure involving any one element or combinations of elements can be included in each aspect of the present disclosure. This disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following detailed description or illustrated in the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1a shows a schematic representation of a linearized nucleic acid that is used as a template for production of sa-mRNA. The definitions of the abbreviations in the nucleotide sequence map are as follows: L1 is a first linker, Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker, Pr2 is a second promoter sequence, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, L3 is a third linker, GOI is one or more gene or genes of interest, L4 is a fourth linker, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail. Note that any one or more of the illustrative components of the molecule are optional and the present disclosure includes aspects that contain fewer than all of the illustrated elements. FIG. 1b shows the engineered sa-mRNA constructs of the disclosure (SAM001, SAM002, and SAM003) and the nucleotide and amino acid sequences changed in the nsP region compared to the wildtype SAM001.
- (2) FIG. 2 shows FACS of 4 individual sa-mRNA in 293T cells over time.
- (3) FIG. 3 shows FACS of SAM002 and T7-VEE-GFP sa-mRNA expressing GFP-mRNA in 293T cells at day 1 post transfection.
- (4) FIG. 4 shows a comparison of the nucleotide sequence of the junction region of the T7-Promoter and 5' UTR in T7-VEE-GFP (nucleotides 1819-1840 of SEQ ID NO: 42) and SAM002 (SEQ ID NO: 53).
- (5) FIG. 5 shows sa-mRNA production by in vitro transcription using a microgram template within 30 minutes of in vitro transcription using T7 polymerase.
- (6) FIG. 6 shows the structure prediction of the 3' UTR of wildtype VEE (SEQ ID NO: 54).
- (7) FIG. 7 shows a comparison of the nucleotide sequence of a conserved 19 nucleotide sequence of the 3' UTR of wildtype VEE and modified sequences of the present disclosure, which sequences are labeled as Sam002 (SEQ ID NO: 49), Sam004 (SEQ ID NO: 50), Sam005 (SEQ ID NO: 51), Sam006 (SEQ ID NO: 52).
- (8) FIG. 8 shows a reporter assay of 5 individual sa-mRNA in Raw-ISR-Lucia cells at day 1 post transfection.
- (9) FIG. 9 shows a reporter assay of 4 individual sa-mRNA in Raw-ISR-Lucia cells at day 1 post-transfection, where GFP expression is normalized with nsP3 in comparison to SAM002.
- (10) FIG. 10 shows a schematic representation of a in vivo experiment, which tested the toxicity the E6-C9 LNP formulation by examining changes in bleeding and body weight of Balb/C mice injected intramuscularly with dosage of 10, 5, 2.5, 1.25 μ g mRNA (5 mice/group). At the day 0, 1, 3, and 7 days post injection, the mice were bled and the body weight of the mice were measured.
- (11) FIG. 11 shows body weight changes of Balb/C mice injected intramuscularly with dosage of 10, 5, 2.5, 1.25 μ g mRNA (5 mice/group). At the day 0, 1, 3, and 7 days post injection.
- (12) FIG. 12 shows changes in pro-inflammatory cytokine IFN γ in mice injected intramuscularly with dosage of 10, 5, 2.5, 1.25 μ g mRNA (5 mice/group) at day 0, 1, 3, and 7 post injection as the indicated. The positive control (standard) was labeled as red. Shown are Elisa plots of absorbance (Y-axis) versus dilution factors (X-axis).
- (13) FIG. 13 shows changes in pro-inflammatory cytokine IL6 in mice injected intramuscularly with dosage of 10, 5, 2.5, 1.25 μ g mRNA (5 mice/group) at day 0, 1, 3, and 7 post injection as the indicated. The positive control (standard) was labeled as red. Shown are Elisa plots of absorbance (Y-axis) versus dilution factors (X-axis).
- (14) FIG. 14 shows changes in pro-inflammatory cytokine IL1-beta in mice injected intramuscularly with dosage of 10, 5, 2.5, 1.25 μ g mRNA (5 mice/group) at day 0, 1, 3, and 7 post injection as the indicated. The positive control (standard) was labeled as red. Shown are Elisa plots of absorbance (Y-axis) versus dilution factors (X-axis).
- (15) FIG. 15 shows changes in pro-inflammatory cytokine TNF α in mice injected intramuscularly

with dosage of 10, 5, 2.5, 1.25 μg mRNA (5 mice/group) at day 0, 1, 3, and 7 post injection as the indicated. The positive control (standard) was labeled as red. Shown are Elisa plots of absorbance (Y-axis) versus dilution factors (X-axis).

(16) FIG. **16** shows a schematic representation of released mRNA sequences of BioNTech-Pfizer (BNT162b2) and Moderna vaccine (mRNA-1273) the antigens are 1273 amino acids, including S1 (RBD), S2, transmembrane domain and short cytosolic domain. Since the transmembrane domain leads to the expression of SPIKE antigens on the cell surfaces, the transfected cells could be targeted by immune system, which likely results in hepatitis and myocarditis induced by BNT162b2 and mRNA-1273. This schematic representation also shows the location of 2 proline mutations on S2 (S2P), which are found in the mRNA of covid vaccine BNT162b2, but not in the mRNA of mRNA-1273. This version of the mRNA polypeptide sequence encoding the SPIKE protein is hereinafter referred to as “1273.”

(17) FIG. **17** shows a schematic representation of a truncated, secretion version (1-1208) of the SPIKE protein. As the 1-1208 amino acids were used for structural studies of the SPIKE protein the truncated protein (“1028”). This schematic representation also shows the location of 2 proline mutations on S2 (S2P), which is found in the mRNA of covid vaccine BNT162b2, but not in the mRNA of mRNA-1273.

(18) FIG. **18** shows FACS plots of GFP (a) expression (X-axis) versus live dead dye staining of 7-AAD (Y-axis) of E6-LNP encompassing a sa-mRNA encoding the SARS-COV-2-BA.1-1273, SARS-COV-2-BA.2-1273, SARS-COV-2-BA.1-1273-S2P, and SARS-COV-2-BA.2-1273-S2P transfected into 293 T-cells by lipofectamine. The cells were collected and the SPIKE of Omicron BA.1 and BA.2 were detected by the SPD-M265 bnAb (broad neutralization antibody) using flow cytometer. The results show that S2P is dispensable to stabilize the structure of SPIKE with transmembrane domain for recognition by SPD-M265 bnAb. The cells were analyzed by flow cytometer at day 1 post transfection.

(19) FIG. **19** shows FACS plots of binding of SPIKE expression with its receptor ACE2 versus live dead dye staining of 7-AAD (Y-axis) of E6-LNP encompassing a sa-mRNA encoding the SARS-COV-2-BA.1-1273, SARS-COV-2-BA.2-1273, SARS-COV-2-BA.1-1273-S2P, and SARS-COV-2-BA.2-1273-S2P transfected into 293 T-cells by lipofectamine. The treated cells were collected and the SPIKE of Omicron BA.1 and BA.2 were detected by the ACE2 conjugated with FITC using flow cytometer. The results show that S2P is dispensable to stabilize the structure of SPIKE with transmembrane domain for recognition by SPIKE receptor ACE2 conjugated with FITC. The cells were analyzed by flow cytometer at day 1 post transfection.

(20) FIG. **20** shows ELISA data with absorbance on the Y-axis and dilution factor X-axis comparing S2P effects on secretion version of BA.1-1208, BA.1-1208-S2P, BA.2-1208, and BA.2-1208-S2P by transfecting 293 T cells with different sa-mRNA encoding with BA.1-1208, BA.1-1208-S2P, BA.2-1208, and BA.2-1208-S2P by lipofectamine and detecting absorbance using SPD-M265-bnAb. The data show that S2P is indispensable to stabilize the structure of the secretion SPIKE without transmembrane domain. The supernatants of transfected cells at day 1 post transfection were analyzed by ELISA.

(21) FIG. **21** shows a schematic representation of a mouse experiment where Balb/c mice (10 mice per group) were injected at day 0 and 2WP1 (2 weeks post 1st injection) with E6-C9-LNP encapsulating 2 μg of sa-mRNA encoding a including BA.2-1208, BA.2-1273, BA.2-1208-S2P, BA.2-1273-S2P, or only sa-mRNA not encoding a SARS-COV-2 Omicron variant. Each group of mice were bled at day 0, and 2WP1, and 2WP2 (two week post second injection) to compare vaccine efficacy over time between different populations.

(22) FIG. **22** shows the results of an assay of receptor-binding domain (RBD-specific immunoglobulin G (IgG) binding titers against SARS-COV-2 for a mouse experiment where Balb/c mice (10 mice per group) were injected at day 0 and 2WP1 (2 weeks post 1st injection) with E6-C9-LNP encapsulating 2 μg of sa-mRNA encoding a including BA.2-1208, BA.2-1273, BA.2-

1208-S2P, BA.2-1273-S2P, or only sa-mRNA not encoding a SARS-COV-2 Omicron variant. Each group of mice were bled at day 0, and 2WP1, and 2WP2 (two week post second injection) to compare vaccine efficacy over time between different populations.

(23) FIG. 23 shows a schematic representation of a dual subgenomic promoter sa-mRNA, with expression vectors encoding a SARS-COV-2 antigen and immunomodulators (e.g. cytokines et al) under subgenomic promoter 1 and 2, respectively.

(24) FIG. 24 shows a schematic representation of a dual subgenomic promoter sa-mRNA, with expression vectors encoding immunomodulators (e.g. cytokines et al) and a SARS-CoV-2 antigen under subgenomic promoter 1 and 2, respectively.

(25) FIG. 25 shows transcripts of nsP3 and enhanced green fluorescent protein (eGFP) encoded by sa-mRNA and modified mRNA constructs SAM001, SAM002, SAM003, and modified mRNA from MOD001 normalized with mouse Actin beta (n=3). C2C12 cells were transfected with the P6-LNP encapsulated either SAM001 or SAM002 encoding with GFP. At day 1 post transfection, total RNAs were extracted, and reverse transcribed to cDNA. Then quantitatively polymerase chain reactions (qPCR) were performed using the probes specifically targeting the nsP3 and eGFP. Shown are the fold changes of nsP3 and eGFP that normalized with mouse Actin beta (n=3).

(26) FIG. 26 shows a comparison of transcript expression between SAM001 and SAM002 sa-mRNA constructs encoding Luciferase in vivo. Balb-c mice at 6-8 weeks old were intramuscularly injected at both hind legs with the P6-LNP encapsulated either SAM001 or SAM002 encoding with Luciferase. The mice were intraperitoneally injected with 200 ul Luciferin (30 mg/ml) per mouse and imaged at 5 minutes after injections of Luciferin by in vitro imaging system (Perkin Elmer). Shown are the total flux (photon/second) at the time point indicated (n=10).

(27) FIG. 27 shows a comparison of LLC1 tumor growth that was treated with E2-LNP-SAM001-IL12 and E2-LNP-SAM002-IL12. C57BL6/J mice (n=5) at 6-8 weeks old were subcutaneously injected with 1 million Lewis Lung Carcinoma (LLC1) cells. At day 7 post injections, the mice were intratumorally treated with PBS, E2-LNP-SAM001-IL12 and E2-LNP-SAM002-IL12. Shown are tumor area (Y-axis) versus time (X-axis).

(28) FIG. 28a shows a schematic representation of a linearized SAM002 that is used as a template for production of sa-mRNA. The definitions of the abbreviations in the nucleotide sequence map are as follows: 5UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, Puromycin R is the puromycin resistance gene, 3UTR is a 3' untranslated region and contigs are subgenomic intervals generated as vectors to facilitate sequencing and numbered for the identification of mutations after directed evolution.

(29) FIG. 28b shows the location of nucleotide and amino acid mutations in the non-structural proteins of the linearized nucleic acid after directed evolution. The contig numbers correspond to which region the mutations occurred on the linearized SAM002.

(30) FIG. 28c shows where the mutations occur in the mutants with regards to the SAM002 contigs and the allele number of each mutant.

(31) Note that any one of more of the illustrative components of the molecule are optional and the present disclosure includes aspects that contain fewer than all of the illustrated elements.

DETAILED DESCRIPTION

(32) The disclosure relates to novel nucleic acid constructs and compositions, and methods to deliver one or more biologically active agents to subjects in need thereof and methods involving the same. The disclosure also provide methods of delivering biologically active agents to a cell, specifically delivering a therapeutic, diagnostic and/or prophylactic agent to an organ, producing a sa-mRNA of interest in the cell, and treating a disease or disorder in a subject in need thereof. For example, a method of producing a sa-mRNA of interest in a cell involves contacting a nanoparticle composition comprising a sa-mRNA with a cell, whereby the sa-mRNA may be translated to produce the polypeptide of interest. A method of delivering a biologically active agent to a mammalian cell or organ may involve administration of a nanoparticle composition including the

biologically active agent to a subject, in which the administration involves contacting the cell or organ with the composition, whereby the biologically active agent is delivered to the cell or organ.

(33) It is important to note that while many of the approaches described in this specification and the examples given are focused on vaccine development, they are equally applicable to sa-mRNA for other intended uses, such as for gene therapy or gene regulation.

(34) Although the present disclosure is described in detail below, it is to be understood that this disclosure is not limited to the particular methodologies, protocols and reagents described herein as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the present disclosure. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

(35) The present disclosure includes constructs of the sa-mRNAs and variations thereof as shown and described, and methods of making and using the constructs. The present disclosure includes noncytopathic and cytopathic versions of the sa-mRNAs and variations thereof. The present disclosure includes sequences and engineering of conjugations between elements of the constructs as shown and described. The present disclosure includes self-amplifying mRNAs that reduce the transcription numbers of sa-mRNA (e.g., nsP3) and subgenome (e.g., eGFP) to make less-cytopathic versions of the sa-mRNA. The present disclosure includes methods and constructs for expressions of payload genes that encode various desired payloads for therapeutic, prophylactic, and/or diagnostic uses. The present disclosure includes methods and constructs including structure-based engineering to control replication rate and interferon responses of sa-mRNAs. The present disclosure includes methods for directed evolutions to identify mutations on sa-mRNA by encoding puromycin resistant genes in the subgenome. The present disclosure includes sa-mRNAs having identified mutations according to the present disclosure, including contigs having identified mutations and combinations thereof. The present disclosure includes designs of antigens in combinations of immunomodulators, such as cytokines and chemokine, in combination with antigens from infectious diseases and tumors.

(36) Additionally, several documents are cited throughout the text of this specification. Each of the documents cited herein (including all patents, patent applications, scientific publications, manufacturer's specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety.

(37) Definitions

(38) As used herein, the terms “gene of interest,” “genes of interest,” “gene or genes of interest,” “GOI,” or “coding region” refers to the nucleotide sequence which encode the amino acids found in polypeptides and proteins as a result of translation of a mRNA molecule, including from a sa-mRNA. A GOI, for the purposes of this disclosure, include, but is not limited to, polynucleotides encoding antigens (such as SARS-COV2 Omicron variants) and immunomodulators (such as IL12 and IL21).

(39) As used herein, “nucleotide” is a term of art that refers to a molecule that contains a nucleoside or deoxynucleoside, and at least one phosphate. A nucleoside or deoxynucleoside contains a single 5 carbon sugar moiety (e.g., ribose or deoxyribose) linked to a nitrogenous base, which is either a substituted pyrimidine (e.g., cytosine (C), thymine (T) or uracil (U)) or a substituted purine (e.g., adenine (A) or guanine (G)). A “polynucleotide” refers to a series or sequence of nucleotides.

(40) As used herein, the terms “modified nucleotide” refers to a nucleotide that contains one or more chemical modifications (e.g. substitutions) in or on the nitrogenous base of the nucleoside (e.g., cytosine (C), thymine (T) or uracil (U)), adenine (A) or guanine (G)). A nucleotide analog can contain further chemical modifications in or on the sugar moiety of the nucleoside (e.g., ribose, deoxyribose, modified ribose, modified deoxyribose, six-membered sugar analog, or open-chain sugar analog), or the phosphate. There are more than 96 naturally occurring modified nucleosides found on mammalian RNA. See, e.g., Limbach et al, Nucleic Acids Research, 22 (12): 2183-2196

(1994). The preparation of nucleotides and modified nucleotides and nucleosides are well-known in the art, e.g. from U.S. Pat. Nos. 4,373,071, 4,458,066, 4,500,707, 4,668,777, 4,973,679, 5,047,524, 5,132,418, 5,153,319, 5,262,530, 5,700,642 all of which are incorporated by reference in their entirety herein, and many modified nucleosides and modified nucleotides are commercially available.

(41) As used herein, “nucleic acid” refers a nucleic acid molecule. According to the present disclosure, nucleic acids comprise genomic DNA, cDNA, RNA, recombinantly prepared and chemically synthesized molecules. According to the present disclosure, a nucleic acid may be in the form of a single-stranded or double stranded and linear or covalently closed circular molecule. The nucleic acid of the present disclosure may also containing non-natural nucleotides and modified nucleotides. “Nucleic acid” also refers to a consecutive list of abbreviations, letters, characters or words, which represent nucleotides. In some aspects, a “nucleic acid template” refers to a nucleic acid that is capable of transcription into RNA (e.g. self-amplifying mRNA).

(42) As used herein, the term “contig” refers to contiguous regions of DNA sequence. “Contigs” can be determined by any number methods known in the art, such as, by comparing sequencing reads for overlapping sequences, and/or by comparing sequencing reads against databases of known sequences in order to identify which sequencing reads have a high probability of being contiguous. Contigs are often assembled from individual sequence reads or previously assembled sequence information in combination with sequence reads having overlapping end or edge sequence. Generally but not exclusively, contigs comprise overlapping sequence reads that assemble into a larger sequence grouping, in many cases without intervening gaps or regions of undetermined sequence, or alternately without regions of known sequence and unknown length.

(43) The term “allele” refers to alternative forms of a gene, a reference nucleic acid or portions thereof. Alleles occupy the same locus or position on homologous chromosomes. Alleles of a specific gene or reference can differ from each other in a single nucleotide, or several nucleotides, and can include substitutions, deletions, and insertions of nucleotides. An allele can also be a form of a reference nucleic acid containing a mutation.

(44) In one aspect, a library includes a collection of nucleic acid members, e.g., a collection of whole genomes, subgenomic fragments, cDNA, cDNA fragments, RNA (e.g., mRNA or sa-mRNA), RNA fragments, or a combination thereof. “Member” or “library member” or other similar term, as used herein, refers to a nucleic acid molecule, e.g., a DNA, RNA, or a combination thereof that is the member of a library. The data of each library member may comprise the number of each nucleoside in an amplicon that would be generated for each allele using each primer or the nucleotide sequence of each member. In this aspect of populating the database, a nucleic acid with a particular allele is selected and a primer pair is used to generate an amplicon. The amplicon's nucleotide sequence can be determined using a method known in the art, such as BAC clone sequencing, physical maps, and Sanger sequencing. An entry in the database is made to associate the base composition with the allele, contig or library member.

(45) As used herein, the term “selectable marker” refers to a nucleotide sequence encoding a gene product that allow for the selection of bacterial cells that have been transformed. Selectable markers can be expressed in the bacterial host and may include genes which render bacteria resistant to drugs such as ampicillin, chloramphenicol, erythromycin, kanamycin (neomycin), and tetracycline (Davies et al., *Ann. Rev. Microbiol.*, 32:469 (1978)). Selectable markers may also include biosynthetic genes, such as those in the histidine, tryptophan, and leucine biosynthetic pathways.

(46) As used herein, the term “regulatory element” refers to a nucleotide sequence that controls, at least in part, the transcription of a gene or genes of interest. Regulatory elements may include promoters, enhancers, and other nucleic acid sequences (e.g., polyadenylation signals) that control or help to control nucleic acid transcription or translation. Examples of transcription regulatory elements are described, for example, in Goeddel, *Gene Expression Technology: Methods in*

Enzymology 185 (Academic Press, San Diego, Calif., 1990).

(47) As used herein, the term “non-coding” refers to nucleotide sequences that do not encode a polypeptide or an expressed protein. Non-coding sequences include but are not limited to introns, enhancers, promoter regions, 3' untranslated regions, 5' untranslated regions, linkers and GOI which encode regulatory structures.

(48) As used herein, the term “operably linked” refers to a first molecule joined to a second molecule, wherein the molecules are so arranged that the first molecule affects the function of the second molecule. The two molecules may or may not be part of a single contiguous molecule and may or may not be adjacent. For example, a promoter is operably linked to a GOI if the promoter modulates transcription of said GOI in a cell. Additionally, two portions of a transcription regulatory element are operably linked to one another if they are joined such that the transcription-activating functionality of one portion is not adversely affected by the presence of the other portion. Two transcription regulatory elements may be operably linked to one another by way of a linker nucleic acid (e.g., an intervening non-coding nucleic acid) or may be operably linked to one another with no intervening nucleotides present.

(49) As used herein, the term “linker” refers to a nucleotide sequence added between two nucleotide sequences to connect said two nucleotide sequences. There is no particular limitation regarding the linker sequence.

(50) As used herein, the term “subgenomic promoter,” is a promoter that can be used to transcribe the subgenome of alphaviruses encoding structural proteins by RNA dependent RNA polymerase encoded by nsP. When two or more subgenomic promoters are present in a nucleic acid comprising multiple expression units, the promoters can be the same or different. In certain aspects, subgenomic promoters can be modified using techniques known in the art in order to increase or reduce viral transcription of the proteins, see e.g. U.S. Pat. No. 6,592,874, which is incorporated by reference in their entirety herein.

(51) As used herein, the term “expression unit” as used herein mean a nucleotide sequence capable of directing expression of a particular GOI in an appropriate cell, comprising a promoter functional in said cell and a coding region. If translation is required, it also typically comprises sequences required for proper translation of the nucleotide sequence. The GOI may code for a protein or polypeptide of interest but may also code for a regulatory structure of interest, for example siRNA, or any other noncoding regulatory RNA. A nucleic acid may contain a plurality of expression units. The expression unit comprising the nucleotide sequence of interest may be chimeric, meaning that at least one of its components is heterologous with respect to at least one of its other components. The expression unit may also be one which is naturally occurring but has been obtained in a recombinant form useful for heterologous expression. Typically, however, the expression unit is heterologous with respect to the host, i.e., the particular DNA or RNA sequence of the expression unit does not occur naturally in the host cell and must have been introduced into the host cell or an ancestor of the host cell by a transformation event. The expression of the nucleotide sequence in the expression unit may be under the control of a constitutive promoter or of an inducible promoter, which initiates transcription only when the host cell is exposed to some particular external stimulus.

(52) As used herein, the term “genomic DNA” is referring to the heritable genetic information of a host organism. Said genomic DNA comprises the DNA of the nucleus (also referred to as chromosomal DNA) but also the DNA of the other cellular organelles (e.g., mitochondria). In some aspects, the term genomic DNA refers to the chromosomal DNA of the nucleus.

(53) As used herein, the terms “polypeptide,” “peptide,” “oligopeptide,” “gene product,” “expression product” and “protein” are used interchangeably herein to refer to a polymer or oligomer of consecutive amino acid residues. The terms “gene product” and “expression product” can also refer to regulatory structures.

(54) As used herein, an “effective amount” of a sa-mRNA refers to an amount sufficient to elicit

expression of a detectable amount of an antigen or protein, e.g., an amount suitable to produce a desired therapeutic, diagnostic or prophylactic effect.

(55) As used herein, the term “naked” as used herein refers to nucleic acids that are substantially free of other macromolecules, such as lipids, polymers, and proteins. A “naked” nucleic acid, such as a plasmid or a sa-mRNA, is not formulated with other macromolecules to improve cellular uptake. Accordingly, a naked nucleic acid is not encapsulated in, absorbed on, or bound to a liposome, a microparticle or nanoparticle, a cationic emulsion, and the like.

(56) As used herein, the term “transfection” or “transformation” refers to introducing one or more nucleic acids into an organism or into a host cell. Various methods may be employed in order to introduce nucleic acids into cells in vitro or in vivo. Such methods include transfection of nucleic acid-CaPO₄ precipitates, transfection of nucleic acids associated with DEAE, transfection of infection with viruses carrying the nucleic acids of interest, liposome mediated transfection, lipid nanoparticle (LNP) mediated transfection, lipofectamine and the like.

(57) As used herein, the term “reporter” relates to a molecule, typically a peptide or protein, which is encoded by a reporter gene and measured in a reporter assay. Existing systems usually employ an enzymatic reporter (e.g. GFP or Luciferase) and measure the activity of said reporter.

(58) As used herein, “encapsulation efficiency” refers to the amount of a biological agent that becomes part of a nanoparticle composition, relative to the initial total amount of biologically active agent used in the preparation of a nanoparticle composition. For example, if 97 mg of biologically active agent are encapsulated in a nanoparticle composition out of a total 100 mg of biologically active agent initially provided to the composition, the encapsulation efficiency may be given as 97%. As used herein, “encapsulation” may refer to complete, substantial, or partial enclosure, confinement, surrounding, or encasement.

(59) As used herein, a “nanoparticle composition” or “LNP formulation” is a composition comprising one or more lipids. Nanoparticle compositions are typically sized on the order of micrometers or smaller and may include a lipid bilayer. Nanoparticle compositions encompass lipid nanoparticles (LNPs), liposomes (e.g., lipid vesicles), and lipoplexes. For example, a nanoparticle composition may be a liposome having a lipid bilayer with a diameter of 500 nm or less. For example, the lipid component of a nanoparticle composition may include one or more cationic/ionizable, PEGylated, structural, or other lipids, such as phospholipids.

(60) As used herein, a “lipid component” is that component of a nanoparticle composition that includes one or more lipids. For example, the lipid component may include one or more cationic/ionizable, PEGylated, structural, or other lipids, such as phospholipids.

(61) As used herein, the terms “PEG lipid” or “PEGylated lipid” refer to a lipid comprising a polyethylene glycol component. For example, a PEG lipid may be selected from the following non-limiting group: PEG-modified phosphatidylethanolamines, PEG-modified phosphatidic acids, PEG-modified ceramides, PEG-modified dialkylamines, PEG-modified diacylglycerols, PEG-modified dialkylglycerols, and mixtures thereof. For example, a PEG lipid may be PEG-c-DOMG, PEG-DMG, PEG-DLPE, PEG-DMPE, PEG-DPPC, or a PEG-DSPE lipid.

(62) As used herein, the terms “phospholipid” or “helper lipid” refer to a lipid that includes a phosphate moiety and one or more carbon chains, such as unsaturated fatty acid chains. A phospholipid may include one or more multiple (e.g., double or triple) bonds (e.g., one or more unsaturations). Particular phospholipids may facilitate fusion to a membrane. For example, a cationic phospholipid may interact with one or more negatively charged phospholipids of a membrane (e.g., a cellular or intracellular membrane). Fusion of a phospholipid to a membrane may allow one or more elements of a lipid-containing composition to pass through the membrane permitting, e.g., delivery of the one or more elements to a cell. In general, phospholipids may include a phospholipid moiety and one or more fatty acid moieties.

(63) Phospholipids useful in the compositions and methods of the present disclosure may be selected from the following non-limiting group: 1,2-distearoyl-sn-glycero-3-phosphocholine

(DSPC), 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-dilinoleoyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dimyristoyl-sn-glycero-phosphocholine (DMPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-diundecanoyl-sn-glycero-phosphocholine (DUPC), 1-palmitoyl-2-olcoyl-sn-glycero-3-phosphocholine (POPC), 1,2-di-O-octadecenyl-sn-glycero-3-phosphocholine (18:0 Diether PC), 1-oleoyl-2-cholesterylhemisuccinoyl-sn-glycero-3-phosphocholine (OChemPC), 1-hexadecyl-sn-glycero-3-phosphocholine (C16 Lyso PC), 1,2-dilinolenoyl-sn-glycero-3-phosphocholine, 1,2-diarachidonoyl-sn-glycero-3-phosphocholine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphocholine, 1,2-diphytanoyl-sn-glycero-3-phosphoethanolamine (ME 16.0 PE), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinoleoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinolenoyl-sn-glycero-3-phosphoethanolamine, 1,2-diarachidonoyl-sn-glycero-3-phosphoethanolamine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphoethanolamine, 1,2-dioleoyl-sn-glycero-3-phospho-rac-(1-glycerol) sodium salt (DOPG), and sphingomyelin. In some embodiments, a nanoparticle composition includes DSPC. In certain embodiments, a nanoparticle composition includes DOPE.

(64) As used herein, “ionizable lipids” are lipids that may have a positive or partial positive charge at physiological pH in addition to a lipid according to Formula (E6) disclosed in PCT Patent Application No. PCT/US2023/017777, which is fully incorporated herein.

(65) As used herein, the terms “stain” or “staining” include methods of detecting subpopulations of cells in a cell sample, and in particular, it relates to methods of detecting dead cells in a cell sample using a membrane permeable nucleic acid binding fluorescent label. The staining method can be used in combination with a cell capture system and/or an optical detection system for detecting the presence of live and or dead cells in a cell sample. For example, dead cells can be detected using fluorescent DNA binding dyes such as propidium iodide and 7-aminoactinomycin D (7-AAD) because they have compromised cell membrane integrity compared to live cells (Lecoeur et al., 2002; Gaforio et al., Cytometry 49:8, 2002; Ormerod et al., Cytometry 14:595, 1993; Schmid et al., J. Immunol. Methods 170:145, 1994; Philpott et al., Blood 87:2244, 1996).

(66) As used herein, the terms “treat,” “treating” or “treatment,” may include alleviating, abating or ameliorating disease or condition symptoms, preventing additional symptoms, ameliorating or preventing the underlying metabolic causes of symptoms, inhibiting the disease or condition, e.g., arresting the development of the disease or condition, relieving the disease or condition, causing regression of the disease or condition, relieving a condition caused by the disease or condition, or stopping the symptoms of the disease or condition. The terms “treat,” “treating” or “treatment”, may include, but are not limited to, prophylactic, diagnostic and/or therapeutic treatments.

(67) As used herein, the terms “associated with,” “conjugated,” “linked,” “attached,” and “tethered,” when used with respect to two or more moieties, means that the moieties are physically associated or connected with one another, either directly or via one or more additional moieties that serves as a linking agent, to form a structure that is sufficiently stable so that the moieties remain physically associated under the conditions in which the structure is used, e.g., physiological conditions.

(68) As used herein, the phrase “biologically active” refers to a characteristic of any substance that has activity in a biological system and/or organism. The term “biologically active agent” refers to any agent that, when administered to a subject, has a therapeutic, diagnostic, and/or prophylactic effect and/or elicits a desired biological and/or pharmacological effect. Such agents include, but are not limited to, cytotoxins, radioactive ions, chemotherapeutic agents, small molecule drugs, proteins, and nucleic acids.

(69) As used herein, the term “payload” refers to a moiety whose biological activity is desired to be delivered (in) to and/or localize at a cell or tissue. Payloads include, but are not limited to biologically active agents, and the like. In some aspects, the payload may be a nucleic acid that encodes a protein or polypeptide. In some aspects, the payload may include or encode a cytokine, a

chemokine, an antibody or antibody fragment, a receptor or receptor fragment, an enzyme, an enzyme inhibitor, a hormone, a lymphokine, a plasminogen activator, a natural or modified immunoglobulin or a fragment thereof, an antigen, a chimeric antibody receptor, variable or hypervariable regions of light and/or heavy chains of an antibody (V.sub.L, V.sub.H), variable fragments (Fv), Fab' fragments, F(ab')₂ fragments, Fab fragments, single chain antibodies (scAb), single chain variable regions (scFv), complementarity determining regions (CDR), domain antibodies (dAbs), single domain heavy chain immunoglobulins of the BHH or BNAR type, single domain light chain immunoglobulins, or other polypeptides known in the art containing an AB capable of binding target proteins or epitopes on target proteins, or any other desired biological macromolecule.

(70) Cytokines of the present disclosure may include but are not limited to an interferon, an interleukin, GM-CSF, G-CSF, LIF, OSM, CD154, LT- β , TNF- α , TNF- β , 4-1BBL, APRIL, CD70, CD153, CD178, GITRL, LIGHT, OX40L, TALL-1, TRAIL, TWEAK, TRANCE, TGF- β 1, TGF- β 1, TGF- β 3, Epo, Tpo, Flt-3L, SCF, M-CSF, and MSP, optionally wherein the CP1 and/or the CP2 is independently selected from IL-2, IL-7, IL-8, IL-10, IL-12, IL-15, IL-18, IL-17, IL-21, an IFN- α , an IFN β , an IFN γ , GM-CSF, TGF- β , LIGHT, GILTR-L, CD40L, CD27L, 4-1BB-L, OX40, and OX40L.

(71) As used herein, the term "conserved" refers to a nucleic acid sequence that occur unaltered in the same position of two or more related sequences being compared. Nucleic acid sequences that are relatively conserved are those that are conserved amongst more related sequences than nucleic acid sequences appearing elsewhere in the sequences. In some aspects, two or more sequences are said to be conserved if they are 100% identical to one another. In some aspects, two or more sequences are said to be conserved if they are about 95% identical, about 98% identical, or about 99% identical to one another.

(72) As used herein, the term "transcription" comprises "in vitro transcription" wherein the term "in vitro transcription" relates to a method in which RNA, in particular sa-mRNA, is synthesized in vitro in a cell-free manner.

(73) As used herein, "expression" of a nucleic acid refers to one or more of the following events: (1) production of an RNA template from a DNA sequence (e.g., by transcription); and (2) translation of an RNA into a polypeptide or protein.

(74) As used herein, two nucleic acids are substantially homologous when the nucleotide sequences have at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. Sequence homology for nucleic acids, which can also be referred to as percent sequence identity, is typically measured using sequence analysis software. See, e.g., the Sequence Analysis Software Package of the Genetics Computer Group (GCG), University of Wisconsin Biotechnology Center, 910 University Avenue, Madison, Wis. 53705. A typical algorithm used comparing a molecule sequence to a database containing a large number of sequences from different organisms is the computer program BLAST (Altschul, 1990; Gish, 1993; Madden, 1996; Altschul, 1997; Zhang, 1997), especially blastp or tblastn (Altschul, 1997).

(75) As used herein, the term "contacting" means establishing a physical connection between two or more entities. For example, contacting a cell with a nanoparticle composition comprising a sa-mRNA means that the mammalian cell and a nanoparticle are made to share a physical connection. Methods of contacting cells with external entities both in vivo and ex vivo are well known in the biological arts.

(76) As used herein, the term "delivering" means providing an entity to a destination. For example, delivering a biologically active agent to a subject may involve administering a nanoparticle composition comprising sa-mRNA including the biologically active agent to the subject (e.g., by an

intravenous, intranasal, intratracheal, intracerebral, intratumoral, intraperitoneal, intramuscular, intradermal, or subcutaneous route).

(77) As used herein, the term “in vitro” refers to events that occur in an artificial environment, e.g., in a test tube or reaction vessel, in cell culture, in a Petri dish, etc., rather than within an organism (e.g., animal, plant, or microbe).

(78) As used herein, the term “in vivo” refers to events that occur within an organism (e.g., animal, plant, or microbe).

(79) As used herein, the term “in situ” refers to events that occur in its original place, or in its natural context.

(80) As used herein, the terms “isolated” refers to a substance or entity that has been (1) separated from at least some of the components with which it was associated when initially produced (whether in nature or in an experimental setting), and/or (2) produced, prepared, and/or manufactured in vitro. Isolated substances and/or entities may be separated from at least about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or more of the other components with which they were initially associated. In some aspects, isolated agents are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. As used herein, a substance is “pure” if it is substantially free of other components.

(81) As used herein, the term “substantially” refers to the qualitative condition of exhibiting total or near-total extent or degree of a characteristic or property of interest.

(82) As used herein, the term “polypeptide” or “polypeptide of interest” refers to a polymer of amino acid residues typically joined by peptide bonds that can be produced naturally (e.g., isolated or purified) or synthetically.

(83) As used herein, “size” or “mean size” in the context of nanoparticle composition refers to the mean diameter of a nanoparticle composition.

(84) As used herein, the term “subject” or “patient” refers to any organism to which a composition in accordance with the disclosure may be administered, e.g., for experimental, diagnostic, prophylactic, and/or therapeutic purposes. Typical subjects include animals (e.g., mammals such as mice, rats, rabbits, non-human primates, and humans) and/or plants.

(85) As used herein, the term “therapeutically effective amount” means an amount of an agent to be delivered (e.g., nucleic acid, drug, composition, therapeutic agent, diagnostic agent, prophylactic agent, etc.) that is sufficient, when administered to a subject suffering from or susceptible to an infection, disease, disorder, and/or condition, to treat, improve symptoms of, diagnose, prevent, and/or delay the onset of the infection, disease, disorder, and/or condition.

(86) As used herein, “methods of administration” may include intravenous, intranasal, intratracheal, intracerebral, intratumoral, intraperitoneal, intramuscular, intradermal, subcutaneous, or other methods of delivering a composition to a subject. A method of administration may be selected to target delivery (e.g., to specifically deliver) to a specific region or system of a body.

(87) As used herein, “modified” means non-natural. For example, an RNA may be a modified RNA. That is, an RNA may include one or more nucleobases, nucleosides, nucleotides, or linkers that are non-naturally occurring. A “modified” species may also be referred to herein as an “altered” species. Species may be modified or altered chemically, structurally, or functionally. For example, a modified nucleobase species may include one or more substitutions that are not naturally occurring.

(88) As used herein, “naturally occurring” means existing in nature without artificial aid.

(89) As used herein, the terms “subgenomic” or “subgenome” refers to a nucleotide sequence (e.g. RNA or DNA) of a length or size which is smaller than the genomic nucleotide sequence from which it was derived. For example, a subgenome can be a region encoding VEE structural proteins, subgenomic RNA can be transcribed from the subgenome using an internal subgenomic promoter, whose sequences reside within the genomic viral RNA or its complement. Transcription of a

subgenome may be mediated by viral-encoded polymerase(s) associated with host cell-encoded proteins (e.g. nsP1-4). In some aspects of the present disclosure, the subgenomic sa-mRNA is produced from a modified alphavirus replicon (e.g. a modified VEE replicon) as disclosed herein and encodes or expresses one or more gene or genes of interest (GOI).

(90) The phrase “pharmaceutically acceptable” is used herein to refer to those compounds, materials, compositions, and/or dosage forms which are, reasonably suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication.

(91) “Pharmaceutically acceptable compositions” may also include salts of one or more compounds. Salts may be pharmaceutically acceptable salts. As used herein, “pharmaceutically acceptable salts” refers to derivatives of the disclosed compounds wherein the parent compound is altered by converting an existing acid or base moiety to its salt form. Generally, such salts can be prepared by reacting the free acid or base forms of these compounds with a stoichiometric amount of the appropriate base or acid in water or in an organic solvent, or in a mixture of the two. In some aspects, a nonaqueous media like ether, ethyl acetate, ethanol, isopropanol, or acetonitrile may be used. Lists of suitable salts are found in Remington's Pharmaceutical Sciences, 17th ed., Mack Publishing Company, Easton, Pa., 1985, p. 1418, Pharmaceutical Salts: Properties, Selection, and Use, P. H. Stahl and C. G. Wermuth (eds.), Wiley-VCH, 2008, and Berge et al., Journal of Pharmaceutical Science, 66, 1-19 (1977), each of which is incorporated herein by reference in its entirety.

(92) As used herein, the term “innate immune response” includes a cellular response to exogenous single stranded nucleic acids, generally of viral or bacterial origin, which involves the induction of cytokine expression and release, particularly the interferons, and cell death. Protein synthesis is also reduced during the innate cellular immune response. The present disclosure provides modified self-amplifying mRNAs that substantially reduce the immune response, including interferon signaling. In some aspects, the immune response is the interferon response of a host cell is 2, 3, 4, 5 or 6 times lower as compared to the immune response induced by a corresponding unmodified nucleic acid. Such a reduction can be measured by expression or activity level of Type 1 interferons or the expression of interferon-regulated genes such as the toll-like receptors (e.g., TLR3, TLR7 and TLR8), and RIG-I like receptors (e.g., RIG1, MDA5, and LGP2). Reduction of innate immune response can also be measured by decreased cell death following one or more administrations of modified RNAs to a cell population; e.g., cell death is the interferon response of a host cell is 2, 3, 4, 5 or 6 times lower than the cell death frequency observed with a corresponding unmodified nucleic acid.

(93) As used herein, the term “immunomodulator” includes cytokines, stem cell growth factors, lymphotoxins, such as tumor necrosis factor (TNF), and hematopoietic factors, such as interleukins (e.g., interleukin-1 (IL-1), IL-2, IL-3, IL-6, IL-10, IL-12, IL-18 and IL-21), colony stimulating factors (e.g., granulocyte-colony stimulating factor (G-CSF) and granulocyte macrophage-colony stimulating factor (GM-CSF)), interferons (e.g., interferons- α , - β and - γ), the stem cell growth factor designated “S1 factor”, erythropoietin, thrombopoietin, and the like.

(94) A cytokine, for the purposes of this disclosure, include any cytokines including but not limited to IL-1, IL-2, IL-3, IL-6, IL-10, IL-12, IL-18, IL-21, interferon- α , interferon- β , and interferon- γ . It may also be a colony stimulating factor, such as GM-CSF, G-CSF, M-CSF, erythropoietin, thrombopoietin, and the like.

(95) As used herein, the terms “approximately” and “about,” as applied to one or more values of interest, refer to a value that is $\pm 20\%$ of the recited value.

(96) Self-Amplifying mRNA

(97) In some aspects, the biologically active agent of the present disclosure is one or more sa-mRNA molecules.

(98) Sa-mRNAs of the disclosure have the ability to self-replicate in cells and, thus, can be used to

induce expression of encoded gene products, such as proteins (e.g., antigens) and regulatory structures (e.g. siRNA, miRNA, saRNA, tRNA, and lincRNA) encoded by the sa-mRNA. In addition, sa-mRNAs are generally based on the genome of an RNA virus (e.g. a Group IV positive single strand RNA virus), and therefore are foreign nucleic acids that can stimulate the innate immune system (e.g. induce an interferon response). This can lead to undesired consequences and safety concerns, such as rapid inactivation and clearance of the RNA, injection site irritation and/or inflammation and/or pain.

(99) The sa-mRNAs of the present disclosure contain modified structures and have reduced capacity to stimulate the innate immune system, which will lead to rapid decay of the sa-mRNA and its associated gene products. Rapid decay of the sa-mRNA and its associated gene products will lead to increased frequency of administration, which is associated with safety concerns and reduced therapeutic efficacy. Thus one aspect of the invention is sa-mRNAs that have reduced cytotoxic effects on the host cell or subject. This provides for enhanced safety of the sa-mRNAs of the present disclosure and provides additional advantages. For example, an advantage of a sa-mRNA with low cytotoxicity allows for administration of a large dose of the sa-mRNAs to produce high expression levels of the encoded gene product with reduced risk of undesired effects, such as injection site irritation and or pain. In addition, because sa-mRNAs of the disclosure have reduced capacity to stimulate the innate immune system, they are well suited to use as vaccines to boost immunity.

(100) One suitable system for producing a sa-mRNA of the present disclosure is to use an alphavirus-based RNA replicon. Alphavirus-based replicons are positive (+)-single stranded replicons that can be translated after delivery to a cell to give a replicase (or replicase-transcriptase). The replicase is translated as a polyprotein which auto-cleaves to provide a replication complex, comprising plurality of non-structural replicase domain sequences, which creates genomic (–)-strand copies of the (+)-strand delivered RNA. These (–)-strand transcripts can themselves be transcribed to give further copies of the (+)-stranded parent RNA and also to give an mRNA transcript which encodes the desired gene product. Translation of the subgenomic transcript thus leads to in situ expression of the desired gene product by the infected cell.

(101) A sa-mRNA may encode (i) a RNA-dependent RNA polymerase which can replicate RNA from sa-mRNA and transcribe (ii) a GOI of the subgenome. The polymerase can be an alphavirus replicase e.g. comprising alphavirus nonstructural proteins 1, 2, 3, and 4.

(102) Whereas natural alphavirus genomes encode structural proteins in addition to the non-structural replicase, in one aspect, an alphavirus based sa-mRNA does not encode alphavirus structural proteins. Thus the sa-mRNA can lead to the production of RNA copies of itself in a cell, but not to the production of RNA-containing alphavirus virions. The inability to produce these virions means that, unlike a wild-type alphavirus, the sa-mRNA cannot perpetuate itself in infectious form. The alphavirus structural proteins which are necessary for perpetuation in wild-type viruses are absent from self-amplifying mRNAs and their place is taken by the GOI, such that the sa-mRNA transcript encodes the desired gene product rather than the structural alphavirus virion proteins.

(103) Thus, the sa-mRNA of the present disclosure may have more than one coding region. The first (5') coding region encodes a plurality of non-structural replicase domain sequences; the second (3') coding region encodes a gene of interest operably linked to a subgenomic promoter. In some aspects the sa-mRNA may have additional (downstream) coding regions e.g. that encode other desired gene products. A coding region molecule can have a 5' sequence which is compatible with the encoded replicase.

(104) The sa-mRNA of the present disclosure may be derived from or based on a virus other than an alphavirus, including but not limited to a Group IV positive-single stranded RNA virus, for example, picornaviridae, togaviridae, coronaviridae, hepeviridae, caliciviridae, flaviviridae, and astroviridae. Suitable wild-type alphavirus sequences are well-known and are available from

sequence depositories, such as the American Type Culture Collection, Rockville, Md.

Representative examples of suitable alphaviruses include Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiyama virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus.

(105) Sa-mRNAs as described herein can amplify themselves and initiate expression of heterologous gene products in the host cell. Sa-mRNAs of the present disclosure, unlike mRNA, use their own encoded viral polymerase to amplify itself. Particular sa-mRNA, such as those based on Group IV RNA viruses such as alphaviruses, generate large amounts of subgenomic mRNAs from which large amounts of proteins (or regulatory structures) can be expressed.

(106) Advantageously, the host cell's own machinery is used by sa-mRNAs to generate an exponential increase of encoded gene products (such as proteins, antigens, or regulatory structures) which can accumulate in the cells or be secreted from the cells. Increased of proteins or antigens by self-amplifying mRNAs takes advantage of the immunostimulatory adjuvant effects, including stimulation of toll-like receptors (TLR) 3, 7 and 8 and non TLR pathways (e.g, RIG-I like receptor, RIG-I, MDA-5, LGP2) by the products of RNA replication and amplification, and translation which induces apoptosis of the transfected cell.

(107) The sa-mRNA of the disclosure may encode any desired gene product, such as a regulatory structure, a polypeptide, a protein or a polypeptide or a fragment of a protein or polypeptide. Additionally, the sa-mRNA of the disclosure may encode a single polypeptide or, optionally, two or more of sequences linked together in a way that each of the sequences retains its identity (e.g., linked in series) when expressed as an amino acid sequence. The polypeptides generated from the sa-mRNAs of the disclosure may then be produced as a fusion protein or engineered in such a manner to result in separate polypeptide or peptide sequences.

(108) The sa-mRNAs of the disclosure may encode one or more immunogenic polypeptides that contain a range of epitopes. In some aspects, such epitopes are capable of eliciting either a helper T-cell response or a cytotoxic T-cell response or both.

(109) The sa-mRNAs described herein may be engineered to express multiple GOI, from two or more coding regions, thereby allowing co-expression of proteins and or regulatory structures, such as a two or more antigens together with cytokines or other immunomodulators, which can enhance the generation of an immune response. Such a sa-mRNA might be particularly useful, for example, in the production of various gene products (e.g., proteins) at the same time, for example, as a bivalent or multivalent vaccine, or in gene therapy applications.

(110) Exemplary gene products that can be encoded by sa-mRNA of the disclosure include proteins and peptides from pathogens, such as bacteria, viruses, fungi and parasites, including any antigenic viral protein (e.g., proteins or peptides from coronavirus, cytomegalovirus, parvovirus, flaviviruses, picornaviruses, norovirus, influenza virus, rhinovirus, yellow fever virus, human immunodeficiency virus (HIV), and the like). Additional exemplary gene products that can be encoded by the sa-mRNAs of the disclosure include any desired eukaryotic polypeptide such as, for example, a mammalian polypeptide such as an enzyme, an enzyme inhibitor, a hormone, a lymphokine, a cytokine, a chemokine, a plasminogen activator, a natural or modified immunoglobulin or a fragment thereof, green fluorescence protein, or any desired combinations of the foregoing. Further exemplary gene products that can be encoded by the sa-mRNA of the disclosure include regulatory structures, such as siRNA, miRNA, gRNA, saRNA, tRNA, and lincRNA, which can be used to regulate expression of endogenous host genes.

(111) The sa-mRNA may also comprise at least one modified nucleic acid and can be prepared using any suitable method. The modification may include a compound selected from the following non-limiting group: pyridin-4-one ribonucleoside, 5-aza-uridine, 2-thio-5-aza-uridine, 2-

thiouridine, 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyl-uridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine, 1-taurinomethyl-4-thio-uridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxyuridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, 5-aza-cytidine, pseudoisocytidine, 3-methyl-cytidine, N4-acetylcytidine, 5-formylcytidine, N4-methylcytidine, 5-hydroxymethylcytidine, 1-methyl-pseudoisocytidine, pyrrolo-cytidine, pyrrolo-pseudoisocytidine, 2-thio-cytidine, 2-thio-5-methyl-cytidine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-cytidine, 2-methoxy-5-methyl-cytidine, 4-methoxy-pseudoisocytidine, 4-methoxy-1-methyl-pseudoisocytidine, 2-aminopurine, 2,6-diaminopurine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-aminopurine, 7-deaza-8-aza-2-aminopurine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diaminopurine, 1-methyladenosine, N6-methyladenosine, N6-isopentenyladenosine, N6-(cis-hydroxyisopentenyl)adenosine, 2-methylthio-N6-(cis-hydroxyisopentenyl)adenosine, N6-glycynylcarbamoyladenosine, N6-threonylcarbamoyladenosine, 2-methylthio-N6-threonylcarbamoyladenosine, N6,N6-dimethyladenosine, 7-methyladenine, 2-methylthio-adenine, 2-methoxy-adenine, inosine, 1-methyl-inosine, wyosine, wybutosine, 7-deaza-guanosine, 7-deaza-8-aza-guanosine, 6-thio-guanosine, 6-thio-7-deaza-guanosine, 6-thio-7-deaza-8-aza-guanosine, 7-methyl-guanosine, 6-thio-7-methyl-guanosine, 7-methylinosine, 6-methoxy-guanosine, 1-methylguanosine, N2-methylguanosine, N2,N2-dimethylguanosine, 8-oxo-guanosine, 7-methyl-8-oxo-guanosine, 1-methyl-6-thio-guanosine, N2-methyl-6-thio-guanosine, N1-Methylpseudouridine-5'-Triphosphate, and N2,N2-dimethyl-6-thio-guanosine. In another aspect, the modifications are independently selected from: 5-methylcytosine, pseudouridine and 1-methylpseudouridine. In one aspect, a modification may be located on a nucleobase of the modified nucleic acid molecule. The modification on the nucleobase may be selected from the group consisting of cytosine, guanine, adenine, thymine and uracil. The modification on the nucleobase may be selected from the group consisting of deaza-adenosine and deaza-guanosine, and a linker may be attached at a C-7 or C-8 position of said deaza-adenosine or deaza-guanosine. The modified nucleobase may be selected from the group consisting of cytosine and uracil, and the linker may be attached to the modified nucleobase at an N-3 or C-5 position. The linker attached to the nucleobase may be selected from the group consisting of diethylene glycol, dipropylene glycol, triethylene glycol, tripropylene glycol, tetraethylene glycol, tetraethylene glycol, divalent alkyl, alkenyl, alkynyl moiety, ester, amide, and ether moiety. In one aspect, two modifications of the nucleic acid molecule may be located on nucleosides of the modified nucleic acid molecule. The modified nucleosides may be selected from 5-methylcytosine and pseudouridine.

(112) Several suitable methods are known in the art for producing RNA molecules that contain modified nucleotides. For example, as described and exemplified herein, a sa-mRNA that contains modified nucleotides can be prepared by transcribing (e.g., in vitro transcription) a nucleic acid that encodes the sa-mRNA using a suitable DNA-dependent RNA polymerase, such as: T7 phage RNA polymerase, SP6 phage RNA polymerase, T3 phage RNA polymerase, T5 phage RNA polymerase, RNA polymerase III, RNA polymerase II, Taq polymerase, Vent polymerase, and the like, or mutants of these polymerases, which allow efficient incorporation of modified nucleotides into RNA molecules. The transcription reaction will contain nucleotides and modified nucleotides, and other components that support the activity of the selected polymerase, such as a suitable buffer, and suitable salts. The incorporation of modified nucleotide into a sa-mRNA may be engineered, for example, to alter the stability of such RNA molecules, to increase resistance against RNases, to

establish replication after introduction into appropriate host cells (“infectivity” of the RNA), and/or to induce or reduce innate and adaptive immune responses.

(113) In one aspect, the sa-mRNA of the disclosure comprise a polynucleotide sequence selected from: a) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 1 (BA.1-1273); b) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 2 (BA.1-1273-S2P); c) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 3 (BA.2-1273); d) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 4 (BA.2-1273-S2P); e) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 5 (BA.1-1208); or f) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 6 (BA.1-1208-S2P). g) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 7 (BA.2-1208); or h) a polynucleotide encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 8 (BA.2-1208-S2P).

(114) In one aspect of the present disclosure, the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-Ag-SGP2-IM wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), or SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P). SGP2 is the second subgenomic promoter, and IM is the immunomodulator.

(115) In one aspect, the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-IM-SGP2-AG wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, IM is the immunomodulatory, SGP2 is the second subgenomic promoter, and Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), or SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P).

(116) In some aspects, the IM encodes one or more cytokines, chemokines, immune stimulators or inhibitors. In one aspect, the IM is selected from IL12 and IL21. In one aspect, the IM encodes one or more cytokines selected from SEQ ID NOs: 22 (hIL12-P40), 24 (hIL12-P35), 15 (mIL12 P40), 17 (mIL12-P35), and 21 (mIL21). In one aspect, SGP1 is SEQ ID NO: 9 (SGP1). In one aspect, SGP2 is SEQ ID NO: 11 (SGP2). In one aspect, IM is selected from SEQ ID NO: 13 (IM1), and SEQ ID NO: 20 (IM2).

(117) In another aspect, the present disclosure includes sa-mRNA comprising the following operably linked nucleic acid sequence from 5' to 3': SP-IL12P40-L1-IL12P35-L2-IL21 wherein SP is a signal peptide, IL12-P40 is interleukin-12 comprising heavy chain p40, L1 is linker 1, IL12 P35 is interleukin-12 comprising light chain p35, L2 is linker 2, and IL21 is interleukin-21.

(118) In some aspects, SP is selected from SEQ ID NO: 14 (MSP) and SEQ ID NO: 21 HSP. In some aspects, IL12-P40 is selected from SEQ ID NO: 15 (mIL12-P40) and SEQ ID NO: 22 (hIL12-P40). In some aspects, L1 is selected from SEQ ID NO: 16 (L(a)) and SEQ ID NO: 23 (L(c)). In some aspects, IL12-P35 is selected from SEQ ID NO: 17 (mIL12-P35) and SEQ ID NO: 24 (hIL12-P35). In some aspects, L2 is selected from SEQ ID NO: 18 (L(b)) and SEQ ID NO: 25 (L(d)). In some aspects, IL12-P40 is selected from SEQ ID NO: 19 (mIL21) and SEQ ID NO: 26 (hIL21).

(119) In some aspects, at least one non-structural replicase domain sequence comprise sequences selected from Group IV RNA viruses, selected from Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, and Astroviridae. In some aspects, at least one non-

structural replicase domain sequence comprise sequences selected from Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiya virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. In yet another aspect, at least one non-structural replicase domain sequence is obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). In some aspects, the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4).

(120) In some aspects, SGP1 is a viral promoter that is recognized by viral RNA dependent RNA polymerase (RdRP). In some aspects, SGP2 is a viral promoter that is recognized by viral RNA dependent RNA polymerase (RdRP). In some aspects, SGP1 and SGP2 are different subgenomic promoters.

(121) In some aspects, the sa-mRNA of the disclosure comprises one or more linkers. In some aspects, the linkers are selected from the group SEQ ID Nos: 16 (L(a)), 18 (L(b)), 23 (L(c)), and 25 (L(d)).

(122) In some aspects, the sa-mRNA of the present disclosure comprises a polynucleotide encoding a modified SARS-COV-2 spike protein. In some aspects, the polynucleotide encoding a modified SARS-COV-2 spike protein comprise a nucleic sequence selected from the group SEQ ID NO: 1 (BA.1-1273), SEQ ID NO: 2 (BA.1-1273-S2P), SEQ ID NO: 3 (BA.2-1273), SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), and SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), and SEQ ID NO: 8 (BA.2-1208-S2P).

(123) Sa-mRNAs of the present disclosure can be introduced into target cells or subjects using any suitable technique, e.g., by direct injection, microinjection, electroporation, lipofection, biolistics, and the like. The sa-mRNA may also be introduced into cells by way of receptor-mediated endocytosis. See e.g., U.S. Pat. No. 6,090,619; Wu and Wu, *J. Biol. Chem.*, 263:14621 (1988); and Curiel et al, *Proc. Natl. Acad. Sci. USA*, 88:8850 (1991).

(124) The sa-mRNAs of the present disclosure can be delivered into cells via amphiphiles. See e.g., U.S. Pat. No. 6,071,890. Typically, a nucleic acid molecule may form a complex with the cationic amphiphile. Mammalian cells contacted with the complex can readily take it up.

(125) The sa-mRNAs of the present disclosure can be delivered as naked RNA (e.g. merely as an aqueous solution of RNA) but, to enhance entry into cells and also subsequent intercellular effects, the sa-mRNA may be administered in combination with a delivery system, such as a particulate or emulsion delivery system. A large number of delivery systems are well known to those of skill in the art. Such delivery systems include, for example lipid nanoparticle based delivery (Debs and Zhu (1993) WO 93/24640; Mannino and Gould-Fogerite (1988) *BioTechniques* 6 (7): 682-691; Rose U.S. Pat. No. 5,279,833; Brigham (1991) WO 91/06309; and Feigner et al (1987) *Proc. Natl. Acad. Sci. USA* 84:7413-7414), as well as use of viral vectors {e.g., adenoviral (see, e.g., Berns et al (1995) *Ann. NY Acad. Sci.* 772:95-104; Ali et al (1994) *Gene Ther.* 1:367-384; and Haddada et al. (1995) *Curr. Top. Microbiol. Immunol.* 199 (Pt 3): 297-306 for review), papillomaviral, retroviral (see, e.g., Buchscher et al. (1992) *J. Virol.* 66 (5) 2731-2739; Johann et al. (1992) *J. Virol.* 66 (5): 1635-1640 (1992); Sommerfelt et al, (1990) *Viol.* 176:58-59; Wilson et al. (1989) *J. Virol.* 63:2374-2378; Miller et al, *J. Virol.* 65:2220-2224 (1991); Wong-Staal et al, PCT/US94/05700, and Rosenberg and Fauci (1993) in *Fundamental Immunology*, Third Edition Paul (ed) Raven Press, Ltd., New York and the references therein, and Yu et al, *Gene Therapy* (1994) *supra.*), and adeno-associated viral vectors (see, West et al (1987) *Virology* 160:38-47; Carter et al (1989) U.S. Pat. No. 4,797,368; Carter et al WO 93/24641 (1993); Kotin (1994) *Human Gene Therapy* 5:793-801; Muzyczka (1994) *J. Clin. Invest.* 94:1351 and Samulski (*supra*) for an overview of AAV vectors; see also, Lebkowski, U.S. Pat. No. 5,173,414; Tratschin et al (1985) *Mol. Cell. Biol.* 5 (11): 3251-

3260; Tratschin, et al (1984) Mol. Cell. Biol, 4:2072-2081; Hermonat and Muzyczka (1984) Proc. Natl. Acad. Sci. USA, 81:6466-6470; McLaughlin et al (1988) and Samulski et al (1989) J. Virol, 63:03822-3828), and the like.

(126) Three particularly useful delivery systems are (i) LNPs (ii) non-toxic and biodegradable polymer microparticles (iii) cationic submicron oil-in-water emulsions. In one aspect, the sa-mRNA of the present disclosure is delivered using LNPs.

(127) In one aspect, a sa-mRNA of the disclosure encodes two separated expression units, the first expression unit comprising a polynucleotide encoding a modified antigen, wherein the polynucleotide encoding the modified antigen is truncated to not include nucleotides encoding a transmembrane domain and short cytosolic domain amino acids of the antigen, operably linked to a first subgenomic promoter; and the second expression unit encoding immunomodulators (IM) that are operably linked to a second subgenomic promoter. The polynucleotide encoding a modified antigen comprise a sequence that is 90%, 95%, 98%, 99% or 100% identical to SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), SEQ ID NO: 8 (BA.2-1208-S2P). In addition, in some aspects, any of SEQ ID NO: 1, 3, or 5 wherein "T" is replaced by "U". Those of skill in the art will recognize that, due to the degenerate nature of the genetic code, a variety of codons differing in their nucleotide sequences can be used to encode a given amino acid. A particular nucleotide sequence encoding a polypeptide described herein are referenced merely to illustrate an aspect of the disclosure, and the disclosure includes nucleic acids of any sequence that encode a polypeptide comprising the same amino acid sequence of the polypeptides and proteins utilized in the methods of the disclosure. In similar fashion, a polypeptide can typically tolerate one or more amino acid substitutions, deletions, and insertions in its amino acid sequence without loss or significant loss of a desired activity. The disclosure includes such polypeptides with alternate amino acid sequences, and the amino acid sequences encoded by the RNA or DNA sequences shown herein merely illustrate aspects of the disclosure.

(128) In one aspect, the sa-mRNA of the disclosure comprises, from 5' to 3', alphavirus nonstructural proteins 1-4 (nsP1-4) from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE); a first subgenomic promoter, a first expression unit encoding an antigen selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), SEQ ID NO: 8 (BA.2-1208-S2P), a second subgenomic promoter, and a second expression unit encoding one or more immunomodulator(s). In some aspects, the second expression unit encodes from 5' to 3': a signal peptide, interleukin-12 comprising heavy chain p40 (IL12-P40), linker 1 (L1), interleukin-12 comprising light chain p35 (IL12 P35), linker 2 (L2), and interleukin-21 (IL21).

(129) In one aspect, the sa-mRNA of the disclosure comprises, alphavirus nonstructural proteins 1-4 (nsP1-4) from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE); a first subgenomic promoter, a first expression unit encoding one or more immunomodulator(s), a second subgenomic promoter, and a second expression unit encoding an antigen selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.1-1208), SEQ ID NO: 8 (BA.1-1208-S2P). In some aspects, the second expression unit encodes from 5' to 3': a signal peptide, interleukin-12 comprising heavy chain p40 (IL12-P40), linker 1 (L1), interleukin-12 comprising light chain p35 (IL12 P35), linker 2 (L2), and interleukin-21 (IL21).

(130) In one aspect, the sa-mRNA of the present disclosure can incorporate one or more custom GOI built by synthetic methods known in the art, or cloned from cDNA or a genomic library.

(131) Sa-mRNA of the present disclosure can encode an antigen which can be tested for ability to induce humoral immune responses, as evidenced, for example, by induction of B cell production of antibodies specific for an antigen of interest. These assays can be conducted using, for example, peripheral B lymphocytes from immunized individuals. Such assay methods are known to those of

skill in the art. Other assays that can be used to characterize the sa-mRNA of the present disclosure can involve detecting expression of the encoded antigen by the target cells. For example, FACS can be used to detect antigen expression on the cell surface or intracellularly. Another advantage of FACS selection is that one can sort for different levels of expression; sometimes-lower expression may be desired. Other suitable method for identifying cells which express a particular antigen involve panning using monoclonal antibodies on a plate or capture using magnetic beads coated with monoclonal antibodies.

(132) De Novo Synthesis of Self-Amplifying mRNA

(133) The sa-mRNAs of the present disclosure may be produced from a nucleic acid template in the form of recombinant DNA expression vectors, RNA replicons or plasmids. The nucleic acid template of the present disclosure encodes two expression units comprising: i) an origin of replication sequence (Ori); ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter; and iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter, wherein the first expression unit encodes a selectable marker and the second expression unit encodes a sa-mRNA.

(134) The nucleic acid template may be produced using a suitable synthetic method either alone or in combination with one or more other methods. Such methods are well known in the art, including chemical synthesis using suitable protecting groups such as CEM (Masuda et al., (2007) Nucleic Acids Symposium Series 57:3-4), the β -cyanoethyl phosphoramidite method (Beaucage S L et al. (1981) Tetrahedron Lett 22:1859); nucleoside H-phosphonate method (Garegg P et al. (1986) Tetrahedron Lett 27:4051-4; Frochler B C et al. (1986) Nucl Acid Res 14:5399-407; Garegg P et al. (1986) Tetrahedron Lett 27:4055-8; Gaffney B L et al. (1988) Tetrahedron Lett 29:2619-22). These chemistries can be performed or adapted for use with automated nucleic acid synthesizers that are commercially available. Additional suitable synthetic methods are disclosed in Uhlmann et al. (1990) Chem Rev 90:544-84, and Goodchild J (1990) Bioconjugate Chem 1:165. Nucleic acid synthesis can also be performed using suitable recombinant methods that are well-known and conventional in the art, including cloning, processing, and/or expression of polynucleotides and gene products encoded by such polynucleotides. DNA shuffling by random fragmentation and PCR reassembly of gene fragments and synthetic polynucleotides are examples of known techniques that can be used to design and engineer polynucleotide sequences. Site-directed mutagenesis can be used to alter nucleic acids and the encoded proteins, for example, to insert new restriction sites, alter glycosylation patterns, change codon preference, produce splice variants, introduce mutations and the like. Suitable methods for transcription, translation and expression of nucleic acid sequences are known and conventional in the art. (See generally, Current Protocols in Molecular Biology, Vol. 2, Ed. Ausubel, et al., Greene Publish. Assoc. & Wiley Interscience, Ch. 13, 1988; Glover, DNA Cloning, Vol. II, IRL Press, Wash., D.C., Ch. 3, 1986; Bitter, et al., in Methods in Enzymology 153:516-544 (1987); The Molecular Biology of the Yeast *Saccharomyces*, Eds. Strathern et al., Cold Spring Harbor Press, Vols. I and II, 1982; and Sambrook et al., Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Press, 1989.)

(135) The present disclosure discloses a method of increasing the copy number of the nucleic acid template by transforming the nucleic acid into suitable host cells (e.g. *Escherichia coli* cells); selecting cells that express the selectable marker encoded by the first expression unit; subculturing the selected cells to obtain a population of host cells that express the selectable marker; and propagating the population of selected host cells to increase the copy number of the nucleic acid template.

(136) In one aspect, a nucleic acid template of the disclosure comprise a sequence that is 90%, 95%, 98%, 99% or 100% identical to SAM001 (SEQ ID NO: 35), SAM002 (SEQ ID NO: 36), SAM003 (SEQ ID NO: 37), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), MOD001 (SEQ ID NO: 41), or T7-VEE-GFP (SEQ ID NO: 42). In addition, in some aspects, any of SEQ ID NOs: 35, 36, 37, 38, 39, 40, 41, or 42 wherein "T" is replaced by

“U”. Those of skill in the art will recognize that, due to the degenerate nature of the genetic code, a variety of codons differing in their nucleotide sequences can be used to encode a given amino acid. A particular nucleotide sequence encoding a polypeptide described herein are referenced merely to illustrate an aspect of the disclosure, and the disclosure includes nucleic acids of any sequence that encode a polypeptide comprising the same amino acid sequence of the polypeptides and proteins utilized in the methods of the disclosure. In similar fashion, a polypeptide can typically tolerate one or more amino acid substitutions, deletions, and insertions in its amino acid sequence without loss or significant loss of a desired activity. The disclosure includes such polypeptides with alternate amino acid sequences, and the amino acid sequences encoded by the RNA or DNA sequences shown herein merely illustrate aspects of the disclosure.

(137) In one aspect, SEQ ID NO: 35 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 35 has “T” replaced with “U”. The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the ampicillin resistance (AmpR) promoter; a selectable marker comprising the ampicillin resistance gene (AmpR); a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter, alphavirus nonstructural proteins 1-4 (nsP1-4) from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE); a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(138) In one aspect, SEQ ID NO: 36 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 36 has “T” replaced with “U”. The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(139) In one aspect, SEQ ID NO: 37 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 37 has “T” replaced with “U”. The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(140) In one aspect, SEQ ID NO: 38 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 104 has “T” replaced with “U”. The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 38); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(141) In one aspect, SEQ ID NO: 39 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 105 has “T” replaced with “U”. The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 39); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 112); a 3'

untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(142) In one aspect, SEQ ID NO: 40 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 40 has "T" replaced with "U". The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(143) In one aspect, SEQ ID NO: 41 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 41 has "T" replaced with "U". The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(144) In one aspect, SEQ ID NO: 42 provides a nucleic acid of the disclosure. In another aspect the sequence of SEQ ID NO: 42 has "T" replaced with "U". The nucleic acid comprises, from 5' to 3', a first linker (SEQ ID NO: 43); a first promoter sequence comprising the AmpR promoter; a selectable marker comprising AmpR; a second linker (SEQ ID NO: 44); a second promoter sequence for in vitro transcription comprising the T7 promoter; a 5' untranslated region; a subgenomic promoter; alphavirus nsP1-4 from the TC-83 strain of VEE; a third linker (SEQ ID NO: 45); a subgenomic promoter; the gene of interest comprising GFP; a fourth linker (SEQ ID NO: 46); a 3' untranslated region; and a 3' poly-adenylated tail (poly-A tail).

(145) In one aspect, the sa-mRNA of the present disclosure can incorporate one or more custom GOI built by synthetic methods known in the art, or cloned from cDNA or a genomic library. The GOI, along with promoters, other regulatory elements, optionally one or more linkers, an origin of replication, and a selectable marker are incorporated into a nucleic acid template. Nucleic acid templates of essentially any length and sequence can be produced in high yield in *Escherichia coli*. Sa-mRNA of any desired sequence can be produced from nucleic acid templates by in vitro transcription.

(146) In vitro transcription (IVT) methods permit template-directed synthesis of RNA molecules (including self-amplifying mRNA) of almost any sequence. The size of the RNA molecules that can be synthesized using IVT methods range from short oligonucleotides to long nucleic acid polymers of several thousand bases. IVT methods permit synthesis of large quantities of RNA transcript (e.g., from microgram to milligram quantities) (Beckert et al., Synthesis of RNA by in vitro transcription, Methods Mol Biol. 703:29-41 (2011); Rio et al. RNA: A Laboratory Manual. Cold Spring Harbor: Cold Spring Harbor Laboratory Press, 2011, 205-220; Cooper, Geoffrey M. The Cell: A Molecular Approach. 4th ed. Washington D.C.: ASM Press, 2007. 262-299). Generally, IVT utilizes a nucleic acid template featuring a promoter sequence upstream of a sequence of interest. The promoter sequence is most commonly a bacteriophage promoter (e.g. the T7, T3, SP6, or T5 promoter sequence) but many other promoter sequences can be tolerated (e.g SV40, β -lactamase promoter, *E. coli* galactose promoter, arabinose promoter, alkaline phosphatase promoter, trp promoter, lac promoter, lacUV5 promoter, trc promoter and tac promoter) including those designed de novo. Transcription of the DNA template is typically best achieved by using the RNA polymerase corresponding to the specific promoter sequence. Exemplary RNA polymerases include, but are not limited to T7 phage RNA polymerase, SP6 phage RNA polymerase, T3 phage RNA polymerase, T5 phage RNA polymerase, RNA polymerase III, RNA polymerase II, Taq polymerase, Vent polymerase, and the like, or mutants of these polymerases. IVT is generally

initiated at a dsDNA but can proceed using RNA and/or on a single strand.

(147) Self-Amplifying mRNA Library

(148) One aspect of the present disclosure is a method of generating a sa-mRNA library. In one aspect, the invention is a method for preparing a library of sa-mRNA derived from a reference sa-mRNA comprising: (i) performing directed evolution of a reference sa-mRNA sample comprising the steps of: (a) delivering a reference sa-mRNA sample encoding a selection marker into host cell(s), (b) culturing said host cell(s) over a period of time under conditions that require replication of the reference sa-mRNA sample and permit expression of the selection marker, wherein mutations occur in the replicated sa-mRNA compared to the reference sa-mRNA, (c) selecting cells that express the selectable marker;

(ii) extracting the replicated sa-mRNA from the host cell(s) and sequencing the replicated sa-mRNA; and thereby producing a library of sa-mRNA sequences.

(149) In one aspect, the selection marker is an antibiotic resistance gene. In one aspect, the selection marker is a puromycin resistance gene. In one aspect, the reference sa-mRNA is delivered into a host cell using a delivery mechanism. In one aspect, the delivery system is a lipid nanoparticle. In one aspect, the reference sa-mRNA is selected from a group comprising SEQ ID NOs. 1-8 and SEQ ID NOs 35-42. In one aspect, the conditions that require replication of the reference sa-mRNA sample and permit expression of the selection marker is a culture environment containing an antibiotic. In one aspect, the concentration of the antibiotic affects the rate of mutation of the reference sa-mRNA.

(150) In one aspect, the disclosure provides a method of evaluating mutations of the replicated self-amplifying mRNA, the method comprising: (i) obtaining a group of contig sequences comprising mutation(s) compared to a reference sa-mRNA sample, (ii) sequencing the contig sequences, and (iii) determining the number of mutations in the contig sequences compared to the reference sa-mRNA. In one aspect, the contig sequences are fragments of the replicated sa-mRNA. In certain aspects, the group of contig sequences comprise SEQ ID NOs. 27-34.

(151) Increased In Vitro Transcription

(152) One aspect of the present disclosure is a nucleic acid containing modified promoters and regulatory elements, such as a modified 5' UTR. Said nucleic acid shows an unexpected improvement in transcription efficiency while reducing the amount of truncated single-stranded ribonucleic acid (ssRNA) (e.g., sa-mRNA) transcript produced during an in vitro transcription (IVT) reaction. In a typical IVT reaction, greater than 50% (molarity) of the RNA transcripts produced are truncated abortive products (referred to herein as truncated ssRNA transcripts). Only a small fraction (e.g., 0.2-0.5%) of initiation events lead to full-length "run-off" ssRNA transcripts, which is inefficient and costly for large-scale IVT RNA synthesis systems. Sa-mRNA transcripts in particular are longer than conventional mRNA (larger than 7 kilo nucleotides) and are particularly susceptible to truncated abortive products. Thus, use of the IVT methods of the present disclosure (which include, for example, nucleic acid constructs, modified promoters and/or modified 5'UTR), in some aspects, results in a sa-mRNA transcript yield that is at least 40% greater than the sa-mRNA transcript yield of an IVT method without the modified regulatory elements of the present disclosure.

(153) Preferably, the nucleic acid template of the present disclosure comprise a modified T7 promoter operably linked to nucleic acid comprising a sequence that encodes a modified 5' untranslated region (UTR) a plurality of non-structural replicase domain sequences, one or more gene or genes of interest (GOI), a 3' UTR, and a poly-A tail, wherein the sequence that encodes the T7 promoter and the sequence that encodes the 5' UTR is modified to enhance the binding strength of T7 polymerase to the T7 promoter to increase transcript yield.

(154) In some aspects, a modified T7 promoter comprises at least one insertion at position at the 5' end of the wildtype T7 promoter nucleotide sequence. The modification may be, for example, insertion of a single guanine (G) at the 5' end of the wildtype T7 promoter. In some aspects, the

modified T7 promoter comprises SEQ ID NO: 47 (TAATACGACTCACTATAGG).

(155) In some aspects, a modified 5'UTR comprises at least one insertion at position 3 relative to the 5' end of the wildtype 5'UTR nucleotide sequence. The modification may be, for example, insertion of a single adenine (A) at position 3 of the wildtype 5'UTR of wildtype T7-VEE-GFP (SEQ ID NO: 42). In some aspects, the modified 5'UTR comprises ATAGG (repeating the last 5 nucleotides of T7 promoter).

(156) In one aspect, a nucleic acid of the present disclosure consisting of a nucleotide sequence which is at least 90% identical to SEQ ID NO: 36.

(157) Preferably, the nucleic acid template containing a modified T7 promoter and 5'UTR of the present disclosure will cause a host cell to produce more self-amplifying mRNA, which will translate an increased amount of gene product relative to the amount of gene product produced by the same cell type that contains the corresponding sa-mRNA that does not contain modified nucleotides. Methods of determining translation efficiency are well known in the art, and include, e.g. measuring the activity or amount of an encoded protein (e.g. luciferase and/or GFP), or measuring radioactive label incorporated into the translated protein (See, e.g., Ngosuwana J, Wang N M et al, J Biol Chem 2003; 278 (9): 7034-42).

(158) Immune Response Modulation

(159) One aspect of the present disclosure is a nucleic acid containing regulatory elements, such as a modified 3' UTR. Said nucleic acid is capable of decreasing the immunogenicity and/or immunostimulatory capacity (immune response) of said nucleic acid. In one aspect, the nucleic acid of the present disclosure is a sa-mRNA. In another aspect, the nucleic acid of the present disclosure is a nucleic acid template (e.g. a DNA or RNA template), which encodes a sa-mRNA.

(160) In general, exogenous nucleic acids, particularly of viral origin, induce an innate immune response when introduced into cells, resulting in interferon (IFN) production and cell death. However, it is of great interest for therapeutics, diagnostics, reagents and for biological assays to deliver a nucleic acid, e.g., a ribonucleic acid (RNA) inside a cell, either in vivo or ex vivo, such as to cause intracellular translation of the nucleic acid and production of the encoded protein. Of particular importance is the delivery and function of a non-integrative nucleic acid (e.g. RNA), as nucleic acids characterized by integration into a target cell are generally imprecise in their expression levels, deleteriously transferable to progeny and neighbor cells, and suffer from the substantial risk of mutation. Provided herein are nucleic acids encoding useful gene products capable of modulating a cell's function and/or activity, and methods of making and using these nucleic acids and gene products. As described herein, these nucleic acids are capable of reducing the innate immune response of a population of cells into which they are introduced, thus increasing the efficiency of protein production in that cell population.

(161) The sa-mRNA of the present disclosure encodes at least one gene product, by preferably increasing the adenine (A) content of the 3'UTR. In some aspects, use of the nucleic acid molecules and modified regulatory elements of the present disclosure (which include, for example, nucleic acid constructs, and/or modified 3'UTR), results in interferon responses that are 2 times, 3 times, 4 times, or 5 times lower than the interferon response to sa-mRNAs without the modified regulatory elements of the present disclosure after one day post-transfection.

(162) In some aspects, a modified 3'UTR comprises at least one modification at any one of positions 6, -1, or -2 relative to a conserved 19 nucleotide sequence SEQ ID NO: 49

(GGATTTTGTTTTAAATATTC). In another aspect the sequence of SEQ ID NO: 49 has "T" replaced with "U". The modification may be, for example, a mutant 3'UTR of an alphavirus comprising point mutations at position 6 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. The modification may also be, for example, a mutant 3'UTR of an alphavirus comprising point mutations at positions -1 and -2 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. The modification may also be, for example, a mutant 3'UTR of an alphavirus comprising point

mutations at positions -1, -2 and 6 relative to the conserved 19 nucleotide sequence, SEQ ID NO: 49, of the wild-type 3'UTR of an alphavirus. In some aspects, the modified 3'UTR conserved sequence comprise GGATTTTATTTTAAATATTC (SEQ ID NO: 50), AAATTTTGTTTTAAATATTC (SEQ ID NO: 51), or AAATTTTATTTTAAATATTC (SEQ ID NO: 52). In other aspects the sequence of SEQ ID NO: 50, SEQ ID NO: 51, and SEQ ID NO: 52 has "T" replaced with "U".

(163) Sa-mRNA of the present disclosure can encode an antigen which can be tested for ability to induce humoral immune responses, as evidenced, for example, by induction of B cell production of antibodies specific for an antigen of interest. These assays can be conducted using, for example, peripheral B lymphocytes from immunized individuals. Such assay methods are known to those of skill in the art. Other assays that can be used to characterize the sa-mRNA of the present disclosure can involve detecting expression of the encoded antigen by the target cells. For example, FACS can be used to detect antigen expression on the cell surface or intracellularly. Another advantage of FACS selection is that one can sort for different levels of expression; sometimes-lower expression may be desired. Other suitable method for identifying cells which express a particular antigen involve panning using monoclonal antibodies on a plate or capture using magnetic beads coated with monoclonal antibodies.

(164) Antigens of SARS-COV2 Omicron Variant BA.2

(165) The disclosure also relates to polypeptides encoding a modified SARS-COV2 antigen, wherein the polynucleotide encoding the modified antigen is truncated to not include nucleotides encoding a transmembrane domain and short cytosolic domain amino acids of the antigen. Since the SARS-Coronavirus 2 (SARS-COV2) was firstly identified from Wuhan, China, there has been 526,808,553 cases and 6,280,679 deaths reported in global by May 25, 2022

(<https://coronavirus.jhu.edu/map.html>). Although the first generation of COVID vaccines (e.g. BNT162b2 (BioNTech-Pfizer), mRNA-1273 (Moderna)) are available and the temporary variants of SARS-Coronavirus 2 show gradually mild and lower death rate, the pandemic is still ongoing, giving rise to severe social and economic crisis. Therefore, while many people globally have been vaccinated with one or more shots of the first-generation COVID vaccines, there remains a need to develop COVID booster vaccines to address shortcomings of current COVID vaccines such as induction of hepatitis, and myocarditis; reduction efficiency in dealing with the rapid evolutions of SARS-COV-2; inefficiency on preventing infection; and quick decrease of antibody titer.

(166) In some aspects, the polypeptides of the present disclosure encode secreted versions of the SPIKE protein. The first generation of mRNA COVID vaccines, BNT162b2 and mRNA-1273, comprise 1273 amino acids including: S1 (RBD), S2, transmembrane domain and a short cytosolic domain. Since the transmembrane domain leads to the expression of SPIKE antigens on the cell surface of transfected cells, the transfected cells are targeted by immune system, this likely leads to side effects of hepatitis, and myocarditis that manifest in some individuals vaccinated using BNT162b2 and mRNA-1273. The polypeptides of the present disclosure encode a modified SPIKE protein that is secreted while retaining its native structure. This will prevent the expression of SPIKE antigens on the cell surfaces. The secreted version of the modified SPIKE antigen is able to trigger humoral immune responses and shows comparable BA.2 specific IgG compared to transmembrane SPIKE proteins.

(167) In some aspects, the polypeptides of the present disclosure encode a modified SPIKE protein with 2 proline mutations on S2 (S2P). The S2P mutation keep the conformation of SPIKE protein for induction of neutralization antibodies stabilize the structure of SPIKE for recognition by broad neutralization antibody (bnAb) SPD-M265 and hACE2-FITC (FIG. 18-20).

(168) Pharmaceutical Compositions

(169) The disclosure also relates to pharmaceutical compositions comprising a sa-mRNA of the present disclosure (which optionally contains a modified 3' UTR of the present disclosure), a pharmaceutically acceptable carrier and a suitable delivery system of the present disclosure, as

described herein, such as liposomes, lipid nanoparticles, nanoemulsions, PLG micro- and nanoparticles, lipoplexes, chitosan micro- and nanoparticles and other polyplexes. If desired other pharmaceutically acceptable components can be included, such as excipients and adjuvants.

(170) Pharmaceutical Compositions

(171) The disclosure also relates to pharmaceutical compositions comprising a self-amplifying mRNA (which optionally contains a modified 3' UTR of the present disclosure), a pharmaceutically acceptable carrier and a suitable delivery system as described herein, such as liposomes, nanoemulsions, PLG micro- and nanoparticles, lipoplexes, chitosan micro- and nanoparticles and other polyplexes. If desired other pharmaceutically acceptable components can be included, such as excipients and adjuvants.

(172) Nanoparticle Composition

(173) Preferably, the sa-mRNA of the present disclosure is delivered using a nanoparticle composition comprising one or more cationic and/or ionizable lipids (e.g., lipids that may have a positive or partial positive charge at physiological pH); one or more PEG or PEG-modified lipids (a lipid modified with polyethylene glycol); one or more structural lipids (e.g. cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, tomatine, ursolic acid, alpha-tocopherol, and mixtures thereof); and one or more phospholipids (e.g. (poly)unsaturated lipids).

(174) Adjuvants

(175) In some aspects, a pharmaceutical composition that includes one or more lipids described herein may further include one or more adjuvants, e.g., Glucopyranosyl Lipid Adjuvant (GLA), CpG oligodeoxynucleotides (e.g., Class A or B), poly(I: C), aluminum hydroxide, and Pam3CSK4.

(176) Medical Uses

(177) In some aspects, the sa-mRNA of the present disclosure optionally encodes messenger mRNA (mRNA), small interfering RNA (siRNA), micro-RNA (miRNA), guide RNA (gRNA), self-activating RNA (saRNA), transfer RNA (tRNA), long intergenic non-coding (lincRNA), etc.

(178) In certain aspects, the biologically active sa-mRNA of the present disclosure encodes an mRNA. Said mRNA may encode any polypeptide of interest, including any naturally or non-naturally occurring or otherwise modified polypeptide. A polypeptide encoded by an mRNA may be of any size and may have any secondary structure or activity. In some aspects, a polypeptide encoded by an mRNA may have a therapeutic effect when expressed in a cell. In some aspects, the polypeptide encoded by the mRNA is a modified SPIKE antigen.

(179) In other aspects, the biologically active sa-mRNA of the present disclosure encodes a siRNA or a miRNA. A siRNA or miRNA may be capable of selectively knocking down or down regulating expression of a gene of interest. For example, a siRNA could be selected to silence a gene associated with a particular disease, disorder, or condition upon administration to a subject in need thereof of a nanoparticle composition including the siRNA. A siRNA may comprise a sequence that is complementary to an mRNA sequence that encodes a gene or protein of interest. In some aspects, the siRNA may be an immunomodulatory siRNA.

(180) Formulations

(181) Pharmaceutical compositions may include a biologically active sa-mRNA and one or more additional components, such as a lipid component and one or more additional components. A nanoparticle composition may be designed for one or more specific applications or targets. The elements of a nanoparticle composition may be selected based on a particular application or target, and/or based on the efficacy, toxicity, expense, ease of use, availability, or other feature of one or more elements. Similarly, the particular formulation of a nanoparticle composition may be selected for a particular application or target according to, for example, the efficacy and toxicity of particular combinations of elements.

(182) The sa-mRNA of a pharmaceutical composition may include, for example, a sa-mRNA comprising: an antigen selected from the group SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3

(BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), SEQ ID NO: 8 (BA.2-1208-S2P); immunomodulators selected from the group SEQ ID NO: 13 (IM1), and SEQ ID NO: 20 (IM2); SEQ ID NO: 9 (SGP1); SGP2 is SEQ ID NO: 11 (SGP2); and a nucleotide sequence encoding nsp1-4.

(183) The amount of a biologically active sa-mRNA may depend on the size, composition, desired target and/or application, or other properties of the therapeutic, diagnostic and/or prophylactic. Generally, the size of sa-mRNA is always larger than 7 kilo nucleotides. The relative amounts of the sa-mRNA and other elements (e.g., lipids) in a pharmaceutical composition may also vary. In some aspects, the wt/wt ratio of the lipid component to a sa-mRNA in a nanoparticle composition may be from about 5:1 to about 60:1, such as 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 11:1, 12:1, 13:1, 14:1, 15:1, 16:1, 17:1, 18:1, 19:1, 20:1, 25:1, 30:1, 35:1, 40:1, 45:1, 50:1, and 60:1. For example, the wt/wt ratio of the lipid component to a sa-mRNA may be from about 1:1 to about 40:1. In certain aspects, the wt/wt ratio is about 20:1. The amount of a therapeutic, diagnostic and/or prophylactic in a nanoparticle composition may, for example, be measured using absorption spectroscopy (e.g., ultraviolet-visible spectroscopy).

(184) Pharmaceutical compositions may include one or different therapeutic agents (e.g. sa-mRNA) and delivery systems. Pharmaceutical compositions may further include one or more pharmaceutically acceptable excipients or accessory ingredients such as those described herein. General guidelines for the formulation and manufacture of pharmaceutical compositions and agents are available, for example, in Remington's *The Science and Practice of Pharmacy*, 21st Edition, A. R. Gennaro; Lippincott, Williams & Wilkins, Baltimore, Md., 2006. Excipients and accessory ingredients may be used in any pharmaceutical composition, except insofar as any excipient or accessory ingredient may be incompatible with one or more components of a sa-mRNA delivery system. An excipient or accessory ingredient may be incompatible if its combination with the component may result in any undesirable biological effect or otherwise deleterious effect.

(185) In some aspects, one or more excipients or accessory ingredients may make up greater than 50% of the total mass or volume of a pharmaceutical composition. For example, the one or more excipients or accessory ingredients may make up 50%, 60%, 70%, 80%, 90%, or more of a pharmaceutical composition. In some aspects, a pharmaceutically acceptable excipient is at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% pure. In some aspects, an excipient is approved for use in humans and for veterinary use. In some aspects, an excipient is approved by United States Food and Drug Administration. In some aspects, an excipient is pharmaceutical grade. In some aspects, an excipient meets the standards of the United States Pharmacopoeia (USP), the European Pharmacopocia (EP), the British Pharmacopocia, and/or the International Pharmacopocia.

(186) Relative amounts of the one or more delivery systems, the one or more pharmaceutically acceptable excipients, and/or any additional ingredients in a pharmaceutical composition in accordance with the present disclosure will vary, depending upon the identity, size, and/or condition of the subject treated and further depending upon the route by which the composition is to be administered.

(187) In certain aspects, the pharmaceutical compositions of the disclosure are refrigerated or frozen for storage and/or shipment (e.g., being stored at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C. For example, the pharmaceutical composition comprising the sa-mRNA of the present disclosure is a solution that is refrigerated for storage and/or shipment at, for example, about -20° C., -30° C., -40° C., -50° C., -60° C., -70° C., or -80° C. In certain aspects, the disclosure also relates to a method of increasing stability of pharmaceutical compositions comprising sa-mRNA and a delivery system by storing the pharmaceutical compositions at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C.,

e.g., about -5°C. , -10°C. , -15°C. , -20°C. , -25°C. , -30°C. , -40°C. , -50°C. , -60°C. , -70°C. , -80°C. , -90°C. , -130°C. or -150°C.). For example, the pharmaceutical compositions disclosed herein are stable for about at least 1 week, at least 2 weeks, at least 3 weeks, at least 4 weeks, at least 5 weeks, at least 6 weeks, at least 1 month, at least 2 months, at least 4 months, at least 6 months, at least 8 months, at least 10 months, at least 12 months, at least 14 months, at least 16 months, at least 18 months, at least 20 months, at least 22 months, or at least 24 months, e.g., at a temperature of 4°C. or lower (e.g., between about 4°C. and -20°C.). In one aspect, the formulation is stabilized for at least 4 weeks at about 4°C. In certain aspects, the pharmaceutical composition of the disclosure comprises a sa-mRNA disclosed herein, a nanoparticle composition delivery system, and a pharmaceutically acceptable carrier selected from one or more of Tris, an acetate (e.g., sodium acetate), an citrate (e.g., sodium citrate), saline, PBS, and sucrose. In certain aspects, the pharmaceutical composition of the disclosure has a pH value between about 5 and 8 (e.g., 5, 5.5, 6, 6.5, 6.8 6.9, 7.0, 7.1, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9 or 8.0, or between 7.5 and 8 or between 7 and 7.8). For example, a pharmaceutical composition of the disclosure comprises a sa-mRNA disclosed herein, a nanoparticle composition delivery system, Tris, saline and sucrose, and has a pH of about 7.5-8, which is suitable for storage and/or shipment at, for example, about -20°C. For example, a pharmaceutical composition of the disclosure comprises a sa-mRNA disclosed herein, a nanoparticle composition delivery system, and PBS and has a pH of about 7-7.8, suitable for storage and/or shipment at, for example, about 4°C. or lower. “Stability,” “stabilized,” and “stable” in the context of the present disclosure refers to the resistance of pharmaceutical compositions disclosed herein to chemical or physical changes (e.g., degradation, particle size change, aggregation, change in encapsulation, etc.) under given manufacturing, preparation, transportation, storage and/or in-use conditions, e.g., when stress is applied such as shear force, freeze/thaw stress, etc.

(188) Pharmaceutical compositions of the disclosure may be administered to any patient or subject, including those patients or subjects that may benefit from a therapeutic effect provided by the delivery of a biologically active agent to one or more particular cells, tissues, organs, or systems or groups thereof, such as the renal system. Although the descriptions provided herein of pharmaceutical compositions are principally directed to compositions which are suitable for administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to any other mammal. Modification of compositions suitable for administration to humans in order to render the compositions suitable for administration to various animals is well understood, and the ordinarily skilled veterinary pharmacologist can design and/or perform such modification with merely ordinary, if any, experimentation. Subjects to which administration of the compositions is contemplated include, but are not limited to, humans, other primates, and other mammals, including commercially relevant mammals such as cattle, pigs, horses, sheep, cats, dogs, mice, and/or rats.

(189) A pharmaceutical composition of the present disclosure may be prepared by any method known or hereafter developed in the art of pharmacology. In general, such preparatory methods include bringing the active ingredient into association with an excipient and/or one or more other accessory ingredients, and then, if desirable or necessary, dividing, shaping, and/or packaging the product into a desired single- or multi-dose unit.

(190) A pharmaceutical composition in accordance with the present disclosure may be prepared, packaged, and/or sold in bulk, as a single unit dose, and/or as a plurality of single unit doses. As used herein, a “unit dose” is discrete amount of the pharmaceutical composition comprising a predetermined amount of the active ingredient (e.g., sa-mRNA). The amount of the active ingredient is generally equal to the dosage of the active ingredient which would be administered to a subject and/or a convenient fraction of such a dosage such as, for example, one-half or one-third of such a dosage.

(191) Injectable preparations, for example, sterile injectable aqueous or oleaginous suspensions

may be formulated according to the known art using suitable dispersing agents, wetting agents, and/or suspending agents. Sterile injectable preparations may be sterile injectable solutions, suspensions, and/or emulsions in nontoxic parenterally acceptable diluents and/or solvents, for example, as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are water, Ringer's solution, U.S.P., and isotonic sodium chloride solution. Sterile, fixed oils may be employed as a solvent or suspending medium. For this purpose any bland fixed oil can be employed including synthetic mono- or diglycerides. Fatty acids such as oleic acid can be used in the preparation of injectables.

(192) Injectable formulations can be sterilized, for example, by filtration through a bacterial-retaining filter, and/or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved or dispersed in sterile water or other sterile injectable medium prior to use.

(193) Methods of Producing Polypeptides in Cells

(194) The present disclosure provides methods of producing a sa-mRNA of interest in a mammalian cell. Methods of producing sa-mRNA in a cell involve contacting a cell with sa-mRNA (either as naked RNA, or in combination with a delivery system), encoding one or more gene or genes of interest. Upon contacting the cell, the sa-mRNA may be taken up and translated in the cell to produce the gene product.

(195) In general, the step of contacting a cell with a sa-mRNA encoding a gene or genes of interest may be performed in vivo, ex vivo, in culture, or in vitro. The amount of sa-mRNA, may depend on the type of cell or tissue being contacted, the means of administration, the physiochemical characteristics of the sa-mRNA and delivery system (e.g., size, charge, and chemical composition), and other factors. In general, an effective amount of the sa-mRNA will allow for efficient polypeptide production in the cell. Metrics for efficiency may include polypeptide translation (indicated by polypeptide expression), level of sa-mRNA degradation, and immune response indicators.

(196) The step of contacting a nanoparticle composition containing a sa-mRNA with a cell may involve or cause transfection. A phospholipid including in the lipid component of a nanoparticle composition may facilitate transfection and/or increase transfection efficiency, for example, by interacting and/or fusing with a cellular or intracellular membrane. Transfection may allow for the transcription and translation of the sa-mRNA within the cell.

(197) Methods of Delivering Therapeutic Agents to Cells and Organs

(198) The present disclosure provides methods of delivering a biologically active agent to a cell or organ. Delivery of a biologically active agent to a cell involves administering a delivery system including the biologically active agent to a subject, where administration of the composition involves contacting the cell with the composition. In the instance that a biologically active agent is a sa-mRNA, upon contacting a cell, a translatable sa-mRNA may be translated in the cell to produce a polypeptide of interest. However, sa-mRNA of the present disclosure may encode gene products that are substantially not translatable (e.g. regulatory structures) may also be delivered to cells. Regulatory structures may be useful as vaccines and/or may sequester translational components of a cell to reduce expression of other species in the cell.

(199) In some aspects, a delivery system such as a nanoparticle composition may target a particular type or class of cells (e.g., cells of a particular organ or system thereof). For example, a nanoparticle composition delivering a biologically active agent of interest may be specifically delivered to a mammalian liver, kidney, spleen, femur, or lung. Specific delivery to a particular class of cells, an organ, or a system or group thereof implies that a higher proportion of the therapeutic, diagnostic and/or prophylactic are delivered to the destination (e.g., tissue) of interest relative to other destinations. In some aspects, specific delivery may result in a greater than 2 fold, 5 fold, 10 fold, 15 fold, or 20 fold increase in the amount of therapeutic and/or prophylactic per 1 g of tissue of the targeted destination (e.g., tissue of interest, such as a liver) as compared to another destination (e.g., the spleen). In some aspects, the tissue of interest is selected from the group

consisting of a liver, kidney, a lung, a spleen, a femur, an ocular tissue (e.g., via intraocular, subretinal, or intravitreal injection), vascular endothelium in vessels (e.g., intra-coronary or intra-femoral) or kidney, and tumor tissue (e.g., via intratumoral injection).

(200) As another example of targeted or specific delivery, a sa-mRNA of the present disclosure may encode a protein-binding partner (e.g., an antibody or functional fragment thereof, a scaffold protein, or a peptide) or a receptor on a cell surface. A sa-mRNA may additionally or instead be used to direct the synthesis and extracellular localization of lipids, carbohydrates, or other biological moieties. Alternatively, other biologically active agents (e.g., lipids or ligands) of a delivery system may be selected based on their affinity for particular receptors (e.g., low density lipoprotein receptors) such that a delivery system may more readily interact with a target cell population including the receptors. For example, ligands may include, but are not limited to, members of a specific binding pair, antibodies, monoclonal antibodies, Fv fragments, single chain Fv (scFv) fragments, Fab' fragments, F(ab')₂ fragments, single domain antibodies, camelized antibodies and fragments thereof, humanized antibodies and fragments thereof, and multivalent versions thereof; multivalent binding reagents including mono- or bi-specific antibodies such as disulfide stabilized Fv fragments, scFv tandems, diabodies, tribodies, or tetrabodies; and aptamers, receptors, and fusion proteins.

(201) In some aspects, a ligand may be a surface-bound antibody, which can permit tuning of cell targeting specificity. This is especially useful since highly specific antibodies can be raised against an epitope of interest for the desired targeting site. In one aspect, multiple antibodies are expressed on the surface of a cell, and each antibody can have a different specificity for a desired target. Such approaches can increase the avidity and specificity of targeting interactions.

(202) In certain aspects, compositions in accordance with the present disclosure may be administered at dosage levels sufficient to deliver from about 0.0001 mg/kg to about 10 mg/kg, from about 0.001 mg/kg to about 10 mg/kg, from about 0.005 mg/kg to about 10 mg/kg, from about 0.01 mg/kg to about 10 mg/kg, from about 0.05 mg/kg to about 10 mg/kg, from about 0.1 mg/kg to about 10 mg/kg, from about 1 mg/kg to about 10 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 5 mg/kg to about 10 mg/kg, from about 0.0001 mg/kg to about 5 mg/kg, from about 0.001 mg/kg to about 5 mg/kg, from about 0.005 mg/kg to about 5 mg/kg, from about 0.01 mg/kg to about 5 mg/kg, from about 0.05 mg/kg to about 5 mg/kg, from about 0.1 mg/kg to about 5 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 5 mg/kg, from about 0.0001 mg/kg to about 2.5 mg/kg, from about 0.001 mg/kg to about 2.5 mg/kg, from about 0.005 mg/kg to about 2.5 mg/kg, from about 0.01 mg/kg to about 2.5 mg/kg, from about 0.05 mg/kg to about 2.5 mg/kg, from about 0.1 mg/kg to about 2.5 mg/kg, from about 1 mg/kg to about 2.5 mg/kg, from about 2 mg/kg to about 2.5 mg/kg, from about 0.0001 mg/kg to about 1 mg/kg, from about 0.001 mg/kg to about 1 mg/kg, from about 0.005 mg/kg to about 1 mg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 1 mg/kg, from about 0.1 mg/kg to about 1 mg/kg, from about 0.0001 mg/kg to about 0.25 mg/kg, from about 0.001 mg/kg to about 0.25 mg/kg, from about 0.005 mg/kg to about 0.25 mg/kg, from about 0.01 mg/kg to about 0.25 mg/kg, from about 0.05 mg/kg to about 0.25 mg/kg, or from about 0.1 mg/kg to about 0.25 mg/kg of a therapeutic, diagnostic and/or prophylactic (e.g., a self-amplifying mRNA) in a given dose, where a dose of 1 mg/kg (mpk) provides 1 mg of a biologically active agent per 1 kg of subject body weight. In some aspects, a dose of about 0.001 mg/kg to about 10 mg/kg of a biologically active agent (e.g., self-amplifying mRNA) may be administered. In other aspects, a dose of about 0.005 mg/kg to about 2.5 mg/kg of a biologically active agent may be administered. In certain aspects, a dose of about 0.1 mg/kg to about 1 mg/kg may be administered. In other aspects, a dose of about 0.05 mg/kg to about 0.25 mg/kg may be administered. A dose may be administered one or more times per day, in the same or a different amount, to obtain a desired level of sa-mRNA expression and/or biologically active agent, or imaging effect.

(203) The desired dosage may be delivered, for example, three times a day, two times a day, once a

day, every other day, every third day, every week, every two weeks, every three weeks, or every four weeks. In certain aspects, the desired dosage may be delivered using multiple administrations (e.g., two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, or more administrations). In some aspects, a single dose may be administered, for example, prior to or after a surgical procedure or in the instance of an acute disease, disorder, or condition.

(204) Pharmaceutical compositions including one or more biologically active agents may be used in combination with one or more other biologically active or imaging agents. By “in combination with,” it is not intended to imply that the agents must be administered at the same time and/or formulated for delivery together, although these methods of delivery are within the scope of the present disclosure. For example, one or more pharmaceutical compositions including one or more different biologically active agents may be administered in combination. Compositions can be administered concurrently with, prior to, or subsequent to, one or more other desired therapeutics or medical procedures. In general, each agent will be administered at a dose and/or on a time schedule determined for that agent. In some aspects, the present disclosure encompasses the delivery of compositions, or imaging, therapeutic, diagnostic, or prophylactic compositions thereof in combination with agents that improve their bioavailability, reduce and/or modify their metabolism, inhibit their excretion, and/or modify their distribution within the body.

(205) It will further be appreciated that biologically active or imaging active agents utilized in combination may be administered together in a single composition or administered separately in different compositions. In general, it is expected that agents utilized in combination will be utilized at levels that do not exceed the levels at which they are utilized individually. In some aspects, the levels utilized in combination may be lower than those utilized individually.

(206) The particular combination of therapies to employ in a combination regimen will take into account compatibility of the desired therapeutic, diagnostic and/or prophylactic procedure and the desired biological effect to be achieved. It will also be appreciated that the therapies employed may achieve a desired effect for the same disorder (for example, a composition useful for treating cancer may be administered concurrently with a chemotherapeutic agent), or they may achieve different effects (e.g., control of any adverse effects, such as infusion related reactions).

(207) This disclosure includes the following non-limiting items: 1. A method of increasing the copy number of a nucleic acid comprising: a) contacting cells with a nucleic acid encoding two expression units, the nucleic acid comprising: i) an origin of replication sequence (Ori); ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter; and iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter, wherein the first expression unit encodes a selectable marker and the second expression unit encodes a self-amplifying mRNA (sa-mRNA); b) selecting cells that express the selectable marker; c) subculturing the selected cells to obtain a population of cells that express the selectable marker; and d) propagating the population of cells to increase the copy number of the nucleic acid. 2. The method item 1, wherein the nucleic acid is a recombinant DNA molecule. 3. The method of item 2, wherein the recombinant DNA molecule is a plasmid. 4. The method of item 1, wherein the nucleic acid is a closed circular molecule or a linear molecule. 5. The method of any one of items 1-4, wherein the nucleic acid is suitable for in vitro transcription of RNA after linearization using the nucleic acid as a template. 6. The method of any one of items 1-5, wherein the cell is a bacterium. 7. The method of item 6, wherein the bacterium is *Escherichia coli*. 8. The method of any one of items 1-7, wherein the nucleic acid comprises SAM001 (SEQ ID NO: 35), SAM002 (SEQ ID NO: 36), SAM003 (SEQ ID NO: 37), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), MOD001 (SEQ ID NO: 41), or T7-VEE-GFP (SEQ ID NO: 42). 9. The method of any one of items 1-8, wherein the nucleic acid sequence has at least 90% sequence identity to SAM001 (SEQ ID NO: 35), SAM002 (SEQ ID NO: 36), SAM003 (SEQ ID NO: 37), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), MOD001 (SEQ ID NO: 41), or T7-VEE-GFP (SEQ ID NO: 42). 10. The method of any one of

items 1-9, wherein the first expression unit comprises the following operably linked nucleic acid sequence in a 5' to 3' direction or in a 3' to 5' direction: Pr1-SM wherein Pr1 is the first promoter sequence, and SM is the selectable marker. 11. The method of any one of items 1-10, wherein the first promoter is an ampicillin resistance (AmpR) promoter, a kanamycin resistance (KanR) promoter, a chloramphenicol resistance (CamR) promoter, an erythromycin resistance (ErmR) promoter, and a tetracycline resistance (TetR) promoter. 12. The method of any one of items 1-11, wherein the selectable marker is AmpR, KanR, CamR, ErmR, or TetR. 13. The method of any one of items 1-12, wherein the second expression unit comprises the following operably linked nucleic acid sequence from 5' to 3': Pr2-5'UTR-nsP-SGP-GOI-3'UTR-PolyA wherein Pr2 is the second promoter sequence for in vitro transcription, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, GOI is one or more genes of interest, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail (poly-A tail). 14. The method of item 13, wherein at least one gene of interest (GOI), encodes a therapeutic polypeptide, a prophylactic polypeptide, a diagnostic polypeptide, an antigen, or a non-coding gene that encodes regulatory structures. 15. The method of any one of items 13-14, wherein the regulatory structures are selected from a group comprising small interfering RNA (siRNA), micro-RNA (miRNA), guide RNA (gRNA), self-activating RNA (saRNA), transfer RNA (tRNA), or long intergenic non-coding (lincRNA). 16. The method of any one of items 13-14, wherein at least one GOI encodes an infectious disease antigen, an allergic antigen, or a tumor antigen. 17. The method of item 13, wherein at least one GOI encodes a reporter gene. 18. The method of item 17, wherein the reporter gene is green fluorescent protein (GFP). 19. The method of any one of items 13-18, wherein the plurality of non-structural replicase domain sequences are obtained from a Group IV positive single strand RNA virus selected from the group comprising Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, and Astroviridae. 20. The method of any one of items 13-19, wherein the plurality of non-structural replicase domain sequences are obtained from an alphavirus selected from the group comprising Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiyama virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzyllagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. 21. The method of any one of items 13-20, wherein the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4). 22. The method of any one of items 13-21, wherein the plurality of non-structural replicase domain sequences are obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). 23. The method of any one of items 1-22, wherein the second promoter is selected from the group consisting of T7, T3, SV40, SP6, T5, β -lactamase promoter, *E. coli* galactose promoter, arabinose promoter, alkaline phosphatase promoter, tryptophan (trp) promoter, lactose operon (lac) promoter, lacUV5 promoter, trc promoter and tac promoter. 24. The method of any one of items 1-23, wherein the nucleic acid further comprises one or more linkers. 25. The method of item 24, wherein the nucleic acid sequence comprises from 5' to 3': a) Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-L4-3'UTR-PolyA; b) L1-Ori-SM-Pr1-Pr2-5'UTR-nsP-L3-GOI-L4-3'UTR-PolyA c) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-GOI-L4-3'UTR-PolyA; d) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-3'UTR-PolyA; or e) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-L4-3'UTR-PolyA, wherein L1 is a first linker, Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker, Pr2 is a second promoter sequence, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, L3 is a third linker, SGP is a subgenomic promoter, GOI is one or more genes of interest, L4 is a fourth linker, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail (poly-A tail). 26. The method of item 25, wherein each of L1, L2, L3, and L4 is independently selected from a nucleic acid sequence comprising

(208) TABLE-US-00003 (SEQ ID NO: 43)
CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAAGGCCAGCAAAAAGGCC
AGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT, (SEQ ID NO: 44)
CACATTTCCCCGAAAAGTGCCACCTGAGCTC, (SEQ ID NO: 45)
TTCGAAGGCGCGCCTCTAGAGCCACC, or (SEQ ID NO: 46)
CATCGATGATATCGCGGCCGCATACAGCAGC, or wherein L1 comprises SEQ ID
NO: 43 (CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAAGGCCAGCAAAAAGGC
CAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT); L2 comprises SEQ ID NO: 44
(CACATTTCCCCGAAAAGTGCCACCTGAGCTC); L3 comprises SEQ ID NO: 45
(TTCGAAGGCGCGCCTCTAGAGCCACC); and L4 comprises SEQ ID NO: 46
(CATCGATGATATCGCGGCCGCATACAGCAGC). 27. A self-amplifying mRNA comprising a
nucleic acid sequence from 5' to 3': a) 5'UTR-nsP-L-GOI-L-3'UTR-PolyA b) 5'UTR-nsP-GOI-L-
3'UTR-PolyA; c) 5'UTR-nsP-L-GOI-3'UTR-PolyA; or wherein 5'UTR is a 5' untranslated region,
nsP is a plurality of non-structural replicase domain sequences, L is a linker, SGP is a subgenomic
promoter, GOI is one or more genes of interest, 3'UTR is a 3' untranslated region, and Poly-A is a
3' poly-adenylated tail (poly-A tail). 28. The self-amplifying mRNA of item 27, wherein the GOI is
an antigen or antigen receptor. 29. The self-amplifying mRNA of any one of items 27-28, wherein
the GOI is a viral antigen. 30. The self-amplifying mRNA of any one of items 27-29, wherein the
GOI is a modified SARS-COV-2 spike protein. 31. The self-amplifying mRNA of any one of items
27-30, wherein the immunomodulator is a cytokine, a chemokine, or other immune stimulator or
inhibitor. 32. The self-amplifying mRNA of any one of items 27-31, comprising a polynucleotide
sequence selected from: a) a polynucleotide encoding a modified SARS-COV-2 spike protein
comprising the nucleic sequence set forth in SEQ ID NO: 1 (BA.1-1273); b) a polynucleotide
encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ
ID NO: 2 (BA.1-1273-S2P); c) a polynucleotide encoding a modified SARS-COV-2 spike protein
comprising the nucleic sequence set forth in SEQ ID NO: 3 (BA.2-1273); d) a polynucleotide
encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ
ID NO: 4 (BA.2-1273-S2P); e) a polynucleotide encoding a modified SARS-COV-2 spike protein
comprising the nucleic sequence set forth in SEQ ID NO: 5 (BA.1-1208); or f) a polynucleotide
encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ
ID NO: 6 (BA.1-1208-S2P); g) a polynucleotide encoding a modified SARS-COV-2 spike protein
comprising the nucleic sequence set forth in SEQ ID NO: 7 (BA.2-1208); or h) a polynucleotide
encoding a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ
ID NO: 8 (BA.2-1208-S2P). 33. A self-amplifying mRNA encoding two separated expression units,
the nucleic acid comprising: i) a first expression unit comprising a polynucleotide encoding a
modified antigen, wherein the polynucleotide encoding the modified antigen is truncated to not
include nucleotides encoding a transmembrane domain and short cytosolic domain amino acids of
the antigen, operably linked to a first subgenomic promoter; and ii) a second expression unit
encoding immunomodulators (IM) that are operably linked to a second subgenomic promoter. 34.
The self-amplifying mRNA of item 33, wherein the polynucleotide sequence encoding the
modified antigen comprises replacement of a transmembrane domain of the antigen with a
secretion antigen. 35. The self-amplifying mRNA of item 33 or item 34, wherein the antigen is a
modified SARS-CoV-2 spike protein, wherein the polynucleotide has been truncated to not include
nucleotides encoding a SARS-COV-2 transmembrane domain and short cytosolic domain amino
acids. 36. The self-amplifying mRNA of any one of items 33-35, wherein the polynucleotide
sequence encoding a coronavirus spike protein truncated to not include nucleotides encoding a
SARS-COV-2 transmembrane domain and short cytosolic domain amino acids corresponding to
amino acids 1209-1273 of a nucleotide sequence is SEQ ID NOs: 1 (BA.1-1273) or 3 (BA.2-1273).
37. The self-amplifying mRNA of any one of items 33-36, wherein the sa-mRNA comprises the
following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-Ag-SGP2-IM wherein

nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), or SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P), SGP2 is the second subgenomic promoter, and IM is the immunomodulator. 38. The self-amplifying mRNA of any one of items 33-36, wherein the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-IM-SGP2-AG wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, IM is the immunomodulator, SGP2 is the second subgenomic promoter, and Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), or SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P). 39. The self-amplifying mRNA of any one of items 33-38, wherein the IM encodes one or more cytokines, chemokines, immune stimulators or inhibitors. 40. The self-amplifying mRNA of any one of items 33-39, wherein the IM is IL12 or IL21. 41. The self-amplifying mRNA of any one of items 33-40, wherein the IM encodes one or more cytokines selected from SEQ ID NOs: 22 (hIL12-P40), 24 (hIL12-P35), 26 (hIL21), 15 (mIL12 P40), 17 (mIL12-P35), and 19 (mIL21). 42. The self-amplifying mRNA of any one of items 33-41, wherein SGP1 is SEQ ID NO: 9 (209) TABLE-US-00004 (SGP1) (TAACCTGAATGGACTACGACATAGTCTAGTCCGCCAAG). 43. The self-amplifying mRNA of any one of items 33-42, wherein SGP2 is SEQ ID NO: 11 (210) TABLE-US-00005 (SGP2) (GAACTTCCATCATAGTTATGGCCATGACTACTCTAGCTAGCAGTGTTAA ATCATTGAGCTACCTGAGAGGGGCCCCTATAACTCTCTACGGC). 44. The self-amplifying mRNA of any one of items 33-43, wherein IM is selected from SEQ ID NO: 13 (IM1), and SEQ ID NO: 20 (IM2). 45. The self-amplifying mRNA of any one of items 33-44, comprising the following operably linked nucleic acid sequence from 5' to 3': SP-IL12 P40-L1-IL12 P35-L2-IL21 Wherein SP is a signal peptide, IL12-P40 is interleukin-12 comprising heavy chain p40, L1 is linker 1, IL12 P35 is interleukin-12 comprising light chain p35, L2 is linker 2, and IL21 is interleukin-21. 46. The self-amplifying mRNA of any one of 33-45, wherein SP is selected from SEQ ID NO: 14 (MSP) (ATGACCTCCCGGCTTGTGAGGGTACTGGCTGCTGCTATGCTGGTGGCTGCTG CTGTGAGTGTGGC) and SEQ ID NO: 21 (HSP) (ATGGACTGGACCTGGCGAATACTGTTCTTGGTTGCCGCCGCTACAGGGACTC ACGCA). 47. The self-amplifying mRNA of any one of items 33-46, wherein IL12-P40 is selected from SEQ ID NO: 15 (mIL12-P40) and SEQ ID NO: 22 (hIL12-P40). 48. The self-amplifying mRNA of any one of items 33-47, wherein L1 is selected from SEQ ID NO: 16 (L(a)) and SEQ ID NO: 23 (L(c)). 49. The self-amplifying mRNA of any one of items 33-48, wherein IL12-P35 is selected from SEQ ID NO: 17 (mIL12-P35) and SEQ ID NO: 24 (hIL12-P35). 50. The self-amplifying mRNA of any one of items 33-49, wherein L2 is selected from SEQ ID NO: 18 (L(b)) and SEQ ID NO: 25 (L(d)). 51. The self-amplifying mRNA of any one of items 33-50, wherein IL12-P40 is selected from SEQ ID NO: 19 (mIL21) and SEQ ID NO: 26 (hIL21). 52. The self-amplifying mRNA of any one of items 33-51, wherein the plurality of non-structural replicase domain sequences is obtained from a Group IV RNA virus selected from Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, or Astroviridae. 53. The self-amplifying mRNA of any one of items 33-52, wherein the plurality of non-structural replicase domain sequences are obtained from an alphavirus selected from Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiyama virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus, or Buggy Creek virus. 54. The self-amplifying mRNA of any one

of items 33-53, wherein the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4). 55. The self-amplifying mRNA of any one of items 33-54, wherein the plurality of non-structural replicase domain sequences are obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). 56. The self-amplifying mRNA of any one of items 33-55, wherein SGP1 is a viral promoter that is recognized by viral RNA dependent RNA polymerase. 57. The self-amplifying mRNA of any one of items 33-56, wherein SGP2 is a viral promoter that is recognized by viral RNA dependent RNA polymerase. 58. The self-amplifying mRNA of any one of items 33-57, wherein SGP1 and SGP2 are different subgenomic promoters. 59. The self-amplifying mRNA of any one of items 33-58, wherein the sa-mRNA further comprises one or more linkers. 60. The self-amplifying mRNA of any one of items 33-59, wherein the linkers are selected from the group SEQ ID NOs: 16 (L(a)), 18 (L(b)), 23 (L(c)), and 25 (L(d)). 61. A composition comprising the self-amplifying mRNA of any one of items 33-60 and a pharmaceutically acceptable carrier. 62. The composition of item 61, further comprising a self-amplifying mRNA delivery system. 63. The composition of item 62, wherein the self-amplifying mRNA delivery system is a lipid nanoparticle. 64. A method of expressing a gene in a cell, comprising delivering the self-amplifying mRNA of any one of items 33-60 to the cell, and maintaining the cell under conditions suitable for expression of the gene encoded by the GOI. 65. The method of item 64, wherein the cell is in an animal cell. 66. A method for producing a self-amplifying mRNA, the method comprising: a) performing an in vitro transcription reaction using an initial amount of the nucleic acid produced by the method of any one of items 1-27; and b) producing a self-amplifying mRNA by in vitro transcription, using the nucleic acid as a template. 67. A method of inducing an immune response in an individual, comprising administering to the individual a sa-mRNA produced from the method of item 66. 68. A nucleic acid encoding a self-amplifying mRNA comprising a mutant T7 promoter comprising the nucleotide sequence of SEQ ID NO: 47 (TAATACGACTCACTATAGG) operably linked to a 5' UTR, a plurality of non-structural replicase domain sequences, one or more gene or genes of interest (GOI), a 3' UTR, and a poly-A tail. 69. The nucleic acid of item 68, wherein the 5'UTR comprises the nucleotide sequence ATAGG. 70 The nucleic acid of any one of items 68-69, comprising SAM002 (SEQ ID NO: 36). 71. The nucleic acid of any one of items 68-70, wherein the nucleic acid further comprises one or more linkers. 72 The nucleic acid of any one of items 68-71 comprising the following nucleic acid sequence from 5' to 3': a) Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-L4-3'UTR-PolyA; b) L1-Ori-SM-Pr1-Pr2-5'UTR-nsP-L3-GOI-LA-3'UTR-PolyA c) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-GOI-LA-3'UTR-PolyA; d) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-3'UTR-PolyA; or e) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-L4-3'UTR-PolyA, wherein L1 is a first linker, Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker, T7' is a mutant T7 promoter of SEQ ID NO: 47 (TAATACGACTCACTATAGG), 5'UTR' is a mutant 5' untranslated region of ATAGG, nsP is a plurality of non-structural replicase domain sequences, L3 is a first linker, SGP is a subgenomic promoter, GOI is one or more gene or genes of interest, LA is a second linker, 3'UTR is a 3' untranslated region, and Poly-A is a 3' poly-adenylated tail (poly-A tail). 73. The nucleic acid of any one of items 68-72, wherein each of L1, L2, L3, and L4 is independently selected from a nucleic acid a sequence comprising

(211) TABLE-US-00006 (SEQ ID NO: 43)

CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCC

AGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT, (SEQ ID NO: 44)

CACATTTCCCCGAAAAGTGCCACCTGAGCTC, (SEQ ID NO: 45)

TTCGAAGGCGCGCCTCTAGAGCCACC, or (SEQ ID NO: 46)

CATCGATGATATCGCGGCCGCATACAGCAGC, or wherein L1 comprises SEQ ID NO: 43 (CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGC

CAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT); L2 comprises SEQ ID NO: 44

(CACATTCTCCGAAAGTGCACCTGAGCTC); L3 comprises SEQ ID NO: 45 (TTCGAAGGCGCGCCTCTAGAGCCACC); and L4 comprises SEQ ID NO: 46 (CATCGATGATATCGCGGCCGCATACAGCAGC). 74. The nucleic acid of any one of items 68-73, wherein the first promoter is an ampicillin resistance (AmpR) promoter, a kanamycin resistance (KanR) promoter, a chloramphenicol resistance (CamR) promoter, an erythromycin resistance (ErmR) promoter, or a tetracycline resistance (TetR) promoter. 75. The nucleic acid of any one of items 68-74, wherein the selectable marker is AmpR, KanR, CamR, ErmR, or TetR. 76. The nucleic acid of any one of items 68-75, wherein at least one gene of interest (GOI) encodes a therapeutic polypeptide, a prophylactic polypeptide, a diagnostic polypeptide, an antigen, an antigen receptor, or a non-coding gene that encodes regulatory structures. 77. The nucleic acid of item 76, wherein the regulatory structures are selected from a group comprising small interfering RNA (siRNA), micro-RNA (miRNA), self-activating RNA (saRNA), transfer RNA (tRNA), long intergenic non-coding (lincRNA). 78. The nucleic acid of any one of items 68-77, wherein at least one GOI encodes an infectious disease antigen, an allergic antigen, or a tumor antigen. 79. The nucleic acid of any one of items 68-78, wherein at least one GOI encodes a reporter gene. 80. The nucleic acid of item 79, wherein the reporter gene is green fluorescent protein (GFP). 81. The nucleic acid of any one of items 68-80, wherein the plurality of non-structural replicase domain sequences are obtained from a Group IV RNA virus selected from the group comprising Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, and Astroviridae. 82. The nucleic acid of any one of items 68-81, wherein the plurality of non-structural replicase domain sequences are obtained from an alphavirus selected from the group comprising Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiya virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whartaroa virus, Babanki virus, Kyzyllagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. 83. The nucleic acid of any one of items 68-82, wherein the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4). 84. The nucleic acid of any one of items 68-83, wherein the plurality of non-structural replicase domain sequences are obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). 85. A method for producing a self-amplifying mRNA, the method comprising: a) performing an in vitro transcription reaction using an initial amount of the nucleic acid of any one of items 68-84; and b) producing a self-amplifying mRNA by in vitro transcription, using the nucleic acid as a template. 86. The method of item 85, wherein the amount of self-amplifying mRNAs containing a mutant T7 promoter of SEQ ID NO: 47 (TAATACGACTCACTATAGG) and a mutant 5' untranslated region of ATAGG produced is at least 40% greater compared to the amount of the self-amplifying mRNAs produced from a nucleic acid template with wildtype T7 promoter and 5' UTR. 87. A composition comprising the self-amplifying mRNA produced from the method of item 85 and a pharmaceutically acceptable carrier. 88. The composition of item 87, further comprising a self-amplifying mRNA delivery system. 89. The composition of item 88, wherein the self-amplifying mRNA delivery system is a nanoparticle composition. 90. A method of expressing a gene encoded by a GOI in a cell, comprising delivering the self-amplifying mRNA produced from the method of item 85 or item 86 to a cell, and causing the cell to express the gene encoded by the GOI. 91. The method of item 90, wherein the cell is in an animal cell. 92. A method of inducing an immune response in an individual, comprising administering to the individual a self-amplifying mRNA produced from the method of item 85 or item 86. 93. A nucleic acid encoding a self-amplifying mRNA comprising: a) a mutant 3'UTR of an alphavirus comprising point mutations at position 6 relative to a conserved 19 nucleotide sequence GGATTTTGTTTTAAATATTC (SEQ ID NO: 49) of the wild-type 3'UTR of an alphavirus; b) a mutant 3'UTR of an alphavirus comprising point mutations at positions -1, and -2 relative to a

conserved 19 nucleotide sequence GGATTTTGTTTTAAATATTTC (SEQ ID NO: 49) of the wild-type 3'UTR of an alphavirus; c) a mutant 3'UTR of an alphavirus comprising point mutations at positions -1, -2 and 6 relative to a conserved 19 nucleotide sequence GGATTTTGTTTTAAATATTTC (SEQ ID NO: 49) of the wild-type 3'UTR of an alphavirus; d) a mutant 3'UTR of an alphavirus comprising a sequence selected from a group comprising GGATTTTATTTTAAATATTTC (SEQ ID NO: 50), AAATTTTGTTTTAAATATTTC (SEQ ID NO: 51), or AAATTTTATTTTAAATATTTC (SEQ ID NO: 52); or e) a promoter operably linked to a 5' UTR, a plurality of non-structural replicase domain sequences, one or more gene or genes of interest (GOI), the mutant 3'UTR of any one of SEQ ID NOs: 49-52, and a poly-A tail. 94. The nucleic acid of item 93 comprising a sequence selected from a group comprising SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40). 95. The nucleic acid of item 94, wherein the nucleic acid further comprises one or more linkers. 96 The nucleic acid of any one of items 93-95, wherein the nucleic acid sequence comprises from 5' to 3': a) Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-L4-3'UTR-PolyA; b) L1-Ori-SM-Pr1-Pr2-5'UTR-nsP-L3-GOI-LA-3'UTR-PolyA c) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-GOI-L4-3'UTR-PolyA; d) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-L3-GOI-3'UTR-PolyA; or e) L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-L4-3'UTR-PolyA, wherein, L1 is a first linker, Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker, Pr2 is a second promoter sequence, 5'UTR is a 5' untranslated region, nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter, L3 is a first linker, GOI is one or more gene or genes of interest, LA is a second linker, 3'UTR' is a mutant 3' untranslated region, and Poly-A is a 3' polyadenylated tail (poly-A tail), wherein the 3'UTR' is: a) a mutant 3'UTR of an alphavirus comprising GGATTTTATTTTAAATATTTC (SEQ ID NO: 50); b) a mutant 3'UTR of an alphavirus comprising AAATTTTGTTTTAAATATTTC (SEQ ID NO: 51); or c) a mutant 3'UTR of an alphavirus comprising AAATTTTATTTTAAATATTTC (SEQ ID NO: 52). 97. The nucleic acid of item 96, wherein each of L1, L2, L3, and L4 is independently selected from a nucleic acid a sequence comprising

(212) TABLE-US-00007 (SEQ ID NO: 43)

CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCC

AGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT, (SEQ ID NO: 44)

CACATTTCCCCGAAAAGTGCCACCTGAGCTC, (SEQ ID NO: 45)

TTCGAAGGCGCGCCTCTAGAGCCACC, or (SEQ ID NO: 46)

CATCGATGATATCGCGGCCGCATACAGCAGC, or wherein L1 comprises SEQ ID NO: 43 (CGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCC CAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTT); L2 comprises SEQ ID NO: 44 (CACATTTCCCCGAAAAGTGCCACCTGAGCTC); L3 comprises SEQ ID NO: 45 (TTCGAAGGCGCGCCTCTAGAGCCACC); and L4 comprises SEQ ID NO: 46 (CATCGATGATATCGCGGCCGCATACAGCAGC). 98. The nucleic acid of any one of items 96-97, wherein the first promoter is an ampicillin resistance (AmpR) promoter, a kanamycin resistance (KanR) promoter, a chloramphenicol resistance (CamR) promoter, an erythromycin resistance (ErmR) promoter, and a tetracycline resistance (TetR) promoter. 99. The nucleic acid of any one of items 96-98, wherein the selectable marker is AmpR, KanR, CamR, ErmR, or TetR. 100. The nucleic acid of any one of items 96-99, wherein at least one gene of interest (GOI), encodes a therapeutic polypeptide, a prophylactic polypeptide, a diagnostic polypeptide, an antigen, or a non-coding gene that encodes regulatory structures. 101. The nucleic acid of item 100, wherein the regulatory structures are selected from a group comprising small interfering RNA (siRNA), microRNA (miRNA), self-activating RNA (saRNA), transfer RNA (tRNA), long intergenic non-coding (lincRNA). 102. The nucleic acid of item 101, wherein at least one GOI encodes an infectious disease antigen, an allergic antigen, or a tumor antigen. 103. The nucleic acid of any one of items 96-102, wherein at least one GOI encodes a reporter gene. 104. The nucleic acid of item 103,

wherein the reporter gene is green fluorescent protein (GFP). 105. The nucleic acid of any one of items 96-104, wherein the plurality of non-structural replicase domain sequences are obtained from a Group IV RNA virus selected from the group comprising Picornaviridae, Togaviridae, Coronaviridae, Hepeviridae, Caliciviridae, Flaviviridae, and Astroviridae. 106. The nucleic acid of any one of items 96-105, wherein the plurality of non-structural replicase domain sequences are obtained from an alphavirus selected from the group comprising Eastern Equine Encephalitis virus (EEE), Venezuelan Equine Encephalitis virus (VEE), Everglades virus, Mucambo virus, Pixuna virus, Western Equine Encephalitis virus (WEE), Sindbis virus, Semliki Forest virus, Middelburg virus, Chikungunya virus, O'nyong-nyong virus, Ross River virus, Barmah Forest virus, Getah virus, Sagiya virus, Bebaru virus, Mayaro virus, Una virus, Aura virus, Whataroa virus, Babanki virus, Kyzylagach virus, Highlands J virus, Fort Morgan virus, Ndumu virus and Buggy Creek virus. 107. The nucleic acid of any one of items 96-106, wherein the plurality of non-structural replicase domain sequences are alphavirus nonstructural proteins 1-4 (nsP1-4). 108. The nucleic acid of any one of items 96-107, wherein the plurality of non-structural replicase domain sequences are obtained from the TC-83 strain of Venezuelan Equine Encephalitis virus (VEE). 109. A method for producing a self-amplifying mRNA, the method comprising: a) performing an in vitro transcription reaction using an initial amount of the nucleic acid of any one of items 96-108; and b) producing a self-amplifying mRNA by in vitro transcription, using the nucleic acid as a template. 110. A composition comprising the self-amplifying mRNA produced from the method of item 109 and a pharmaceutically acceptable carrier. 111. The composition of item 110, further comprising a self-amplifying mRNA delivery system. 112. The composition of item 111, wherein the self-amplifying mRNA delivery system is a lipid nanoparticle. 113. A method of expressing a gene encoded by a GOI in a cell, comprising delivering the self-amplifying mRNA produced from the method of item 109, and maintaining the cell under conditions suitable for expression of the gene encoded by the GOI. 114. The method of item 113, wherein the cell is in an animal cell. 115. A method of inducing an immune response in an individual, comprising administering to the individual a self-amplifying mRNA produced from the method of item 109. 116. A method for decreasing the interferon response of a host cell compared to the interferon response of the host cell where a self-amplifying mRNA containing a wild-type 3'UTR of an alphavirus is introduced, comprising introducing the self-amplifying mRNA produced from the method of item 109 into the host cell. 117. The method according to item 116, wherein the interferon response of a host cell is 2, 3, 4, 5 or 6 times lower than the amount of interferon response of the host cell to the introduction of a self-amplifying mRNA containing a wild-type 3'UTR of an alphavirus. 118. A method of de novo synthesizing a construct for making a self-amplifying nucleic acid comprising: a) contacting *Escherichia coli* cells with a nucleic acid encoding two expression units, the nucleic acid comprising: i) an origin of replication sequence (Ori); ii) a first expression unit encoding a first nucleotide sequence that is operably linked to a first promoter that for expressing selectable marker; and iii) a second expression unit encoding a second nucleotide sequence that is operably linked to a second promoter for in vitro transcriptions of self-amplifying nucleic acids, wherein the first expression unit encodes a selectable marker and the second expression unit encodes a self-amplifying nucleic acid; b) selecting *Escherichia coli* cells that express the selectable marker; and c) subculturing the selected *Escherichia coli* cells to obtain a population of *Escherichia coli* cells that express the selectable marker; d) propagating the population of cells; and e) performing in vitro transcription of the second expression unit to produce the self-amplifying nucleic acid. 119. A polynucleotide encoding: a) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 1 (BA.1-1273); b) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 2 (BA.1-1273-S2P); c) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 3 (BA.2-1273); d) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 4 (BA.2-1273-S2P); e) a modified SARS-COV-2 spike protein comprising the nucleic

sequence set forth in SEQ ID NO: 5 (BA.1-1208); f) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 6 (BA.1-1208-S2P); g) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 7 (BA.2-1208); or h) a modified SARS-COV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 8 (BA.2-1208-S2P). 120. A method for preparing a library of self-amplifying mRNA derived from a reference self-amplifying mRNA comprising: (i) performing directed evolution of a reference self-amplifying mRNA sample comprising the steps of: (a) delivering a reference self-amplifying mRNA sample encoding a selection marker into host cell(s), (b) culturing said host cell(s) over a period of time under conditions that require replication of the reference self-amplifying mRNA sample and permit expression of the selection marker, wherein mutations occur in the replicated self-amplifying mRNA compared to the reference self-amplifying mRNA, (c) selecting cells that express the selectable marker; and (ii) extracting the replicated self-amplifying mRNA from the host cell(s) and sequencing the replicated self-amplifying mRNA; thereby producing a library of self-amplifying mRNA sequences. 121. The method of item 120, wherein the selection marker is an antibiotic resistance gene. 122. The method of any one of items 120-121, wherein the selection marker is a puromycin resistance gene. 123. The method of any one of items 120-122, wherein the reference self-amplifying mRNA is delivered into a host cell using a delivery mechanism. 124. The method of item 123, wherein the delivery system is a lipid nanoparticle. 125. The method of any one of items 120-124, wherein the reference self-amplifying mRNA is selected from a group comprising SEQ ID Nos. 1-8 and SEQ ID NOs 35-42. 126. The method of any one of items 120-125, wherein the conditions that require replication of the reference self-amplifying mRNA sample and permit expression of the selection marker is a culture environment containing an antibiotic. 127. The method of any one of items 120-126, wherein the concentration of the antibiotic affects the rate of mutation of the reference self-amplifying mRNA. 128. A method of evaluating mutations of the replicated self-amplifying mRNA produced from the method of any one of items 120-127, the method comprising: (i) obtaining a group of contig sequences comprising mutation(s) compared to a reference self-amplifying mRNA sample; (ii) sequencing the contig sequences; and (iii) determining the number of mutations in the contig sequences compared to the reference self-amplifying mRNA. 129. The method of item 128, wherein the contig sequences are fragments of the replicated self-amplifying mRNA. 130. The method of item 128, wherein the contig sequence is SEQ ID NO: 27. 131. The method of item 128, wherein the contig sequence is SEQ ID NO: 28. 132. The method of item 128, wherein the contig sequence is SEQ ID NO: 29. 133. The method of item 128, wherein the contig sequence is SEQ ID NO: 30. 134. The method of item 128, wherein the contig sequence is SEQ ID NO: 31. 135. The method of item 128, wherein the contig sequence is SEQ ID NO: 32. 136. The method of item 128, wherein the contig sequence is SEQ ID NO: 33. 137. The method of item 128, wherein the contig sequence is SEQ ID NO: 34. 138. A method of identifying self-amplifying mRNA with reduced cytotoxic effects as part of a therapeutic product comprising: (i) obtaining a group comprising a plurality of self-amplifying mRNA; (ii) quantifying the relative gene product expression of each self-amplifying mRNA over a period of time; and (iii) identifying self-amplifying mRNA(s) showing stable gene product expression over the period of time compared to other self-amplifying mRNA of the group; wherein the self-amplifying mRNA(s) that show sustained gene product expression over the period of time show reduced cytotoxic effects as part of a therapeutic product compared to the other self-amplifying mRNA of the group. 139. The method of item 138, wherein the period of time is between 12 hours to 10 days. 140. The method of any one of items 138-139, wherein the period of time is 1 day. 141. The method of any one of items 138-140, wherein the self-amplifying mRNA(s) that show sustained gene product expression over the period of time show a change in expression of less than 20 fold. 142. A therapeutic product comprising SAM002 (SEQ ID NO: 36), wherein the therapeutic product shows reduced cytotoxic effects as part of a therapeutic product compared to a therapeutic product comprising SAM001 (SEQ ID NO: 35).

EXAMPLES

(213) The biologically active agent of the disclosure may be delivered using a nanoparticle composition comprising a ionizable lipid; one or more PEG lipids; one or more structural lipids (e.g. cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, tomatine, ursolic acid, alpha-tocopherol, and mixtures thereof); and one or more phospholipids.

Example 1: LNP Delivery System

(214) One suitable system for delivering the sa-mRNA of the disclosure is using a lipid nanoparticle (LNP) delivery system. A method of increasing transfection efficiency and decreasing cytotoxicity of a LNP formulation is by using a novel ionizable lipid described in PCT Patent Application No. PCT/US2023/017777, which is fully incorporated herein. The ionizable lipid in an LNP formulation play a key role in the uptake of LNP by cells and the release of LNP from the endosome. Formula E6 (1,1',1''-(1,3,5-triazinane-1,3,5-triyl)tris(3-(ditridecylamino) propan-1-one)) is a novel ionizable lipid described in PCT Patent Application No. PCT/US2023/017777 and shown below:

(215) ##STR00004##

(216) The structure of E6 was confirmed by ¹H NMR spectroscopy and mass spectrometry.

.sup.1H_NMR (400 MHz, CDCl₃, δ) 0.88 (t, J=0.8 Hz, 18H), 1.23-1.32 (m, 132H), 1.42-1.44 (m, 12H), 2.43 (s, 12H), 2.67 (s, 6H), 2.81 (s, 6H), 5.26 (s, 6H). MS (m/z): [M+2H].sup.2+ calcd. for C.sub.90H.sub.182N.sub.6O.sub.3, 697.7; found, 697.9.

(217) E2 (1,1',1''-(1,3,5-triazinane-1,3,5-triyl)tris(3-(dinonylamino) propan-1-one)) and P6 (N-(2-(cyclohex-1-en-1-ylamino)-1-(1-ethylpiperidin-4-yl)-2-oxoethyl)-N-(heptadecan-9-yl)palmitamide) are novel ionizable lipids described in PCT Patent Application No. PCT/US2023/017777 and shown below:

(218) ##STR00005##

(219) The structure of E2 was confirmed by ¹H NMR spectroscopy and mass spectrometry.

.sup.1H-NMR (400 MHz, CDCl₃, δ) 5.26 (s, 6H), 2.79 (t, J=8 Hz, 6H), 2.65 (t, J=8 Hz, 6H), 2.41 (t, J=8 Hz, 12H), 1.42-1.26 (m, 84H), 0.88 (t, J=8 Hz, 18H). MS (ESI) m/z 529.6 [M+2H].sup.2+.

(220) The structure of P6 was confirmed by ¹H NMR spectroscopy and mass spectrometry.

.sup.1H-NMR (400 MHz, Me-OD, δ) 6.05 (t, J=8 Hz, 1H), 4.26 (s, 1H), 3.26-3.17 (m, 3H), 2.49-2.42 (m, 5H), 2.19-2.01 (m, 6H), 1.67-1.57 (m, 14H), 1.31-1.16 (m, 48H), 0.92-0.87 (m, 12H). MS (APCI) m/z 742.7 [M+H].sup.+.

(221) The nucleic acids of the present disclosure may be delivered using a LNP delivery system wherein the LNP is formulated with ionizable lipid, helper lipid, cholesterol, and PEG-lipid. In one aspect, the LNP of the disclosure has a molar ratio of about 2-60% ionizable lipid, about 5-40% helper lipid, about 30-80% cholesterol and about 0.5-30% PEG-lipid. In one aspect, the LNP of the disclosure has a molar ratio of about 2-10% ionizable lipid, about 5-15% helper lipid, about 40-80% cholesterol and about 0.5-3% PEG-lipid. In one aspect of the disclosure, the ionizable lipid is E6. In one aspect of the disclosure, the helper lipid is independently selected from DOPE (2-dioleoyl-sn-glycero-3-phosphoethanolamine), DSPC (1,2-distearoyl-sn-glycero-3-phosphocholine), and POPE (1-palmitoyl-2-oleoyl-sn-glycero-3-phosphoethanolamine). In one aspect, nucleic acids of the present disclosure may be delivered using a LNP delivery system wherein the ionizable lipid is E6, the helper lipid is DOPE and the PEG-lipid is DMG-PEG2000. In one aspect of the present disclosure, the LNP is composed of E6, DOPE, cholesterol, and DMG-PEG-2000. In one aspect, the LNP has a molar ratio of 50% ionizable lipid, 10% helper lipid, 38.5% cholesterol, and 1.5% PEG-lipid. In another aspect, the LNP has a molar ratio of 7.5% ionizable lipid, 15% helper lipid, 75% cholesterol, and 2.5% PEG-lipid. In another aspect, the LNP of the disclosure has a molar ratio of 5% ionizable lipid, 10% helper lipid, 50% cholesterol, and 1.5% PEG-lipid. In some aspects, the ionizable lipid may include E6, the helper lipid may include DOPE and the PEG-lipid may include DMG-PEG2000.

(222) E6, was synthesized and formulated into LNPs with DOPE, cholesterol and DMG-PEG2000. Transfection efficacy and cell cytotoxicity of the LNPs were assayed in 293T cells using a self-amplifying mRNA encoding GFP. The LNPs of the present disclosure were found to effectively transfect each type of cell in vitro and demonstrated comparable or increased encapsulation efficiency and comparable or decreased cytotoxicity between the secreted version of the modified SPIKE protein of the present disclosure and the SPIKE protein encoded by mRNA currently approved by FDA.

Example 2: Self-Amplifying mRNA

(223) The sa-mRNA of the present disclosure were synthesized using the TC-83 strain of VEE, a subclass of alphavirus, wherein the sa-mRNA of the present disclosure is derived from wildtype TC-83 replicon without the alphavirus structural proteins; SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), 4 (BA.2-1273-S2P), 5 (BA.1-1208), SEQ ID NO: 6 (BA.1-1208-S2P), 7 (BA.2-1208), SEQ ID NO: 8 (BA.2-1208-S2P) are modified variants of the SPIKE antigen as shown in FIGS. **16-17**; SEQ ID NO: 13 (IM1) and SEQ ID NO: 20 (IM2) encode immunomodulators SEQ ID NO: 15 (mIL12 P40), SEQ ID NO 17. (mIL12-P35) and SEQ ID NO 19 (mIL21); SEQ ID NO: 9 encodes subgenomic promotor 1; and SEQ ID NO: 11 encodes subgenomic promotor 2 as shown in FIGS. **23-24**.

(224) The modified SPIKE proteins encoded by the sa-mRNAs of the present disclosure form correct conformation of SPIKE protein for vaccinations and can be recognized by the broad neutralization antibodies when transfected into 293 T cells using lipofectamine as shown in FIGS. **18-19**. Data show that 293 T cells transfected with SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), 4 (BA.2-1273-S2P) can be recognized by broad neutralization antibody (bnAb) SPD-M265 and hACE2-FITC measured using FACS plots of GFP (FIG. **18**) or ACE2 (FIG. **19**) expression on the X-axis and live dead dye staining of 7-AAD on the Y-axis. The modified SPIKE protein encoded by the sa-mRNA of the present disclosure can trigger humoral immune responses.

Example 3: Sa-mRNA Encoding a Modified SPIKE Protein

(225) As shown in FIG. **20**, sa-mRNA encoding a modified SPIKE protein containing a S2P modification stabilizes the structure of secretion SPIKE. In this study, 293 T cells transfected with sa-mRNAs with and without the S2P modification and absorbance was measured on the Y-axis using SPD-M265 bnAb and dilution factor was measured on the X-axis. Furthermore, the secreted version of the modified SPIKE antigen (SEQ ID NO: 6 (BA.1-1208-S2P)) show comparable BA.2 specific IgG as SEQ ID Nos: 1 (BA.1-1273) and 2 (BA.1-1273-S2P), which are more similar to the transmembrane SPIKE proteins encoded by the mRNA of the first generation COVID vaccines, as shown in FIG. **22**. The serum of the mice injected with sa-mRNA encoding 3 (BA.2-1273), 4 (BA.2-1273-S2P), 7 (BA.2-1208), and 8 (BA.2-1208-S2P) delivered using the LNP of the present disclosure were collected at day 0, day 14 (2WP1, 2 weeks post 1st injection), and day 28 (2WP2, 2 weeks post 2nd injection) and assayed for titer of antigen specific IgG. As shown in FIGS. **21-22**, the data demonstrated comparable immune responses between the secreted version of the modified SPIKE protein of the present disclosure and the SPIKE protein encoded by mRNA currently approved by FDA.

Example 4: Characterization of the Sa-mRNA of the Disclosure

(226) A study was conducted to compare the cytotoxicity of different sa-mRNA constructs of the disclosure (SAM001, SAM002, and SAM003, and modified mRNA from MOD001) in HEK293 cells. While sa-mRNA advantageously produce an increased number of mRNA transcripts and/or higher expression of gene products, these characteristics could have cytotoxic effects on the host cell or organism. Therefore, it is important for a therapeutic sa-mRNA construct to sustain increased intracellular mRNA transcripts compared to conventional mRNA, while not increasing the transcript and gene product expression levels so much as to harm the host cell. Sustained intracellular mRNA transcript levels allow for a higher initial delivery of the therapeutic sa-mRNA

payload with less cytopathic effects because the transcript and gene product expression levels will not increase dramatically or decay rapidly over time. The results of this study show that SAM001 is a cytopathic construct and SAM002 is a non-cytopathic construct.

(227) As shown in FIG. 2, SAM002 showed the most stable reporter gene expression compared to the other tested sa-mRNA constructs, which suggests that it would be more suitable for vaccinations against infectious diseases or gene replacement therapies. In comparison, SAM001 reporter gene expression decreased dramatically, which suggests that it would be more suitable for cancer therapies requiring increased release of tumor associated antigens.

(228) A study was also conducted to characterize the replication rate of each sa-mRNA construct of the disclosure SAM001, SAM002, and SAM003, and modified mRNA from MOD001). As shown in FIG. 25, the relative expression of nsP3 and eGFP indicate the copies of sa-mRNA and subgenomic transcripts, respectively. SAM002 showed approximately 3 times lower transcript copies of both nsP3 or eGFP compared to SAM001. This result indicates that there is a higher rate of replication of sa-mRNA and transcription of the subgenomic eGFP reporter of SAM001 compared to SAM002, which indicates that there is an increased rate of decay of SAM001 and its transcripts.

(229) As shown in FIG. 26, the expression of the SAM001 encoding Luciferase consistently showed much lower in vivo expression compared to SAM002 when each construct is delivered intramuscularly using a P6-LNP delivery method. This result suggest that SAM002 is more suitable for delivery of therapeutic payloads in higher levels with less cytopathic effects compared to the SAM001 construct. Though the more cytopathic of SAM001 showed shorter and lower expressions of payload genes in FIGS. 2 and 26, those cytopathic effects can be helpful to increase therapeutic efficacy in cancer therapies that function through a release of tumor associated antigens, as demonstrated in FIG. 27. Thus, each of SAM001 and SAM002 has utility as a result of its unique biological activity.

Example 5: Directed Evolution of Sa-mRNA

(230) A study was conducted to produce sa-mRNA constructs. Directed evolution of sa-mRNA was performed on SAM002 at different concentrations (1 µg/ml or 10 µg/ml) of puromycin in C2C12 cells for 60 days. RNA dependent RNA polymerases are known to have a high error rate, which will cause mutants of SAM002 to appear over time. The SAM002 construct as shown in FIG. 28a was divided into 8 contigs, marked 1-8 on FIG. 28a, comprising SEQ ID NOs. 27-34. The 8 overlapping contigs facilitate cloning to a vector form suitable for Sanger DNA sequencing. As shown in FIG. 28b, only 1 mutation was found at nsP4 in 1 µg/ml puromycin. In contrast, there were 6 mutations found in nsP2, nsP3, and nsP4 at 10 µg/ml puromycin. The 6 mutations were in the 2nd, 3rd, 4th, 5th contigs. The 6 mutations of SAM002 is numbered as alleles 2, 4, 2, 2 respectively, which could form 32 variants, Shown in Tables 2-13 below:

(231) TABLE-US-00008 TABLE 1 No. % of GFP normalized at day 3

1	a1BCD-Sam002	93.60967
85.55008	92.43697479	2
a1BCD-Sam001	26.3245	26.94064
24.56418384	3	Ab1CD-Sam002
84.06015	83.50669	80.09049774
4	Ab1CD-Sam001	20.36517
21.08844	21.40921409	5
Ab2CD-Sam002	88.83792	89.1369
91.12781955	6	Ab2CD-Sam001
35.46667	36.71875	32.23593964
7	Ab3CD-Sam002	42.59259
41.76991	40.625	8
Ab3CD-Sam001	7.362637	10.01764
8.549747049	9	ABc1D-Sam002
75.67568	74.12224	72.98701299
10	ABc1D-Sam001	25.06739
19.02174	21.32253711	11
ABCd1-Sam002	88.4273	84.55657
85.80246914	12	ABCd1-Sam001
34.73282	31.71355	36.14609572
13	a1b1CD-Sam002	82.53968
75.84803	83.81962865	14
a1b1CD-Sam001	24.43325	22.82472
18.86075949	15	a1b2CD-Sam002
90.82707	85.01441	85.89928058
16	a1b2CD-Sam001	43.10345
43	41.4004914	17
a1b3CD-Sam002	47.98808	51.11441
51.30674003	18	a1b3CD-Sam001
13.7415	11.36182	14.97975709
19	a1Bc1D-Sam002	68.26265
70.17544	73.82550336	20
a1Bc1D-Sam001	24.86842	26.52005
24.66843501	21	a1BCd1-Sam002
81.64557	85.66929	87.59936407
22	a1BCd1-Sam001	23.58621
28.09587	30.1759134	

(232) TABLE-US-00009 TABLE 2 No. % of MFI normalized at day 3

1	a1BCD-Sam002	26.70408
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24.67162 28.17502279 2 a1BCD-Sam001 92.66064 71.8318 69.01040305 3 Ab1CD-Sam002
29.60338 27.95987 28.80892183 4 Ab1CD-Sam001 54.84388 72.12059 59.69650342 5 Ab2CD-
Sam002 25.2417 22.28458 24.65648054 6 Ab2CD-Sam001 79.93834 75.08098 69.93131997 7
Ab3CD-Sam002 62.02842 59.2609 53.425 8 Ab3CD-Sam001 72.97279 46.89674 60.26471613 9
ABc1D-Sam002 44.37571 45.86502 40.95282024 10 ABc1D-Sam001 80.15279 73.20619
82.89277065 11 ABCd1-Sam002 19.66246 19.21909 18.82683057 12 ABCd1-Sam001 64.61213
70.72639 68.27440117 13 a1b1CD-Sam002 26.06985 28.32976 26.02497234 14 a1b1CD-Sam001
63.195 56.67116 53.00767428 15 a1b2CD-Sam002 21.32604 21.28533 21.99453112 16 a1b2CD-
Sam001 69.66121 71.22326 72.73373474 17 a1b3CD-Sam002 60.85935 58.26164 56.95274307
18 a1b3CD-Sam001 61.42157 45.29323 38.56780888 19 a1Bc1D-Sam002 49.57803 46.91416
44.24096222 20 a1Bc1D-Sam001 68.03935 78.72531 75.14674158 21 a1BCd1-Sam002 18.37446
19.01418 19.03699816 22 a1BCd1-Sam001 61.15648 70.99497 64.04494382
(233) TABLE-US-00010 TABLE 3 No. nsP3 transcripts normalized with Actin 1 a1BCD-Sam002
9.447941 12.38052 10.056107 2 a1BCD-Sam001 51.62507 58.08123 49.8665331 3 Ab1CD-
Sam002 14.42001 24.59 25.4571675 4 Ab1CD-Sam001 84.44851 119.4282 119.428223 5 Ab2CD-
Sam002 14.92853 16.91229 14.825409 6 Ab2CD-Sam001 63.11889 64.89341 62.682899 7
Ab3CD-Sam002 36.75835 41.35529 41.9325889 8 Ab3CD-Sam001 177.294 199.4661 160.897712
9 ABc1D-Sam002 19.42712 17.5087 16.336194 10 ABc1D-Sam001 66.71781 57.68003
72.5045687 11 ABCd1-Sam002 11.71269 10.48315 13.4543426 12 ABCd1-Sam001 60.96883
58.89201 85.6273635 13 a1b1CD-Sam002 17.75311 17.02992 24.7610399 14 a1b1CD-Sam001
148.0561 121.9377 150.122874 15 a1b2CD-Sam002 8.75435 8.815241 11.1579493 16 a1b2CD-
Sam001 7.568461 6.453134 9.44794129 17 a1b3CD-Sam002 56.88593 47.83518 61.3929036 18
a1b3CD-Sam001 182.2784 178.5272 163.14376 19 a1Bc1D-Sam002 36.50444 37.27147
36.0018715 20 a1Bc1D-Sam001 120.2589 124.4998 119.428223 21 a1BCd1-Sam002 10.77787
11.47164 13.6421583 22 a1BCd1-Sam001 27.09585 22.3159 27.09585
(234) TABLE-US-00011 TABLE 4 No. eGFP transcripts normalized with Actin 1 a1BCD-Sam002
5.133704 4.958831 4.40762046 2 a1BCD-Sam001 37.79177 40.50421 30.4844159 3 Ab1CD-
Sam002 7.110741 10.05611 11.1579493 4 Ab1CD-Sam001 62.6829 72.50457 66.7178087 5
Ab2CD-Sam002 6.062866 5.61778 5.46416103 6 Ab2CD-Sam001 41.64294 49.86653 44.323503
7 Ab3CD-Sam002 21.55574 22.47112 24.2514651 8 Ab3CD-Sam001 90.50967 92.41147
81.5718801 9 ABc1D-Sam002 8.514961 6.868523 8.63382589 10 ABc1D-Sam001 67.64915
56.49299 67.1818678 11 ABCd1-Sam002 3.41054 3.24901 3.58010028 12 ABCd1-Sam001
24.93327 26.72281 34.7755156 13 a1b1CD-Sam002 6.543216 6.147501 7.3615012 14 a1b1CD-
Sam001 86.82268 76.10926 74.5429495 15 a1b2CD-Sam002 3.116658 2.8481 3.07375036 16
a1b2CD-Sam001 2.789487 2.584706 3.07375036 17 a1b3CD-Sam002 25.81254 23.91759
25.281322 18 a1b3CD-Sam001 51.62507 33.3589 41.3552906 19 a1Bc1D-Sam002 12.81712
11.15795 10.5560633 20 a1Bc1D-Sam001 50.21338 50.91434 47.176615 21 a1BCd1-Sam002
2.694467 3.052518 3.97236998 22 a1BCd1-Sam001 3.97237 3.917681 4.72397065
(235) TABLE-US-00012 TABLE 5 No. % of GFP normalized at day 3 23 Ab1c1D-Sam002
57.94271 55.04711 58.37696335 24 Ab1c1D-Sam001 14.20749 14.68531 12.45390071 25
Ab2c1D-Sam002 70.09736 68.3844 68.66096866 26 Ab2c1D-Sam001 20.82067 23.27965
21.19309262 27 Ab3c1D-Sam002 34.7651 33.24397 35.53162853 28 Ab3c1D-Sam001 13.96648
20.9589 13.76404494 29 Ab1Cd1-Sam002 74.92308 76.68232 80.26101142 30 Ab1Cd1-Sam001
15.24476 20.0569 13.22727273 31 Ab2Cd1-Sam002 84.28571 82.7957 84.78964401 32 Ab2Cd1-
Sam001 33.70474 30.24251 27.28592163 33 Ab3Cd1-Sam002 44.31138 39.58333 44.50704225
34 Ab3Cd1-Sam001 19.94498 10.28037 10.68181818 35 ABc1d1-Sam002 65.26868 67.29798
67.59847522 36 ABc1d1-Sam001 22.30483 21.73397 23.50791717 37 a1b1c1D-Sam002 51.07731
52.57069 51.30890052 38 a1b1c1D-Sam001 18.37769 16.13723 15.2866242 39 a1b2c1D-Sam002
63.92496 56.13772 57.55725191 40 a1b2c1D-Sam001 18.06167 17.48148 16.79389313 41
a1b3c1D-Sam002 18.5 19.86755 21.06109325 42 a1b3c1D-Sam001 7.973761 10.42735

9.6771654 43 a1b1Cd1-Sam002 73.34315 73.23308 75.65485362 44 a1b1Cd1-Sam001

14.30536 13.49501 15.88652482

(236) TABLE-US-00013 TABLE 6 No. % of MFI normalized at day 3 23 Ab1c1D-Sam002 50.962 48.41584 49.06840699 24 Ab1c1D-Sam001 45.2422 52.21293 80.05868412 25 Ab2c1D-Sam002 40.1987 36.34074 41.85967826 26 Ab2c1D-Sam001 77.60194 64.59525 72.00743858 27 Ab3c1D-Sam002 75.13296 69.43731 88.85059056 28 Ab3c1D-Sam001 82.07372 41.10864 70.3764424 29 Ab1Cd1-Sam002 21.75325 22.44224 21.7947142 30 Ab1Cd1-Sam001 56.84459 57.05329 47.73401939 31 Ab2Cd1-Sam002 16.99651 16.89642 15.9159919 32 Ab2Cd1-Sam001 54.45878 52.79225 51.64851339 33 Ab3Cd1-Sam002 47.26541 48.89045 41.55331882 34 Ab3Cd1-Sam001 57.46982 80.52686 74.81101512 35 ABc1d1-Sam002 38.89447 36.95033 39.74566762 36 ABc1d1-Sam001 73.61894 61.88906 78.83554648 37 a1blc1D-Sam002 48.99976 52.12974 55.4643059 38 a1b1c1D-Sam001 36.22922 41.67375 47.0700924 39 a1b2c1D-Sam002 40.58023 43.91632 45.09061489 40 a1b2c1D-Sam001 77.31859 68.19837 66.24362946 41 a1b3c1D-Sam002 83.91999 67.79338 66.58917612 42 a1b3c1D-Sam001 47.01905 34.42404 37.22906315 43 a1b1Cd1-Sam002 25.29739 23.40211 23.24178645 44 a1b1Cd1-Sam001 83.93086 70.15867 53.45713157

(237) TABLE-US-00014 TABLE 7 No. nsP3 transcripts normalized with Actin 23 Ab1c1D-Sam002 50.21338 43.71329 46.5271206 24 Ab1c1D-Sam001 79.89316 81.57188 79.3412928 25 Ab2c1D-Sam002 36.50444 35.0174 37.5307184 26 Ab2c1D-Sam001 115.3601 106.8913 107.634741 27 Ab3c1D-Sam002 37.27147 41.06963 41.6429394 28 Ab3c1D-Sam001 69.55103 77.1717 62.2499166 29 Ab1Cd1-Sam002 27.85762 29.65082 27.09585 30 Ab1Cd1-Sam001 163.1438 172.4459 178.527189 31 Ab2Cd1-Sam002 34.29675 32.89964 33.1284776 32 Ab2Cd1-Sam001 166.5718 155.4169 148.056088 33 Ab3Cd1-Sam002 0.752623 0.779165 0.83508792 34 Ab3Cd1-Sam001 1.536875 1.231144 1.26575659 35 ABc1d1-Sam002 2.770219 2.496661 2.37841423 36 ABc1d1-Sam001 147.0334 132.5139 142.024892 37 a1b1c1D-Sam002 91.13921 96.33579 95.0095085 38 a1b1c1D-Sam001 191.3407 207.9366 196.720023 39 a1b2c1D-Sam002 15.45498 14.6213 19.6983106 40 a1b2c1D-Sam001 101.1253 101.8287 113.771863 41 a1b3c1D-Sam002 0.447513 0.397768 0.42337266 42 a1b3c1D-Sam001 116.1625 124.4998 133.435617 43 a1b1Cd1-Sam002 22.7848 22.62742 24.5900029 44 a1b1Cd1-Sam001 210.8393 210.8393 207.936613

(238) TABLE-US-00015 TABLE 8 No. eGFP transcripts normalized with Actin 23 Ab1c1D-Sam002 16.44982 16 16.2233517 24 Ab1c1D-Sam001 38.05463 41.64294 44.323503 25 Ab2c1D-Sam002 11.47164 12.90627 14.5203065 26 Ab2c1D-Sam001 49.86653 48.50293 51.9841534 27 Ab3c1D-Sam002 9.063071 6.19026 6.58872814 28 Ab3c1D-Sam001 21.55574 20.39297 19.0273138 29 Ab1Cd1-Sam002 3.41054 3.5801 2.65737163 30 Ab1Cd1-Sam001 26.72281 32.89964 37.7917652 31 Ab2Cd1-Sam002 3.340352 3.530812 3.36358566 32 Ab2Cd1-Sam001 21.40684 17.75311 18.5070109 33 Ab3Cd1-Sam002 0.005486 0.005839 0.03564887 34 Ab3Cd1-Sam001 0.00982 0.008729 0.00872881 35 ABc1d1-Sam002 0.011438 0.010167 0.01045256 36 ABc1d1-Sam001 10.41073 7.727491 10.6294865 37 a1b1c1D-Sam002 24.42015 20.39297 35.2609637 38 a1b1c1D-Sam001 86.22295 89.88447 72.5045687 39 a1b2c1D-Sam002 0.406126 0.539614 0.70710678 40 a1b2c1D-Sam001 68.11969 76.10926 69.0706071 41 a1b3c1D-Sam002 0.004944 0.004072 0.00315094 42 a1b3c1D-Sam001 134.3637 140.0696 138.141214 43 a1b1Cd1-Sam002 4.890561 4.563055 5.16941132 44 a1b1Cd1-Sam001 50.21338 49.18001 46.5271206

(239) TABLE-US-00016 TABLE 9 No. % of GFP normalized at day 3 45 a1b2Cd1-Sam002 86.72087 85.7337 85.2367688 46 a1b2Cd1-Sam001 29.46429 27.00922 31.53846154 47 a1b3Cd1-Sam002 37.07025 36.80124 37.57668712 48 a1b3Cd1-Sam001 5.844828 6.109091 6.763110307 49 Ab1c1d1-Sam002 42.32633 37.85124 41.06870229 50 Ab1c1d1-Sam001 9.556314 5.738832 7.123966942 51 Ab2c1d1-Sam002 60.93294 60 60.34732272 52 Ab2c1d1-Sam001 22.2964 16.52893 20.51282051 53 Ab3c1d1-Sam002 30.11583 30.46272 22.75132275 54 Ab3c1d1-Sam001 9.871959 9.257362 9.638242894 55 a1Bc1d1-Sam002 53.46535 49.91334 58.52842809

56 a1Bc1d1-Sam001 11.69697 11.47692 13.8742573 57 a1b1c1d1-Sam002 45.08929 40.68554
38.14102564 58 a1b1c1d1-Sam001 12.20979 12.54795 21.51724138 59 a1b2c1d1-Sam002
68.81579 70.96774 67.578125 60 a1b2c1d1-Sam001 21.31783 21.50259 25.89641434 61
a1b3c1d1-Sam002 26.46675 24.3807 29.17214192 62 a1b3c1d1-Sam001 9.371795 10.38119
17.31984829 63 ABCD-Sam001 42.87516 41.27182 47.11779449 64 ABCD-Sam002 88.08777
85.36585 87.59231905 65 ABCd2-Sam002 76.58897 82.50433 90.74117236 66 ABCd2-Sam001
25.07269 23.83486 27.81926965

(240) TABLE-US-00017 TABLE 10 No. % of MFI normalized at day 3 45 a1b2Cd1-Sam002
21.59581 21.38896 21.2904946 46 a1b2Cd1-Sam001 80.67199 62.76361 60.79245886 47
a1b3Cd1-Sam002 59.51047 50.75597 49.29017604 48 a1b3Cd1-Sam001 37.66094 100.0263
53.2477737 49 Ab1c1d1-Sam002 46.83513 47.13787 50.37907506 50 Ab1c1d1-Sam001 59.35291
51.54369 51.29244805 51 Ab2c1d1-Sam002 49.5077 41.19688 43.92891221 52 Ab2c1d1-Sam001
54.3385 71.99053 72.05443699 53 Ab3c1d1-Sam002 68.90104 77.1418 86.83062969 54 Ab3c1d1-
Sam001 97.9473 55.35554 49.14956012 55 a1Bc1d1-Sam002 47.22042 38.97181 40.27795818 56
a1Bc1d1-Sam001 55.91852 77.0978 55.41929666 57 a1b1c1d1-Sam002 45.95797 40.88739
55.26597644 58 a1b1c1d1-Sam001 39.12734 47.70419 30.85771948 59 a1b2c1d1-Sam002
36.97674 36.20338 35.07727652 60 a1b2c1d1-Sam001 61.17269 58.38926 53.09734513 61
a1b3c1d1-Sam002 77.3002 69.05956 44.25943546 62 a1b3c1d1-Sam001 54.34123 134.8006
109.557945 63 ABCD-Sam001 75.08005 63.53236 58.12901796 64 ABCD-Sam002 25.37244
25.65748 24.13470845 65 ABCd2-Sam002 22.24231 25.40835 20.73913043 66 ABCd2-Sam001
87.19298 79.77941 87.31617647

(241) TABLE-US-00018 TABLE 11 No. nsP3 transcripts normalized with Actin 45 a1b2Cd1-
Sam002 41.06963 36.75835 35.2609637 46 a1b2Cd1-Sam001 128 137.187 121.937664 47
a1b3Cd1-Sam002 49.18001 51.98415 48.8402947 48 a1b3Cd1-Sam001 136.2394 146.0178
165.421162 49 Ab1c1d1-Sam002 60.54769 63.55792 70.5219274 50 Ab1c1d1-Sam001 257.7806
243.8753 266.871235 51 Ab2c1d1-Sam002 38.58585 43.71329 42.2242531 52 Ab2c1d1-Sam001
62.24992 69.55103 71.5063768 53 Ab3c1d1-Sam002 56.49299 60.96883 60.5476894 54 Ab3c1d1-
Sam001 139.1021 141.0439 141.043855 55 a1Bc1d1-Sam002 26.90869 31.55945 31.3414495 56
a1Bc1d1-Sam001 58.48521 64 62.682899 57 a1b1c1d1-Sam002 29.04061 34.29675 39.3966212
58 a1b1c1d1-Sam001 103.2501 109.1373 135.298309 59 a1b2c1d1-Sam002 27.47409 26.53823
28.0513831 60 a1b2c1d1-Sam001 60.12946 74.02804 66.2569551 61 a1b3c1d1-Sam002 51.26847
55.71524 58.4852128 62 a1b3c1d1-Sam001 128.8903 129.7868 132.51391 63 ABCD-Sam001
69.07061 61.3929 63.1188931 64 ABCD-Sam002 13.92881 13.8326 9.64646262 65 ABCd2-
Sam002 47.17662 47.50475 44.0173382 66 ABCd2-Sam001 13.08643 13.64216 15.1369223

(242) TABLE-US-00019 TABLE 12 No. eGFP transcripts normalized with Actin 45 a1b2Cd1-
Sam002 1.931873 1.292353 1.86606598 46 a1b2Cd1-Sam001 30.90996 28.44297 23.7523771 47
a1b3Cd1-Sam002 16.56424 17.14838 13.3614067 48 a1b3Cd1-Sam001 48.50293 56.49299
61.8199251 49 Ab1c1d1-Sam002 14.3204 15.24221 15.5624792 50 Ab1c1d1-Sam001 80.44886
81.57188 75.5835303 51 Ab2c1d1-Sam002 9.189587 10.05611 8.9382971 52 Ab2c1d1-Sam001 32
35.50622 35.0173984 53 Ab3c1d1-Sam002 28.05138 26.17287 28.6408023 54 Ab3c1d1-Sam001
49.18001 36.50444 48.1678959 55 a1bc1d1-Sam002 14.22148 12.04197 11.7941537 56 a1Bc1d1-
Sam001 37.01402 36.25228 30.6964518 57 a1b1c1d1-Sam002 13.36141 14.52031 18.6357374 58
a1b1c1d1-Sam001 61.3929 64 75.0614368 59 a1b2c1d1-Sam002 9.646463 8.815241 8.6938789
60 a1b2c1d1-Sam001 32.89964 34.77552 33.8245773 61 a1b3c1d1-Sam002 22.16175 23.75238
25.281322 62 a1b3c1d1-Sam001 71.01245 64.89341 71.0124462 63 ABCD-Sam001 35.50622
22.94328 27.857618 64 ABCD-Sam002 14.22148 5.897077 6.45313407 65 ABCd2-Sam002
44.6318 40.78594 30.4844159 66 ABCd2-Sam001 6.868523 7.310652 8.45614432

(243) Tables 1-12 show characterizations of 66 sa-mRNA mutants. To generate these mutants,
C2C12 cells were transfected with SAM002 encoding with puromycin by P6-LNP. The transfected
cells were cultured for 2 months at 1 or 10 µg/ml puromycin. At 2 months post transfection, the

total RNA of selected cells was extracted and reverse transcribed. The specific primers covering contigs from 1 to 6 were for amplicons and sub-cloning. For each contig, 8 clones were cultured and isolated using a Mini-Prep procedure to isolate small plasmid DNA from bacteria while limiting contaminating proteins and genomic DNA for Sanger Sequencing. The contig sequences comprise SEQ ID NOs. 27-34, which correspond to contigs 1-8, respectively. The identified mutations could make 66 combinations and were further engineered into SAM001 or SAM002 at the specified location for each mutation. Thus, this study identified 67 constructs of cytopathic and non-cytopathic sa-mRNA, including the mutation at 1 µg/ml puromycin and the 66 mutants shown in Tables 1-12.

(244) A characterization study was conducted to study the expression level of the 66 sa-mRNA variants identified using the method described above. The 66 sa-mRNA variants were transcribed in vitro and transfected to C2C12 cells using a LNP comprising an ionizable lipid P6 as defined in PCT Patent Application No. PCT/US2023/017777 (P6-LNP), which is fully incorporated herein. The transfected cells were performed by fluorescence-activated cell sorting (FACS) at day 1 and 3 post transfection. The percentages of GFP and mean fluorescent intensities (MFI), representing gene product expression of each variant, were analyzed. The percentage and MFI of GFP at day 3 were normalized compared to the data from day 1. Total RNA of the transfected cells was extracted and reverse transcribed as complementary DNA for quantification polymerase chain reaction (qPCR) by specific probes nsP3 and eGFP.

(245) To characterize the identified variants over time, the variants were transfected using P6-LNP into mouse myoblast C2C12 cells, analyzed by flow cytometer at day 1 and 3 post transfection, and quantified the transcript number of each sa-mRNA construct using nsP3 specific probes. The subgenomic transcripts were quantified using GFP specific probes. The decrease of GFP cells and intensity of GFP ranged broadly between the tested variants.

(246) Thus, the present disclosure includes a sa-mRNA library that is useful for various specialized indications, such as mRNA medicines against infectious diseases, cancers, autoimmune diseases, and rare diseases.

Example 4: Characterization of De Novo Synthesized Sa-mRNA of the Disclosure

(247) The nucleic acids of the present disclosure were synthesized using the TC-83 strain of VEE, a subclass of alphavirus, wherein SAM001 (SEQ ID NO: 35) is derived from wildtype TC-83 replicon without the alphavirus structural proteins, and SAM002 (SEQ ID NO: 36), SAM003 (SEQ ID NO: 37), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), and MOD001 (SEQ ID NO: 41) are modified according to the present disclosure. T7-VEE-GFP (SEQ ID NO: 42) was derived from wildtype TC-83 replicon and comprised a wildtype T7 promoter and a GOI encoding GFP. In vitro transcription efficacy of the nucleic acid template into sa-mRNAs and immune response to the sa-mRNAs were assayed in Raw-264.7, and 293T cells. The nucleic acid templates of the present disclosure were found to effectively transcribe into sa-mRNAs. The data demonstrated better or comparable in vitro transcription yields and decreased immune responses than therapeutic mRNA currently approved by FDA.

(248) DNA fragments of sa-mRNA (sa-mRNA) were de novo synthesized using a strain of Venezuelan equine encephalitis (VEE) virus TC-83, which is a subclass of alphavirus, with deletions of genes encoding structural proteins. The DNA fragments were assembled under T7 promoter as shown in FIG. 1 with other components including linker 1, origin of replication sequences (Ori), Ampicillin resistance gene (SM), Promoter of SM (SM-Pro), linker 2, 5'UTR, nsP1-4, linker 3, reporter genes or genes of interests (GOI), linker 4, 3'UTR, polyadenine (polyA). Based on the wild-type version SAM001, mutations for non-cytopathic versions of SAM002 (C5830T, Pro to Ser) and SAM003 (A5729T, Gln to Leu) were engineered.

(249) A comparison of the stability of different sa-mRNA versions from SAM001, SAM002, and SAM003, and modified mRNA from MOD001 as seen in as shown in FIG. 2 show that SAM002 is more stable compared to the other tested sa-mRNAs, including the modified mRNA. This data

indicates that SAM002 is more suitable for vaccinations against infectious diseases or gene replacements of gene editing.

(250) As can be seen in FIG. 2, expression of SAM001 decreased dramatically over time, which indicates that SAM001 be suitable for transient expressions or cancer immunotherapy as well as therapeutic cancer vaccines.

(251) Comparisons of different versions of sa-mRNA and modified mRNA were conducted. 293 T cells were transfected with different sa-mRNA (sa-mRNA) from SAM001, SAM002, SAM003, and modified mRNA from MOD001 encoding GFP by lipofectamine. The cells were analyzed by flow cytometer at day 1, 3, and 5. Decrease of GFP expression were normalized with the percentage of GFP at day 1. Statistical analyses were performed by one-way ANOVA. The modified sa-mRNA of the disclosure showed better GFP expression over time compared to SAM001 (SEQ ID NO: 35) as shown in FIG. 2.

(252) As the size of sa-mRNA is always larger than 7 kilo nucleotides, it is one of the challenges for mRNA production in limited time by in vitro transcription. As shown in FIG. 4, SAM002 (TAATACGACTCACTATAGGATAGG) (SEQ ID NO: 53) has unique repeating sequences of ATAGG. mRNA productions of SAM002 increased 46% than the T7-VEE-GFP as shown in FIG. 5. Thus, the sa-mRNA of the present disclosure has increased ability for transcription and higher yields, suitable for transcription of large fragments of mRNA and manufacture of high amount of mRNA. In a related experiment, the modified sa-mRNA comprising modified T7 and 5' UTR (SAM002 (SEQ ID NO: 36)) of FIG. 4 showed a higher yield at 30 minutes of in vitro transcription compared to the control (T7-VEE-GFP), as can be seen in FIG. 5.

(253) Interferon responses are innate reactions of host cells to exotic RNAs and materials as well as pathogens, which significantly restrict the half-life of mRNA in cells and give rise to side effects, such as fever. It is a medical challenge to manipulate interferon responses using mRNA.

Sequencing and functional analysis showed that a conserved 19 nucleotides fragment in the 3'UTR of alphavirus is critical to the repair of alphavirus. FIG. 6 shows the structure prediction of 3'UTR of wildtype VEE with the probability of the structure indicated according to the scale. The modified 3'UTR of the disclosure (SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), and SAM006 (SEQ ID NO: 40)) showed reduced or comparable interferon responses than self-amplified mRNA transcribed from nucleic acid templates containing 3'UTR of wildtype VEE (SAM002 (SEQ ID NO: 36) and MOD001 (SEQ ID NO: 41)). Based on the model shown in FIG. 6 of the 3'UTR and Poly-A of the VEE, the G at position 6 and the C at position 19 form a G::C pair to lock a loop, two GG at position minus 1 and 2 form GG::CC pair to build up a stem. A mutation of G to A at position 6, GG to AA at position minus 1, and 2, or both generated SAM004, SAM005, and SAM006 based on SAM002 as shown in FIG. 7.

(254) The sa-mRNA from SAM004, SAM005, and SAM006, and modified mRNA from MOD001 were transfected to Raw-ISR-Lucia cells, an interferon reporter cell developed by Invivogen. As shown in FIG. 8, at day 1 post transfection, SAM004 showed more than 5.1- and 2.1-times lower interferon responses than SAM002 and modified mRNA MOD001 in an interferon simulation assay.

(255) FIG. 8 shows a reporter assay of 5 individual self-amplifying mRNAs produced from nucleic acid templates SAM002 (SEQ ID NO: 36), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), SAM006 (SEQ ID NO: 40), and MOD001 (SEQ ID NO: 41) expressing GFP in Raw-ISR-Lucia cells at day 1 post-transfection. The modified 3'UTR of the disclosure (SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), and SAM006 (SEQ ID NO: 40)) showed reduced or comparable interferon responses than self-amplified mRNA transcribed from nucleic acid templates containing 3'UTR of wildtype VEE (SAM002 (SEQ ID NO: 36) and MOD001 (SEQ ID NO: 41)).

(256) FIG. 9 shows a reporter assay of 4 individual sa-mRNAs produced from nucleic acid templates SAM002 (SEQ ID NO: 36), SAM004 (SEQ ID NO: 38), SAM005 (SEQ ID NO: 39), and SAM006 (SEQ ID NO: 40) expressing GFP in Raw-ISR-Lucia cells at day 1 post-transfection

where GFP expression is normalized with nsP3 in comparison to SAM002. In the GFP mRNA transcripts in SAM004, SAM005, SAM006, and SAM002, GFP level is 2 times lower in SAM004 and even 1.8 times higher in SAM005, than it in SAM002.

(257) Nucleic Acid Sequences

(258) In the following exemplary sequences, represent exemplary GOI that may be replaced with any other GOI. Persons skilled in the art will recognize that these sequences are exemplary and not limiting disclosures that support and serve as proof of the concepts disclosed and claimed herein.

(259) TABLE-US-00020 TABLE 13 Description SEQ ID NO. Sequence Subgenomic SEQ ID NO: 9 TAACCTGAATGGACTACGACATAGTCTAGTC promoter 1 CGCCAAG (SGP1)

Cloning site for SEQ ID NO: 10 TTCGAAGGCGCGCCTCTA SGP1 Subgenomic SEQ ID NO: 11 GAACTTCCATCATAGTTATGGCCATGACTACT promoter 2

CTAGCTAGCAGTGTTAAATCATTACAGCTACCT (SGP2)

GAGAGGGGCCCCTATAACTCTCTACGGC Cloning site for SEQ ID NO: 12

ATCGATGATATCGCGGCCGCATACAGCAGC SGP2 Murine signal SEQ ID NO: 14

ATGACCTCCCGGCTTGTGAGGGTACTGGCTG peptide (MSP)

CTGCTATGCTGGTGGCTGCTGCTGTGAGTGTG GC Murine SEQ ID NO: 15

ATGTGGGAGCTTGAAAAAGACGTCTATGTAG interleukin-12

TAGAAGTGGACTGGACACCTGATGCTCCTGG comprising heavy chain p40 (mIL12)

GAGGAAGATGATATCACCTGGACTTCTGACC P40)

AGAGACACGGGGTGATTGGGAGCGGCAAAA

CCCTGACGATCACTGTGAAGGAGTTTCTGGA

CGCCGGCCAGTATACCTGTCACAAGGGGGGG

GAGACCCTGAGTCATAGCCACCTGTTGCTGC

ACAAGAAGGAGAATGGCATCTGGTCTACAGA

GATCCTGAAGAACTTTAAGAACAAGACCTTC

CTGAAGTGTGAAGCACCAAACTACAGTGGTC

GCTTTACCTGCAGCTGGCTGGTCCAAAGAAA

CATGGACCTGAAATTTAATATAAAGAGTAGC

TCTTCGAGTCCTGATTCCAGGGCCGTGACGT

GCGGCATGGCAAGCCTTTCAGCCGAAAAAGT

CACGCTGGATCAGCGAGACTATGAGAAGTAC

AGCGTTAGCTGTCAGGAGGACGTAACCTTGCC

CGACTGCCGAGGAGACTCTGCCCATAGAGCT

CGCTCTGGAGGCCAGGCAGCAGAACAAATAT

GAGAATTACAGCACTAGTTTCTTTATTAGAG

ACATCATCAAACCCGACCCACCCAAGAATCT

GCAGATGAAGCCGCTGAAGAATAGTCAGGTC

GAGGTTTCCTGGGAATATCCAGATTCATGGT

CCACTCCGCATTCTTATTTTCTTAAAATTC

TTTGTTAGGATTCAGCGGAAAAAAGAAAAGA

TGAAAGAGACGGAGGAAGGGTGCAACCAGA

AGGGGGCCTTCCTGGTGGAAAAGACAAGCAC

TGAGGTCCAATGTAAGGGTGGGAACGTTTGC

GTGCAGGCTCAGGATCGCTACTACAACAGCA

GTTGCTCTAAGTGGGCCTGCGTACCTTGTCGC GTCAGGAGT Linker (L(a)) SEQ ID

NO: 16 GGAGGGGGGGTCAGGGGGTGGCTCAGGCGGC GGCAGTGGGGGCAGC Murine

SEQ ID NO: 17 AGGGTGATCCCAGTGTCTGGGCCGGCCCGTT interleukin-12

GCTTGTCTCAATCCAGAAACCTCCTCAAGAC comprising light

CACTGACGATATGGTAAAGACTGCCCCGAGAG chain p35

ATAGTAACCTTGTACAGCTGAAG (mIL12-P35)
ATATAGACCATGAGGATATAACACGGGACCA
GACCTCTACTCTGAAAACCTGTCTGCCTCTTG
AGCTGCACAAGAACGAGTCCTGTCTGGCTAC
CCGCGAAACCTCAAGCACAACCAGAGGTAGT
TGCCTGCCCCCACAAGACATCGCTTATGA
TGACCTTGTGTCTGGGATCTATTTATGAGGAC
CTGAAGATGTACCAAACCTGAGTTCCAGGCAA
TAAATGCTGCTCTCCAGAATCACAAATCATCA
ACAAATCATCCTTGATAAGGGGATGCTGGTC
GCAATCGACGAGCTCATGCAATCGCTGAACC
ACAATGGGGAAACCCTCAGGCAGAAACCAC
CGGTGGGAGAGGCCGACCCCTACCGTGTTAA
AATGAAGTTGTGTATTCTTTTGCATGCATTCT
CTACAAGAGTCGTTACCATCAATCGCGTCAT GGGGTACCTGTCATCAGCC Linker (L(b))
SEQ ID NO: 18 GGCGGTAGTGGTGGTGGGAGC Murine SEQ ID NO: 19
GGGTACCTGTCATCAGCCGGCGGTAGTGGTG interleukin-21
GTGGGAGCCACAAGTCCTCCCCCAGGGTCC (mIL21)
GGATCGGCTCTTGATCAGACTGAGACATCTG
ATTGATATTGTGCGAGCAGTTGAAGATCTATG
AGAATGACCTCGATCCTGAGTTACTGAGTGC
CCCACAGGACGTTAAAGGGCACTGTGAACAC
GCCGCATTTGCTTGTTTTTCAGAAGGCCAAGCT
GAAACCTTCTAATCCCGGGAATAACAAAACCT
TTCATTATCGATCTCGTCGCGCAGCTGAGGC
GGCGACTTCCTGCACGGCGGGGGGGGAAAA
AGCAAAAGCACATCGCAAAGTGTCCCTCATG
CGACTCTTACGAGAAACGTACCCCTAAGGAG
TTCCTTGAAAGACTCAAATGGCTGCTGCAAA AGATGATCCACCAGCATCTCAGC
Human signal SEQ ID NO: 21 ATGGACTGGACCTGGCGAATACTGTTCTTGG
peptide (HSP) TTGCCGCCGCTACAGGGACTCACGCA Human SEQ ID NO: 22
ATATGGGAGCTGAAGAAGGACGTGTATGTCG interleukin-12
TGGAGCTGGACTGGTACCCAGATGCTCCTGG comprising heavy
CGAAATGGTGGTTTTAACATGTGATAACCCC chain p40 (hIL12
GAGGAGGACGGCATCACATGGACTCTGGACC P40)
AGAGTTCTGAGGTGCTGGGGTCCGGCAAGAC
TCTGACAATCCAGGTAAAGGAGTTCGGCGAC
GCAGGACAGTACACTTGTCACAAGGGAGGTG
AGGTGCTTTCTCACAGCCTGTTGCTGCTCCAT
AAGAAGGAAGACGGTATTTGGTCAACCGACA
TCCTCAAGGACCAGAAGGAGCCCCAAAACA
AGACCTTTCTGAGATGTGAGGCCAAGAATTA
CAGCGGTAGATTCACCTTGTTGGTGGCTCACC
ACCATATCCACAGACTTGACCTTCAGTGTCA
AAAGTTCACGAGGGAGCTCAGATCCTCAAGG
CGTTACCTGTGGCGCAGCGACGCTGTCCGCA
GAAAGAGTCAGGGGAGACAACAAGGAATAC
GAGTACTCTGTGAGTGCCAGGAGGATTCCG
CCTGTCCGGCCGAGAGGAGTCTTTACCTATT
GAGGTGATGGTCGATGCCGTGCACAAGCTTA

AGTACGACATATACATCAAGTTTTTTCATC
CGCGACATCATTAACCTGATCCACCAAAGA
ACCTGCAGCTCAAGCCTCTGAAGAATAGCAG
GCAGGTCGAGGTAAGCTGGGAGTATCCTGAT
ACCTGGTCCACCCCCCACAGTTATTTTCAGCCT
CACCTTCTGCGTCCAAGTCCAGGGAAAGAGC
AAGAGAGAGAAGAAGGATAGGGTGTTCACA
GATAAGACTTCAGCTACTGTGATCTGCAGAA
AGAA_tGCGTCTATCTCTGTGCGAGCACAAAGAC
AGGTACTACAGTTCTAGCTGGAGCGAGTGGG CATCAGTCCCCTGCAGT Linker (L(c))
SEQ ID NO: 23 GGTGGCGGAAGCGGAGGGGGCAGCGGAGGT
GGGAGCGGAGGGAGC Human SEQ ID NO 24
AGGAACCTCCCAGTTGCTACACCTGACCCGG interleukin-12
GAATGTTTTCCATGCCTCCACCATTCCCAGAAT comprising light
CTCCTCCGAGCCGTGTCCAATATGCTGCAAA chain p35 (hIL12-
AGGCTCGGCAGACCTTGGAGTTTTACCCTTG P35)
CACCTCAGAAGAAATCGATCATGAGGATATC
ACAAAGGATAAGACGAGCACTGTTGAGGCAT
GCCTGCCCCCTGGAGCTAACTAAGAATGAGTC
TTGCCTGAACAGCAGGGAGACTTCCTTCATT
ACCAACGGTAGCTGTCTTGCCAGCAGGAAGA
CATCTTTTATGATGGCCCTGTGTCTATCTAGC
ATATATGAAGACCTGAAGATGTACCAGGTGG
AATTCAAAACCATGAATGCTAAGCTTCTCAT
GGATCCCAAGAGGCCAAATCTTCTGACCAG
AATATGCTTGCTGTCATAGATGAACTGATGC
AGGCGTTGAATTTTAACAGCGAGACGGTGCC
TCAAAAAAGCTCACTGGAAGAACCTGATTTT
TATAAAACGAAGATCAAGCTGTGTATTTTAC
TACACGCCTTTAGAATCCGCGCTGTTACCATC
GACAGAGTAATGTCCTACCTAAATGCTTCA Linker (L(d)) SEQ ID NO 25
GGAGGGTCAGGAGGAGGATCC Human SEQ ID NO 26
CAGGACAGGCATATGATCCGGATGCGGCAGC interleukin-21
TGATCGATATTGTAGACCAGTTGAAGAATTA (hIL21)
TGTGAACGACTTAGTGCCGGAATTCCTCCCC
GCCCCCGAGGACGTGGAGACTAATTGTGAGT
GGTCTGCATTCTCATGCTTCCAAAAAGCACA
GCTGAAGAGTGCCAATACCGGCAATAACGAA
AGGATCATCAATGTAAGTATAAAGAAGTTAA
AACGCAAACCGCCCAGTACCAACGCTGGACG
CAGGCAAAAACACAGGCTGACATGCCCCCTCG
TGTGATTCTGACGAAAAAAAAACCTCCAAAGG
AATTCCTGGAAAGGTTCAAGTCCTTATTACA
GAAAATGATTCACCAGCACCTGAGTAGTAGG ACCCACGGATCCGAAGACTCC Linker
1 SEQ ID NO: 43 CGCGTGATAACGCAGGAAAGAACATGTGAG
CAAAAGGCCAGCAAAAGGCCAGGAACCGTA AAAAGGCCGCGTTGCTGGCGTT
Linker 2 SEQ ID NO: 44 CACATTTCCCCGAAAAGTGCCACCTGAGCTC Linker 3
SEQ ID NO: 45 TTCGAAGGCGCGCCTCTAGAGCCACC Linker 4 SEQ ID NO: 46
CATCGATGATATCGCGGCCGCATACAGCAGC T7' SEQ ID NO: 47
TAATACGACTCACTATAGG 5'UTR' intentionally ATAGG skipped sequence 3'UTR conserved

SEQ ID NO: 49 GGATTTTGTGTTTAAATATTT sequence 3'UTR' SEQ ID NO: 50
GGATTTTATTTTAAATATTTTC 3'UTR' SEQ ID NO: 51 AAATTTTGTTTTAAATATTTTC
3'UTR' SEQ ID NO: 52 AAATTTTATTTTAAATATTTTC
(260) TABLE-US-00021 SARS-COV-2, Omicron BA.1-1273 (SEQ ID NO: 1)
ATGTTTGTGTTCTTGGTGTGCTTCCACTGGTCAGTTCCCAATGCGTTAATCTCACCA
CCCGAACTCAACTCCCACCCGCATATACAAATTCCTTCACCAGAGGAGTGTACTATC
CTGACAAAGTGTTCGGTCAAGTGTCTTCCACTCTACTCAGGACCTCTTTCTGCCTTT
CTTTTCTAACGTTACATGGTTTCATGTGATCTCTGGGACAAACGGCACCAAACGCTT
CGACAACCCTGTATTGCCATTCAATGATGGGGTGACTTTGCCTCCATCGAGAAATC
CAACATCATTGAGGATGGATTTTCGGGACTACTCTGGACTCAAAGACACAGAGCCT
GCTGATCGTTAACAACGCCACAAACGTTGTCATCAAAGTGTGCGAATTCCAGTTTTG
CAATGATCCCTTCCTGGACCACAAGAATAACAAGTCCTGGATGGAGAGCGAATTTTC
GGGTCTACAGCAGCGCAAACAACACTGCACCTTCGAGTACGTGAGTCAACCCTTTCTGA
TGGACCTGGAAGGGAAACAGGGAAACTTCAAGAACCTGAGAGAGTTTGTCTTTAAG
AACATCGACGGCTATTTTAAGATCTATAGTAAGCATAACGCTATCATTGTAAGGGAG
CCCGAGGATCTTCCCCAGGGCTTTTCAGCCCTGGAACCTTTGGTTGACTTGCCTATTG
GTATCAATATCACCAGATTTTACAGACCCTTCTGGCATTGCATCGGTCTTATCTTACTCC
AGGTGATTCTCCTCCGGGTGGACTGCCGGCGCCGCTGCCTACTATGTCGGCTATCT
GCAACCAAGAACGTTCCCTGCTCAAGTACAACGAAAACGGCACTATTACGGATGCTG
TTGATTGTGCCCTGGACCCTCTGTCTGAGACTAAATGCACCCTCAAGAGCTTTACCG
TTGAGAAGGGGATTTACCAAACCAGTAATTTCCGGGTCCAACCCACCGAAAGCATT
GTGCGGTTCCCAAATATACCAATCTGTGTCCCTTTGATGAAGTGTTCAATGCTACA
AGGTTTGCTTCTGTGTACGCATGGAATAGGAAACGCATCTCCAATTGTGTGCTGAT
TACTCCGTGCTGTACAATCTGGCCCCATTCTTCACCTTCAAGTGTTATGGCGTTTCAC
CTACCAAACCTTAACGACCTGTGCTTCACTAATGTGTATGCCGACTCTTTTGTGATACG
AGGCGATGAAGTGAGACAGATTGCACCAGGGCAGACCGGCAACATTGCCGACTACA
ACTACAAGCTTCCAGATGACTTTACCGGATGTGTTATTGCATGGAACCTCAAACAAGC
TGGATTCCAAGGTGAGCGGCAACTATACTACCTGTATAGACTGTTTCAGGAAATCCA
ACCTGAAACCATTGAGCGAGATATAAGCACAGAAATCTACCAGGCTGGAACAACAA
CCCTGCAACGGCGTGGCTGGGTCAACTGCTACTTCCCATTGCGCAGTTACAGCTTC
AGACCTACATACGGGGTGGGTACCAACCCTATCGTGTGCTAGTCCTGAGTTTTGAG
CTCCTCCATGCCCCAGCCACAGTCTGTGGCCCCAAGAAAAGCACCAATCTGGTGAA
GAACAAATGCGTGAACTTTAACTTTAACGGACTCAAGGGAACCGGCGTATTGACGG
AGAGTAACAAGAAGTTCCTGCCATTCCAGCAGTTCGGTCGCGATATTGCCGACACTA
CCGACGCTGTCCGAGATCCCCAGACATTGGAGATTCTTGATATCACACCCTGTAGTT
TCGGCGGAGTGAGCGTGATTACGCCCCGAACCAATACCAGCAATCAGGTTGCCGTC
CTGTATCAGGGCGTGAATTGCACCGAGGTACCTGTGCGCATCCACGCTGACCAACTT
ACACCCACATGGCGAGTATATTCCACCGGCTCCAACGTCTTTCAGACACGTGCTGGA
TGTCTGATCGGTGCAGAATATGTTAATAATAGCTACGAGTGTGATATCCCCATCGGT
GCTGGAATATGCGCCTCTTATCAAACCTCAAACCAAATCTCACAGGCGGGCACGTAGT
GTAGCATCCCAAAGTATCATTGCCTACACAATGAGCCTCGGTGCTGAGAATTCTGTC
GCCTACAGCAACAACCTCCATTGCTATCCCTACTAACTTCACAATCAGTGTGACAAC
GAAATTCTGCCCCGTATCTATGACCAAAACAAGCGTTGACTGCACCATGTACATCTGT
GGCGATTCTACCGAATGTAGCAATCTCCTCCTGCAATACGGATCATTCTGCACTCAG
CTGAAGCGTGCCCTCACAGGTATTGCAGTTGAGCAGGACAAGAATACGCAGGAAGT
GTTTGCCAGGTGAAGCAAATCTACAAAACCTCCACCCATAAAATACTTTGGCGGATT
CAATTTCTCACAGATCCTGCCCGATCCCTCAAACCCCTCCAAGCGTAGCTTTATCGA
GGATCTGCTCTTCAACAAGGTAACCCTCGCAGATGCCGGTTTCATCAAGCAGTATGG
CGATTGTCTGGGAGACATCGCCGCTCGGGACCTGATCTGTGCACAGAAGTTCAAAG
GACTGACCGTGCTGCCTCCCTTGCTGACCGACGAGATGATAGCCCAATACACTAGCG

CCCTGCTGGCGGCTGGGACCACTTCTGGAGCTTGGCGGCTGGCCTTC
AGATTCTTTTGTATGCAGATGGCCTACCGCTTTAACGGCATCGGTGTGACACAAA
ACGTTCTGTATGAAAACCAGAACTCATCGCCAACCAGTTCAACAGTGCTATCGGTA
AGATACAGGATAGCCTGTCATCCACTGCCAGCGCATTGGGAAAGTTGCAGGATGTA
GTGAACCACAATGCCCAGGCACTTAACACCCTGGTGAAACAGCTCTCTTCAAAGTTT
GGTGCCATTTCTAGCGTGCTGAATGACATATTTAGCCGGTTGGACAAGGTGGAGGCT
GAAGTGCAGATTGATAGGCTGATAACTGGGCGCCTTCAGTCTCTTCAGACCTATGTG
ACCCAGCAGCTCATCCGCGCTGCTGAAATTCGCGCATCCGCTAACCTGGCAGCAACC
AAAATGTCCGAGTGTGTGCTGGGTGAGTCTAAGAGAGTGGACTTTTGCGGGAAGGG
GTATCACCTGATGTCTTTTCTCAGTCTGCACCCCATGGTGTGGTCTTTCTGCACGTG
ACTTATGTCCCAGCTCAGGAAAAGAACTTCACTACAGCCCCAGCCATCTGCCACGAT
GGGAAAGCCCCTTTCCCAGGGAAGGCGTATTCGTGTCCAATGGTACTCATTGGTTC
GTCACTCAGAGAAATTTCTACGAGCCCCAGATTATAACCACTGACAATACATTTGTA
TCCGGCAATTGTGATGTGGTTATCGGGATTGTGAATAATACTGTTTACGATCCTTTGC
AGCCAGAGCTGGACTCCTTCAAGGAGGAGCTTGACAAATATTTTAAGAATCACACA
TCACCTGACGTCGACCTCGGAGATATTTAGGAATCAATGCTTCCGTGGTCAATATT
CAGAAGGAGATAGACAGGCTGAATGAGGTTGCCAAGAACCTCAACGAGTCTCTGAT
CGATCTGCAGGAGTTGGGCAAGTACGAACAGTATATCAAATGGCCTTGGTACATTTG
GCTTGGGTTTATTGCTGGGCTGATAGCTATCGTCATGGTGACAATTATGTTGTGTTGC
ATGACATCCTGCTGTAGTTGTCTGAAGGGCTGCTGCTCATGCGGCAGCTGTTGCAAG
TTCGACGAGGACGATTCTGAGCCCGTGCTGAAGGGCGTGAAACTGCACTACACATGA
SARS-COV-2, Omicron BA.1-1273-S2P (SEQ ID NO: 2)
ATGTTTGTGTTCTTGGTGTGCTTCCACTGGTCAGTTCCCAATGCGTTAATCTCACCA
CCCGAACTCAACTCCCACCCGCATATACAAATTCCTTCACCAGAGGAGTGTACTATC
CTGACAAAGTGTTTCGGTCAAGTGTCCTCCACTCTACTCAGGACCTCTTTCTGCCTTT
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TCGGCTGGATTTAGGCATCCCAGGTGATGTGCCCAAATATGACATAATATTTGTAA
TGTGAGGACCCCATATAAATACCATCACTATCAGCAGTGTGAAGACCATGCCATTAA
GCTTAGCATGTTGACCAAGAAAGCTTGTCTGCATCTGAATCCCGGGCGGAACCTGTGT
CAGCATAGGTTATGGTTACGCTGACAGGGCCAGCGAAAGCATCATTGGTGCTATAG
CGCGGCAGTTCAAGTTTTCCCGGGTATGCAAACCGAAATCCTCACTTGAAGAGACGG
AAGTTCTGTTTGTATTTCATTGGGTACGATCGCAAGGCCCGTACGCACAATCCTTACAA
GCTTTCATCAACCTTGACCAACATTTATACAGGTTCCAGACTCCACGAAGCCGGATG
TGCACCCTCATATCATGTGGTGCGAGGGGATATTGCCACGGCCACCGAAGGAGTGA
TTATAAATGCTGCTAACAGCAAAGGACAACCTGGCGGAGGGGTGTGCGGAGCGCTG
TATAAGAAATTCCCGGAAAGCTTCGATTTACAGCCGATCGAAGTAGGAAAAGCGCG
ACTGGTCAAAGGTGCAGCTAAACATATCATTTCATGCCGTAGGACCAAACCTTCAACA
AAGTTTCGGAGGTTGAAGGTGACAAACAGTTGGCAGAGGCTTATGAGTCCATCGCT
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ATCTTTTCCGGGAACAAAGATCGACTAACCCAATCATTGAACCATTTGCTGACAGCT
TTAGACACCACTGATGCAGATGTAGCCATATACTGCAGGGACAAGAAATGGGAAAT
GACTCTCAAGGAAGCAGTGGCTAGGAGAGAAGCAGTGGAGGAGATATGCATATCCG
ACGACTCTTCAGTGACAGAACCTGATGCAGAGCTGGTGAGGGTGCATCCGAAGAGT
TCTTTGGCTGGAAGGAAGGGCTACAGCACAAGCGATGGCAAAACTTTCTCATATTTG
GAAGGGACCAAGTTTCACCAGGCGGCCAAGGATATAGCAGAAATTAATGCCATGTG
GCCCCGTTGCAACGGAGGCCAATGAGCAGGTATGCATGTATATCCTCGGAGAAAGCA
TGAGCAGTATTAGGTCGAAATGCCCCGTCGAAGAGTCGGAAGCCTCCACACCACCT
AGCACGCTGCCTTGCTTGTGCATCCATGCCATGACTCCAGAAAGAGTACAGCGCCTA
AAAGCCTCACGTCCAGAACAAATTACTGTGTGCTCATCCTTTCCATTGCCGAAGTAT
AGAATCACTGGTGTGCAGAAGATCCAATGCTCCCAGCCTATATTGTTCTCACCGAAA
GTGCCTGCGTATATTCATCCAAGGAAGTATCTCGTGGAACACCACCGGTAGACGA
GACTCCGGAGCCATCGGCAGAGAACCAATCCACAGAGGGGACACCTGAACAACCAC
CACTTATAACCGAGGATGAGACCAGGACTAGAACGCCTGAGCCGATCATCATCGAA
GAGGAAGAAGAGGATAGCATAAGTTTGCTGTCAGATGGCCCGACCCACCAGGTGCT
GCAAGTCGAGGCAGACATTCACGGGCGGCCCTCTGTATCTAGCTCATCCTGGTCCAT
TCCTCATGCATCCGACTTTGATGTGGACAGTTTATCCATACTTGACACCCTGGAGGG
AGCTAGCGTGACCAGCGGGGCAACGTCAGCCGAGACTAACTCTTACTTCGCAAAGA
GTATGGAGTTTCTGGCGCGACCGGTGCCTGCGCCTCGAACAGTATTCAGGAACCCTC
CACATCCCGCTCCGCGCACAAAGAACACCGTCACTTGCACCCAGCAGGGCCTGCTCG
AGAACCAGCCTAGTTTCCACCCCGCCAGGCGTGAATAGGGTGATCACTAGAGAGGA
GCTCGAGGCGCTTACCCCGTCACGCACTCCTAGCAGGTCGGTCTCGAGAACCAGCCT
GGTCTCCAACCCGCCAGGCGTAAATAGGGTGATTACAAGAGAGGAGTTTGAGGCGT
TCGTAGCACAACAACAATGACGGTTTGTATGCGGGTGCATACATCTTTTCCTCCGACA
CCGGTCAAGGGCATTTACAACAAAAATCAGTAAGGCAAACGGTGCTATCCGAAGTG
GTGTTGGAGAGGACCGAATTGGAGATTTCTGTATGCCCCGCGCCTCGACCAAGAAAA
AGAAGAATTACTACGCAAGAAATTACAGTTAAATCCCACACCTGCTAACAGAAGCA
GATACCAGTCCAGGAAGGTGGAGAACATGAAAGCCATAACAGCTAGACGTATTCTG
CAAGGCCTAGGGCATTATTTGAAGGCAGAAGGAAAAGTGGAGTGCTACCGAACCCCT

GCATCTGTTCTTGTATTGTGTAACCGCTTTGTCGAAGCCCCAACGGTC
GCAGTGGAAGCCTGTAACGCCATGTTGAAAGAGAACTTTCCGACTGTGGCTTCTTAC
TGTATTATTCCAGAGTACGATGCCTATTTGGACATGGTTGACGGAGCTTCATGCTGC
TTAGACACTGCCAGTTTTTTGCCCTGCAAAGCTGCGCAGCTTTCCAAAGAAACACTCC
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AACGTCCTGGCAGCTGCCACAAAAAGAAATTGCAATGTCACGCAAATGAGAGAATT
GCCCCGATTGGATTCGGCGGCCCTTTAATGTGGAATGCTTCAAGAAATATGCGTGTA
TAATGAATATTGGGAAACGTTTAAAGAAAACCCCATCAGGCTTACTGAAGAAAACG
TGGTAAATTACATTACCAAATTAAGGACCAAAGCTGCTGCTCTTTTTTGCGAAGA
CACATAATTTGAATATGTTGCAGGACATAACCAATGGACAGGTTTGTAAATGGACTTAA
AGAGAGACGTGAAAGTGACTCCAGGAACAAAACATACTGAAGAACGGCCCAAGGT
ACAGGTGATCCAGGCTGCCGATCCGCTAGCAACAGCGTATCTGTGCGGAATCCACC
GAGAGCTGGTTAGGAGATTAAATGCGGTCCTGCTTCCGAACATTCATACACTGTTTG
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GTGTTCTGGAAACTGACATCGCGTCGTTTGATAAAAGTGAGGACGACGCCATGGCTC
TGACCGCGTTAATGATTCTGGAAGACTTAGGTGTGGACGCAGAGCTGTTGACGCTGA
TTGAGGCGGCTTTTCGGCGAAATTTTCATCAATACATTTGCCCACTAAAATAATTTA
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TTAACATTGTAATCGCAAGCAGAGTGTTGAGAGAACGGCTAACCGGATCACCATGT
GCAGCATTCAATTGGAGATGACAATATCGTGAAAGGAGTCAAATCGGACAAATTAAT
GGCAGACAGGTGCGCCACCTGGTTGAATATGGAAGTCAAGATTATAGATGCTGTGG
TGGGCGAGAAAGCGCCTTATTTCTGTGGAGGGTTTATTTTGTGTGACTCCGTGACCG
GCACAGCGTGCCGTGTGGCAGACCCCTAAAAAGGCTGTTTAAGCTTGGCAAACCT
CTGGCAGCAGACGATGAACATGATGATGACAGGAGAAGGGCATTGCATGAAGAGTC
AACACGCTGGAACCGAGTGGGTATTCTTTTCAGAGCTGTGCAAGGCAGTAGAATCAA
GGTATGAAACCGTAGGAACCTCCATCATAGTTATGGCCATGACTACTCTAGCTAGCA
GTGTTAAATCATTCACTACCTGAGAGGGGGCCCTATAACTCTCTACGGCT**AACCTG**
AATGGACTACGACATAGTCTAGTCCGCCAAGTTTCGAAGGCGCGCCTCTAGAgccaccAT
GGTGAGCAAGGGCGAGGAGCTGTTACCGGGGTGGTGCCCATCCTGGTTCGAGCTGG
ACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGATGCC
ACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGGCAAGCTGCCCCGTGCC
TGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCACTGCTTCAGCCGCTACCCC
GACCACATGAAGCAGCAGCACTTCTTCAAGTCCGCCATGCCCGAAGGCTACGTCCA
GGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGACCCGCGCCGAGGTGA
AGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCGACTTCAAG
GAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACACTACAACAGCCACAACGT
CTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGAACTTCAAGATCCGCC
ACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTACCAGCAGAACACCCCC
ATCGGCGACGGCCCCGTGCTGCTGCCCCGACAACCACTACCTGAGCACCAGTCCGCC
CTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTCCTGCTGGAGTTCGTGAC
CGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGtagcATCGATGATATCGC
GGCCGCATACAGCAGCAATTGGCAAGCTGCTTACATAGAACTCGCGGCGATTGGCA
TGCCGCCTTAAAATTTTTATTTTATTTTCTTTTCTTTTCCGAATCGGATTTTGTTTTA
ATATTTCAAAAAAAAAAAAAAAAAA

(261) The nucleic acid molecule SAM001 comprises, in the 5'.fwdarw.3' direction of transcription: L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-LA-3'UTR-PolyA wherein, L1 is a first linker having the sequence SEQ ID NO: 43 (annotated by underlining), Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker having the sequence SEQ ID NO: 44 (annotated by underlining), Pr2 is a second promoter sequence having the sequence SEQ ID NO: 47 (annotated in bold), 5'UTR is a 5'

ATGAGGAAATAAATACTGATAAAGGAATTGGACAAAGAAATGAAGGAGCTGGCCGCCCGCTCA
TGAGCGACCCTGACCTGGAAACTGAGACTATGTGCCTCCACGACGACGAGTCGTGT
CGCTACGAAGGGCAAGTCGCTGTTTACCAGGATGTATACGCGGTTGACGGACCGAC
AAGTCTCTATCACCAAGCCAATAAGGGAGTTAGAGTCGCCTACTGGATAGGCTTTGA
CACCACCCCTTTTATGTTTAAGAACTTGGCTGGAGCATATCCATCATACTCTACCAAC
TGGGCCGACGAAACCGTGTTAACGGCTCGTAACATAGGCCTATGCAGCTCTGACGTT
ATGGAGCGGTCACGTAGAGGGATGTCCATTCTTAGAAAGAAGTATTTGAAACCATC
CAACAATGTTCTATTCTCTGTTGGCTCGACCATCTACCACGAGAAGAGGGACTTACT
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TCCAGGCCTGTATGGGAAGCCTTCAGGCTATGCTGCTACGATGCACCGCGAGGGATT
CTTGTGCTGCAAAGTGACAGACACATTGAACGGGGAGAGGGTCTCTTTTCCCGTGTG
CACGTATGTGCCAGCTACATTGTGTGACCAAATGACTGGCATACTGGCAACAGATGT
CAGTGCGGACGACGCGCAAAACTGCTGGTTGGGCTCAACCAGCGTATAGTCGTCA
ACGGTCGCACCCAGAGAAACACCAATACCATGAAAAATTACCTTTTGCCCGTAGTG
GCCCAGGCATTTGCTAGGTGGGCAAAGGAATATAAGGAAGATCAAGAAGATGAAA
GGCCACTAGGACTACGAGATAGACAGTTAGTCATGGGGTGTTGTTGGGCTTTTAGAA
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GTGGAGACACCTCGTGGCTTGATAAAGGTTACCAGCTACGATGGCGAGGACAAGAT
CGGCTCTTACGCTGTGCTTTCTCCGCAGGCTGTACTCAAGAGTGAAAAATTATCTTG
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GAGTTCGTAAACAGGTACCTGCACCATATTGCCACACATGGAGGAGCGCTGAACAC
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CCAAGAAAGAAAACTGTGCAGAAATTATAAGGGACGTCAAGAAAATGAAAGGGCT
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GACTCCGGTCTATTTTCTGCACCCACTGTTCCGTTATCCATTAGGAATAATCACTGGG
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CTCGCAGGTACCCACAACCTGCCTCGGGCAGTTGCCACTGGAAGAGTCTATGACATG
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GTCCCAGGCAAAATGGTTGACTGGTTGTCAGACCGGCCTGAGGCTACCTTCAGAGCT
CGGCTGGATTTAGGCATCCCAGGTGATGTGCCCAAATATGACATAATATTTGTTAAT
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CAGCATAGGTTATGGTTACGCTGACAGGGCCAGCGAAAGCATCATTGGTGCTATAG
CGCGGCAGTTCAAGTTTTTCCCGGGTATGCAAACCGAAATCCTCACTTGAAGAGACGG
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TTATAAATGCTGCTAACAGCAAAGGACAACCTGGCGGAGGGGTGTGCGGAGCGCTG
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GACTCTCAAGGAAGCAGTGGCTAGGAGAGAAGCAGTGGAGGAGATATGCATATCCG
ACGACTCTTCAGTGACAGAACCTGATGCAGAGCTGGTGAGGGTGCATCCGAAGAGT
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GGTCTCCAACCCGCCAGGCGTAAATAGGGTGATTACAAGAGAGGAGTTTGAGGCGT
TCGTAGCACAAACAACAATGACGGTTTGATGCGGGTGCATACATCTTTTCTCCGACA
CCGGTCAAGGGCATTTTACAACAAAAATCAGTAAGGCCAAACGGTGCTATCCGAAGTG
GTGTTGGAGAGGACCGAATTGGAGATTTCTGATGCCCCGCGCCTCGACCAAGAAAA
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GATACCAGTCCAGGAAGGTGGAGAACATGAAAGCCATAACAGCTAGACGTATTCTG
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GCAGTGGAAGCCTGTAACGCCATGTTGAAAGAGAACTTTCCGACTGTGGCTTCTTAC
TGTATTATTCCAGAGTACGATGCCTATTTGGACATGGTTGACGGAGCTTCATGCTGC
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TGACCGCGTTAATGATTCTGGAAGACTTAGGTGTGGACGCAGAGCTGTTGACGCTGA
TTGAGGCGGCTTTCGGCGAAATTTTCATCAATACATTTGCCCACTAAAATAATTTA
AATTCGGAGCCATGATGAAATCTGGAATGTTCTCACACTGTTTGTGAACACAGTCA
TTAACATTGTAATCGCAAGCAGAGTGTTGAGAGAACGGCTAACCGGATCACCATGT
GCAGCATTTCATTGGAGATGACAATATCGTGAAAGGAGTCAAATCGGACAAATTAAT
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TGGGCGAGAAAGCGCCTTATTTCTGTGGAGGGTTTATTTTGTGTGACTCCGTGACCG
GCACAGCGTGCCGTGTGGCAGACCCCTAAAAAGGCTGTTTAAGCTTGGCAAACCT
CTGGCAGCAGACGATGAACATGATGATGACAGGAGAAGGGCATTGCATGAAGAGTC
AACACGCTGGAACCGAGTGGGTATTCTTTTCAGAGCTGTGCAAGGCAGTAGAATCAA
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AATGGACTACGACATAGTCTAGTCCGCCAAGTTCTGAAGGCGCGCCTCTAGAGCCA
CCATGGTGAGCAAGGGCGAGGAGCTGTTACCGGGGTGGTGCCCATCCTGGTC
GAGCTGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCG
AGGGCGATGCCACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGGC
AAGCTGCCCCGTGCCCTGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCA
GTGCTTCAGCCGCTACCCCGACCACATGAAGCAGCACGACTTCTTCAAGTCCG
CCATGCCCCGAAGGCTACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGC
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CATCGAGCTGAAGGGCATCGACTTCAAGGAGGACGGCAACATCCTGGGGCACA
AGCTGGAGTACAACACTACAACAGCCACAACGTCTATATCATGGCCGACAAGCAG
AAGAACGGCATCAAGGTGAACCTTCAAGATCCGCCACAACATCGAGGACGGCAG
CGTGCAGCTCGCCGACCACTACCAGCAGAACACCCCCATCGGCGACGGCCCCG
TGCTGCTGCCCCGACAACCACTACCTGAGCACCCAGTCCGCCCTGAGCAAAGAC
CCCAACGAGAAGCGCGATCACATGGTCCTGCTGGAGTTCGTGACCGCCGCCGG
GATCACTCTCGGCATGGACGAGCTGTACAAGTAGCATCGATGATATCGCGGCCG
CATACAGCAGCAATTGGCAAGCTGCTTACATAGAACTCGCGGCGATTGGCATGCCG
CCTTAAAATTTTATTTTATTTTCTTTTCTTTTCCGAATCGGATTTTGTTTTAAATAT
TCAAAAAAAAAAAAAAAAAA

(263) The nucleic acid molecule SAM002 comprises, in the 5'.fwdarw.3' direction of transcription: L1-Ori-SM-Pr1-L2-Pr2-5'UTR-nsP-SGP-L3-GOI-L4-3'UTR-PolyA

wherein, L1 is a first linker having the sequence SEQ ID NO: 43 (annotated by underlining), Ori is an origin of replication sequence, SM is a selectable marker, Pr1 is a first promoter sequence, L2 is a second linker having the sequence SEQ ID NO: 44 (annotated by underlining), Pr2 is a second

promoter sequence having the sequence SEQ ID NO: 47 (annotated in bold), 5'UTR is a 5' untranslated region having the sequence ATAGG (annotated by underlining), nsP is a plurality of non-structural replicase domain sequences, SGP is a subgenomic promoter having the sequence SEQ ID NO: 9 (annotated in bold), L3 is a third linker having the sequence SEQ ID NO: 45 (annotated by underlining), GOI is GFP, LA is a fourth linker having the sequence SEQ ID NO: 46 (annotated by underlining), 3'UTR is a 3' untranslated region having the sequence SEQ ID NO: 49 (annotated by underlining), and Poly-A is a 3' poly-adenylated tail. Wherein the first expression unit comprises, in the 5'.fwdarw.3' direction of transcription L1-Ori-SM-Pr1-L2 (annotated with italics). Wherein the second expression unit comprises, in the 5'.fwdarw.3' direction of transcription Pr2-5'UTR-nsP-SGP-L3-GOI-LA-3'UTR-PolyA. Wherein the sa-mRNA comprises, in the 5'.fwdarw.3' direction of transcription 5'UTR-nsP-SGP-L3-GOI-LA-3'UTR-PolyA. The nucleic acid molecule of the present disclosure is preferably a closed circular molecule prior to cleavage and a linear molecule after cleavage.

(264) TABLE-US-00023 SAM003 (SEQ ID NO: 37)

AAAAAAAAAACGCGTGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAA
AGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCC
CTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGA
CTATAAAGATAACCAGGCGTTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCG
ACCCTGCCGCTTACCGGATACCTGTCCGCCCTTTCTCCCTTCGGGAAGCGTGGCGCTTT
CTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTTCGCTCCAAGCTGG
GCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATC
GTCTTGAGTCCAACCCGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTA
ACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGG
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CCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTTCGTTTCATCCATA
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CCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCA
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CTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAA
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CGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAA
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CCAAGTATCTTCAGCATCTTTTACTTTTACCAGCGTTTCTGGGTGAGCAAAAACAG
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CATACTCTTCCTTTTTTCAATATTATTGAAGCATTATCAGGGTTATTGTCTCATGAGC
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CCCCGAAAAGTGCCACCTGAGCTCTAATACGACTCACTATAGgAtaGGCGGCGCATG
AGAGAAGCCCAGACCAATTACCTACCCAAAATGGAGAAAGTTCAGTTGACATCGA
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AGCCAAGCAGGTCACCTGATAATGACCATGCTAATGCCAGAGCGTTTTTCGCATCTGGC

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[illegible]

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CATTGGTGCTATAGCGCGGCAGTTCAAGTTTTCCCGGGTATGCAAACCGAAATCCTC
ACTTGAAGAGACGGAAGTTCTGTTTGTATTTCATTGGGTACGATCGCAAGGCCCGTAC
GCACAATtCTTACAAGCTTTTCATCAACCTTGACCAACATTTATACAGGTTCCAGACTC
CACGAAGCCGGATGTGCACCCTCATATCATGTGGTGCGAGGGGATATTGCCACGGC
CACCGAAGGAGTGATTATAAATGCTGCTAACAGCAAAGGACAACCTGGCGGAGGGG
TGTGCGGAGCGCTGTATAAGAAATTCGCGGAAAGCTTCGATTTACAGCCGATCGAA
GTAGGAAAAGCGCGACTGGTCAAAGGTGCAGCTAAACATATCATTTCATGCCGTAGG
ACCAAACCTTCAACAAAGTTTCGGAGGTTGAAGGTGACAAACAGTTGGCAGAGGCTT
ATGAGTCCATCGCTAAGATTGTCAACGATAACAATTACAAGTCAGTAGCGATTCCAC
TGTTGTCCACCGGCATCTTTTCCGGGAACAAAGATCGACTAACCCAATCATTGAACC
ATTTGCTGACAGCTTTAGACACCACTGATGCAGATG Contig 5 (SEQ ID NO: 31)

ATTGAACCATTTGCTGACAGCTTTAGACACCACTGATGCAGATGTAGCCATATACTG
CAGGGACAAGAAATGGGAAATGACTCTCAAGGAAGCAGTGGCTAGGAGAGAAGCA
GTGGAGGAGATATGCATATCCGACGACTCTTCAGTGACAGAACCTGATGCAGAGCT
GGTGAGGGTGCATCCGAAGAGTTCTTTGGCTGGAAGGAAGGGCTACAGCACAAGCG
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CATGTATATCCTCGGAGAAAGCATGAGCAGTATTAGGTTCGAAATGCCCCGTCGAAG
AGTCGGAAGCCTCCACACCACCTAGCACGCTGCCTTGCTTGTGCATCCATGCCATGA
CTCCAGAAAGAGTACAGCGCCTAAAAGCCTCACGTCCAGAAACAAATTACTGTGTGC
TCATCCTTTCCATTGCCGAAGTATAGAATCACTGGTGTGCAGAAAGATCCAATGCTCC
CAGCCTATATTGTTCTCACCGAAAGTGCCTGCGTATATTCATCCAAGGAAGTATCTC
GTGGAAACACCACCGGTAGACGAGACTCCGGAGCCATCGGCAGAGAACCAATCCAC
AGAGGGGACACCTGAACAACCACCACTTATAACCGAGGATGAGACCAGGACTAGA

CACGCGGTCTGGACACGCCGGAGAGCGTCGAAGCGGGGGCGGTGTTCCGCCGAGATC
GGCCCGCGCATGGCCGAGTTGAGCGGTTCCCGGCTGGCCGCGCAGCAACAGATGGA
AGGCCTCCTGGCGCCGCACCGGCCCAAGGAGCCCGCGTGGTTCCTGGCCACCGTCG
GCGTCTCGCCCGACCACCAGGGCAAGGGTCTGGGCAGCGCCGTCGTGCTCCCCGGA
GTGGAGGCGGCCGAGCGCGCCGGGGTGCCCGCCTTCCTGGAGACCTCCGCGCCCCG
CAACCTCCCCTTCTACGAGCGGCTCGGCTTCACCGTCACCGCCGACGTCGAGGTGCC
CGAAGGACCGCGCACCTGGTGCATGACCCGCAAGCCCGGTGCCTGACATCGATGAT
ATCGCGGCCGCGCATAAGCAGCAATTGGCAAGCTGCTTACATAGA AACTCGCGGCGAT
TGGCATGCCGCCTTAAAATTTTTATTTTATTTTCTTTTCTTTTCCGAATCGGATTTTG
TTTTTAATATTTCAAAAAAAAAAAAAAAAAA

(265) In addition, it is to be understood that any particular aspect of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Since such aspects are deemed to be part of the whole of the present disclosure, any part of the whole disclosure may be excluded even if the exclusion is not set forth explicitly herein.

(266) It is to be understood that while the present disclosure has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the present disclosure, which is defined by the scope of the appended claims. Other aspects, advantages, and alterations are within the scope of the following claims.

Claims

1. A self-amplifying mRNA comprising from 5' to 3': a) a 5' untranslated region (UTR), nucleotides 1869 to 9441 of SEQ ID NO: 35 with one or more substitutions selected from C4491A/T5177C/A5361C/A5735G/A6452G/G7663A/A7765G, one or more genes of interest (GOI) that optionally comprises a linker (L) at the 5' and/or 3' end, a 3' UTR, and a poly-adenosine (poly-A) tail; or b) a 5' UTR, nucleotides 1869 to 9441 of SEQ ID NO: 36, the GOI that optionally comprises the L at the 5' and/or 3' end, a 3' UTR, and a poly-A tail; or c) a 5' UTR, nucleotides 1869 to 9441 of SEQ ID NO: 36 with one or more substitutions selected from C4491A/T5177C/A5361C/A5735G/A6452G/G7663A/A7765G, the GOI that optionally comprises the L at the 5' and/or 3' end, a 3' UTR, and a poly-A tail.
2. The self-amplifying mRNA of claim 1, wherein the GOI is an antigen or antigen receptor.
3. The self-amplifying mRNA of claim 1, wherein the GOI is a viral antigen.
4. The self-amplifying mRNA of claim 3, wherein the GOI is a modified SARS-CoV-2 spike protein.
5. The self-amplifying mRNA of claim 1, wherein the GOI is a cytokine, a chemokine, or other immune stimulator or inhibitor.
6. The self-amplifying mRNA of claim 1, wherein the GOI is selected from: a) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 1 (BA.1-1273); b) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 2 (BA.1-1273-S2P); c) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 3 (BA.2-1273); d) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 4 (BA.2-1273-S2P); e) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 5 (BA.1-1208); f) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 6 (BA.1-1208-S2P); g) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 7 (BA.2-1208); or h) a polynucleotide encoding a modified SARS-CoV-2 spike protein comprising the nucleic sequence set forth in SEQ ID NO: 8 (BA.2-1208-S2P).

7. The self-amplifying mRNA of claim 1, wherein the GOI encodes two separated expression units, the nucleic acid comprising: i) a first expression unit comprising a polynucleotide encoding a modified antigen, wherein the polynucleotide encoding the modified antigen is truncated to not include nucleotides encoding a transmembrane domain and short cytosolic domain amino acids of the antigen, operably linked to a first subgenomic promoter; and ii) a second expression unit encoding immunomodulators (IM) that are operably linked to a second subgenomic promoter.
8. The self-amplifying mRNA of claim 7, wherein the polynucleotide sequence encoding the modified antigen comprises replacement of a transmembrane domain of the antigen with a secretion antigen.
9. The self-amplifying mRNA of claim 7, wherein the antigen is a modified SARS-CoV-2 spike protein, wherein the polynucleotide has been truncated to not include nucleotides encoding a SARS-CoV-2 transmembrane domain and short cytosolic domain amino acids.
10. The self-amplifying mRNA of claim 7, wherein the polynucleotide sequence encoding a coronavirus spike protein truncated to not include nucleotides encoding a SARS-CoV-2 transmembrane domain and short cytosolic domain amino acids corresponding to amino acids 1209-1273 of a nucleotide sequence is SEQ ID NOs: 1 (BA.1-1273) or 3 (BA.2-1273).
11. The self-amplifying mRNA of claim 7, wherein the sa-mRNA comprises the following operably linked nucleic acid sequence from 5' to 3': nsP-SGP1-Ag-SGP2-IM wherein nsP is a plurality of non-structural replicase domain sequences, SGP1 is the first subgenomic promoter, Ag is a nucleotide sequence selected from SEQ ID NO: 1 (BA.1-1273), 2 (BA.1-1273-S2P), 3 (BA.2-1273), and SEQ ID NO: 4 (BA.2-1273-S2P), SEQ ID NO: 5 (BA.1-1208), or SEQ ID NO: 6 (BA.1-1208-S2P), SEQ ID NO: 7 (BA.2-1208), or SEQ ID NO: 8 (BA.2-1208-S2P), SGP2 is the second subgenomic promoter, and IM is the immunomodulator.
12. A nucleic acid encoding a self-amplifying mRNA comprising a mutant T7 promoter comprising the nucleotide sequence of SEQ ID NO: 47 (TAATACGACTCACTATAGG) operably linked to a sequence encoding the self-amplifying mRNA of claim 1.
13. The nucleic acid of claim 12, wherein the 5' UTR comprises the nucleotide sequence ATAGG.
14. The self-amplifying mRNA of claim 1, wherein at least one GOI encodes: at least one regulatory structure, wherein the at least one regulatory structure is a small interfering RNA (siRNA), a micro-RNA (miRNA), a guide RNA (gRNA), a self-activating RNA (saRNA), a transfer RNA (tRNA), or a long intergenic non-coding (lincRNA); and/or an infectious disease antigen, an allergic antigen, or a tumor antigen.
15. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM001 (SEQ ID NO: 35).
16. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM002 (SEQ ID NO: 36).
17. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM003 (SEQ ID NO: 37).
18. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM004 (SEQ ID NO: 38).
19. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM005 (SEQ ID NO: 39).
20. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA has at least 95% sequence identity to nucleotides 1869 to 10325 of SAM006 (SEQ ID NO: 40).
21. The self-amplifying mRNA of claim 16, wherein the self-amplifying mRNA comprises point mutations C4491A and T5177C.
22. The self-amplifying mRNA of claim 16, wherein the self-amplifying mRNA comprises point mutations C4491A and A6452G.
23. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA comprises from 5' to 3': nucleotides 1869 to 9441 of SEQ ID NO: 36, the GOI, and nucleotides 10171 to 10325 of

SEQ ID NO: 36.

24. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA comprises from 5' to 3': nucleotides 1869 to 9441 of SEQ ID NO: 37, the GOI, and nucleotides 10171 to 10325 of SEQ ID NO: 37.

25. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA comprises from 5' to 3': nucleotides 1869 to 9441 of SEQ ID NO: 38, the GOI, and nucleotides 10171 to 10325 of SEQ ID NO: 38.

26. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA comprises from 5' to 3': nucleotides 1869 to 9441 of SEQ ID NO: 39, the GOI, and nucleotides 10171 to 10325 of SEQ ID NO: 39.

27. The self-amplifying mRNA of claim 1, wherein the self-amplifying mRNA comprises from 5' to 3': nucleotides 1869 to 9441 of SEQ ID NO: 40, the GOI, and nucleotides 10171 to 10325 of SEQ ID NO: 40.
